

claude-windows / 45ea3a91-654d-4972-9cd3-65f35db54caf

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\45ea3a91-654d-4972-9cd3-65f35db54caf\subagents\workflows\wf_2a64bc88-2a8\agent-a470a64c55d7a42a5.jsonl`  
- SHA-256: `8eba2a0758ba29695d955b826acc425b59b6df5f657a0d173ce8347b8f90f14b`  
- Source modified: `2026-06-10T05:41:39+00:00`  
- Imported at: `2026-07-05T16:48:27+00:00`  
- project: `wf_2a64bc88-2a8`  
- session_id: `45ea3a91-654d-4972-9cd3-65f35db54caf`
```

Transcript

1. user (2026-06-10T05:33:14.936Z)

CONTEXT ? two sibling repos on a Windows machine:

- INDEXER: C:/Users/avidu/Projects/badminton-highlight-indexer ? local AI badminton(->racket-sports) highlight indexer + rally detector. FastAPI + SQLite + ffmpeg + GPU CV via WSL; Gemini = paid cloud AI. docs/ tree is rich. Current branch docs/next-steps-multisport (master = main branch).

- COLLECTOR: C:/Users/avidu/Projects/sports-data-collector ? golden-label acquisition/curation/ingestion CLI (system-of-record candidate for training corpus).

Project state (June 2026): scaffolding complete; owned-model roadmap agreed (docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md) but NO platform/owned-model implementation started; next committed platform slice = async job model. Licensing guardrail: MIT/Apache-2.0/BSD only for code+data+weights; paid LLMs are inference-only, never training-data sources. ENVIRONMENT RULES: this dev environment has NO GPU/torch/cv2 ? do NOT attempt to run the video pipeline or import torch/cv2 modules. Test suites are offline/fully-mocked and SAFE to run. You are READ-ONLY: never edit/create/delete repo files; running tests and git read commands is allowed.

QUALITY BAR: cite file paths (and line numbers where possible) for every finding. Report what IS in the code/docs, not what you assume. Be skeptical and concrete. Your final structured output is consumed programmatically by an orchestrator ? return raw data, not prose for a human.

TASK: STRATEGY-DOC CONSISTENCY sweep of the indexer. Read FULLY, in this order: CLAUDE.md (repo root), docs/README.md, docs/NEXT_STEPS.md (new, on current branch), docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md, docs/ROADMAP.md, docs/DEFERRED_HARDENING_PLAN.md, docs/BACKLOG.md, docs/COMMERCIALIZATION.md, docs/PLATFORM_ARCHITECTURE.md, docs/PMF_MARKET_RESEARCH.md, docs/archives/decisions/DECISIONS.md, docs/archives/checkpoints/2026-06-strategy-foundation/README.md, docs/archives/research/OWNED_MODEL_TRAINING_STUDY.md (at least its correction banner + conclusions).

FIND, as findings: (a) CONTRADICTIONS between docs ? e.g. licensing policy wording (CLAUDE.md says 'open Apache-2.0 only' for training teachers/bases; later ratified policy is 'MIT/Apache-2.0/BSD' ? where do docs disagree?); what is 'the next slice' (async job model vs Phase-0 M0.x vs repo rename ? do docs agree on sequencing and on who decides?); v1 scope definitions; Gemma-4 stance (AMBER/bench-only vs candidate); WASB checkpoint provenance stance across docs; conversational-QA punt-vs-build framing. (b) CLAIMS CONTRADICTED BY REPO REALITY ? verify with Grep/Glob: does training.label_candidates still carry Gemini-derived labels the plan says must be purged (check backend + DB-related code/docs for what actually exists ? note output/ DB itself is gitignored so check code paths + docs claims)? does config.json default_segmenter match what docs claim? do doc-referenced files/flags exist (e.g. output/human_lovo_manifest.json is gitignored ? do docs point at artifacts a fresh clone won't have)? does the repo-rename action item appear anywhere actionable? (c) ROADMAP GAPS / missed gotchas ? sequencing conflicts (async-job slice vs Phase-0 both 'next'), undefined ship-gate F1@tIoU target, cost-to-Gen-2 unquantified, per-sport detector blocker vs the new finding that WASB transfers to table tennis (docs/NEXT_STEPS.md) ? is that finding consistently reflected or does it contradict older 'binding blocker' framing?

Each finding must QUOTE both sides (doc A says X at path:line-ish; doc B / code says Y).

Severity: high = a reader following docs would act wrongly; medium = confusing drift; low = cosmetic.

2. user (2026-06-10T05:33:20.644Z)

```
1      # CLAUDE.md ? project memory / agent handoff
2
3      Local, AI-powered badminton (? racket-sports) highlight indexer + rally detector**:
4      FastAPI + SQLite + ffmpeg + GPU CV/AI, with paid Gemini as the cloud AI. Sibling
5      repos: `sports-data-collector`, a VLC-based `rally-annotator`.
6
7      ## Read these first
8      - docs/README.md ? doc index. docs/CODE_MAP.md ? code navigation + data
9        contracts (the cold-start map; read before touching code).
10     - Strategy (post-scaffolding): docs/COMMERCIALIZATION.md,
11     docs/PMF_MARKET_RESEARCH.md,
12     docs/PLATFORM_ARCHITECTURE.md, docs/DEFERRED_HARDENING_PLAN.md. Also
13     docs/ROADMAP.md
14     (offline-segmenter narrative) and docs/BACKLOG.md.
15     - History: docs/archives/ ? ADRs (decisions/DECISIONS.md), research, and
16     checkpoints/ (latest: 2026-06-strategy-foundation/). Archive policy: only move
17     terminal-state (DONE/REJECTED) work; live docs stay at docs/ top level.
18
19     ## Current state (June 2026)
20     - Scaffolding complete; the post-scaffolding strategy foundation is laid and
21     merged
22     (#52/#53/#55/#57/#58). No platform/owned-model implementation has started.
23     - Next slice = async job model (DEFERRED_HARDENING_PLAN.md #16), single-tenant.
24     Do not start implementation without explicit owner approval.
25
26     ## Architecture in one breath
27     - Pluggable registries (ABC + Registry singleton): segmenters / providers /
28     validators
29     / sports / annotations ? see backend/pipeline/segmenters/base.py. Future:
30     StorageBackend + ComputeBackend registries (PLATFORM §3).
31     - video_id = MD5 of the file ? backend/utils/hashing.py::compute_video_id (one
32     helper).
33     - Typed config = backend/config/ (Pydantic). DB schema evolves via the PRAGMA
34     user_version migration ladder in backend/infrastructure/database.py.
35
36     ## Committed guardrails (non-negotiable)
37     - Paid LLMs (Gemini/OpenAI/Anthropic) = inference / serving / fallback ONLY ? never a
38     training data source (terms/copyright). Training teachers/bases = open Apache-2.0
39     only.
40     - Owned-model base: Apache-2.0, video-capable (Qwen-VL / Gemma 4); reject Gemma
41     3
42     (viral license). Exclude minors from the training pool; per-user training data
43     needs a
44     separate explicit opt-in.
45     - v1 = rally + highlight + history + correction loop; gait/biometrics strictly
46     future
47     (GDPR Art. 9).
48     - Generic data schema; coach ? athlete is an optional overlay, not baked in.
49
50     ## Workflow conventions
51     - Branch off master; ONE PR per work item; NO PR stacking. Don't open a PR unless
52     asked.
53     - Tests: python run_all_tests.py (offline, fully mocked). Python 3.8+; ffmpeg
54     required.
55     - Never push to the default branch directly; descriptive commit messages.
56     - ?? This dev container lacks GPU / torch / cv2 / numpy ? the full pipeline can't run
57     here.
58     - Verify pure-logic paths and hand machine-side checks (real video, GPU/WSL suite) to
59     the owner.
```

3. user (2026-06-10T05:33:21.463Z)

```
1      # docs/
2
3      Live documentation for the badminton-highlight-indexer.
4
5      ## For new contributors / ramp-up (read these first)
6      - [CODE_MAP.md](CODE_MAP.md) ? Complete navigation map of the codebase and data
contracts.
7      - [COMMERCIALIZATION.md](COMMERCIALIZATION.md) ? Living tracker for the commercial
dimension (licensing, codec/output economics, hosting, vendoring, packaging) + PMF risks and the
quality/data-phase priorities. Supersedes the archived `COMMERCIAL_READINESS_REVIEW.md`.
8      - [PMF_MARKET_RESEARCH.md](PMF_MARKET_RESEARCH.md) ? Cited market study: sizing,
customer segments, competitor pricing, and the recommended beachhead (feeds COMMERCIALIZATION
§6).
9      - [PLATFORM_ARCHITECTURE.md](PLATFORM_ARCHITECTURE.md) ? Strategy for the multi-user
"assistant coach" platform: decoupled layers + pluggable storage/compute (BYO), multi-tenant
data model, orchestration, and a phased roadmap.
10     - [DEFERRED_HARDENING_PLAN.md](DEFERRED_HARDENING_PLAN.md) ? Design (pending approval)
for the larger deferred items: packaging, train/eval guardrails, and the async job model (#16,
the first infra slice toward the platform).
11     - [BACKLOG.md](BACKLOG.md) ? Deferred items with rationale and revisit triggers.
12     - [ROADMAP.md](ROADMAP.md) ? High-level four-phase product direction.
13     - [VIDEO_RECORDING_GUIDELINES.md](VIDEO_RECORDING_GUIDELINES.md) ? Capture best
practices (referenced by reports).
14     - [HOW_RALLY_DETECTION_WORKS.md](HOW_RALLY_DETECTION_WORKS.md) ? Mental model of the
two-layer detection system.
15     - [EVALUATION.md](EVALUATION.md) ? How we measure and improve quality.
16
17     ## Practical / ongoing
18     - [TRACKNET_WSL_SETUP.md](TRACKNET_WSL_SETUP.md)
19     - Golden set docs (`GOLDEN_SET`)
20     - Evaluation harness and calibration drivers (in `backend/eval/`)
21
22     ## Historical documents
23     Completed plans, old research, ADRs, and checkpoints live in [`archives/`](archives/).
24
25     See `archives/README.md` for the umbrella policy and `archives/past_projects/README.md`
for classification of small tech debt / starter projects.
26
```

4. user (2026-06-10T05:33:26.220Z)

```
1      # Next steps (live) ? owned-model + multisport
2
3      **Updated:** 2026-06-09 (end of the quality-iteration + multisport-recon session).
**Authoritative roadmap:**
4      `docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md`. **Latest blow-by-blow:** memory `current-
state.md` / `session-handoff.md`.
5      This file = the prioritized "what to do next," refreshed each session.
6
7      ## The recommendation (one paragraph)
8      Build the **quality-first extraction harness** before anything else: a learned **TAL
head** (ActionFormer/ASFormer, MIT)
9      over trajectory features is the robust path ? it **decouples from the multi-court
"background games" confound** that the
10     unsupervised heuristic (and audio, and spatial court-gating) can't escape (all three
tested + failed; it's a *temporal*
11     rally-vs-non-rally problem). **Framework = ms-swift, base = Qwen3-VL** (Gemma-4
AMBER/bench) come **later** for the
12     conversational/grounding upgrade ? start with the head. **On-device is deferred**
(export-clean ops now so it stays
13     reachable). Everything stays **MIT/Apache-2.0/BSD** (code + data + weights). Bottleneck
= **data + per-sport trajectory
14     detectors**, not model architecture.
```

```

15
16     ## Latest findings (2026-06-09)
17     - **6-match badminton LOVO:** learned prototype **0.583** vs heuristic **0.480**
    (learned leads since n=3, AUC 0.83?0.92,
18     decoupled from multi-court). The "contrast" metric (in-rally÷dead-time visibility)
    predicts heuristic F1 monotonically.
19     - **?? Multisport (Roora pickleball + TT):** labels **ingest cleanly** (`curate convert-
    cvat` ? 6 rallies/sport, sane).
20     **WASB-transfer SURPRISE (directional, n=6 each ? TINY):** the badminton shuttle
    detector segments **TABLE TENNIS at
21     F1 0.909** out-of-the-box (!) but **pickleball only 0.308**. ? **The per-sport-
    detector blocker is SPORT-DEPENDENT:** TT may
22     be servable with WASB short-term; pickleball needs a proper (MIT/Apache) ball
    detector. Confirm/deny with more videos.
23     - **Owner field notes (2026-06-09):** camera height/distance **varied on purpose**
    (great generalization data); **multi-court
24     adjacent-court SOUND dominates the signal** ? empirically reconfirms audio-foreground
    isolation won't work + the
25     robustness-first learned-model path. Multi-court is the realistic (academy)
    environment, so this *is* the target distribution.
26
27     ## Prioritized next steps
28     1. **[owner greenlight] Phase-0 (M0.x), $0/local:**
29     - **M0.4 ? Gen-0 TAL harness** (first-class): productionize `rally_seq_proto.py` ?
    one-command train?eval?***ship-gate** over
30     ActionFormer/ASFormer, reusing `metrics.match_intervals`. **Define the ship-gate
    target** (explicit F1@tIoU; floor = beat
31     heuristic LOVO 0.480) ? "very high P/R" must be operationalized, not just "beat the
    floor."
32     - **M0.1 ? corpus spine + the per-video event-index contract** (the conversational
    seam: extract ? event store ? text-to-SQL).
33     - **M0.3 ? compliance:** purge Gemini-derived training labels
    (`training.label_candidates`); WASB **C3** own-weights/clean-swap (gates ship).
34     2. **Multisport thread:**
35     - Ingest the Roora pickleball/TT rally CSVs into the corpus (collector `import-
    local`).
36     - **Get more pickleball + TT videos** ? re-run the WASB-transfer eval to confirm/deny
    TT?0.91, pickleball?0.31 (n=6 is too few).
37     - Once each sport has ?3 matches, run a **merged prototype LOVO across badminton + TT
    (+ pickleball)** ? the "how would a
38     merged model do" experiment. Identify a clean MIT/Apache ball detector for sports
    where WASB doesn't transfer (pickleball).
39     3. **Housekeeping:** **rename repo off "badminton" (multisport); confirm collector =
    corpus/event-store system-of-record.
40
41     ## Pointers / artifacts
42     - Roadmap: `docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md` · Study:
    `docs/archives/research/OWNED_MODEL_TRAINING_STUDY.md`
43     (2026-06-09 banner) · Quality: `docs/archives/quality-iterations/2026-06-multivideo-
    lovo/`.
44     - Eval corpus: `output/human_lovo_manifest.json` (badminton n=6). Roora CSVs:
    `%TEMP%\roora\{pickleball,tabletennis}_rallies.csv`;
45     WASB-transfer trajectories: `%TEMP%\{pickleball,tabletennis}_traj.csv`. Code:
    `backend/eval/{eval_partial_labels,rally_seq_proto,calibrate_local}.py`.
46     - ? No owned-model implementation until owner greenlights a milestone. Committed next
    PLATFORM slice = async job model (single-tenant).
47

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5. user (2026-06-10T05:33:26.547Z)

```

1     # Owned-model implementation plan (roadmap) ? DRAFT for owner review
2
3     **Status:** PLAN / roadmap. No owned-model *implementation* starts until the owner
    greenlights a specific
4     milestone. Paid Gemini stays the always-ready inference/fallback throughout (never a

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training source).
5      **Grounded in:** `docs/archives/research/OWNED_MODEL_TRAINING_STUDY.md`, the n=6 LOVO +
learned prototype
6      (`docs/archives/quality-iterations/2026-06-multivideo-lovo/`), and a verified strategy
workflow (2026-06-09).
7
8      ## Design principle (owner directive): *envision now, build later*
9      Every deferred piece (conversational-QA, on-device, multisport, the VLM) must be
**architecturally envisioned
10     now** so we never hit a mammoth refactor. We **design the seams** (data contracts,
export boundaries, the
11     event store) up front and **implement behind them** when each milestone is greenlit.
Punting a *feature* is fine;
12     punting its *seam* is not.
13
14     ## North Star (reframed per PR #61 review)
15     - **Immediate deliverable:** a **training framework + harness** that produces a model
with **very high precision/
16     recall for extracting highlights + rallies from a clip**. *Quality is the first-order
objective.*
17     - **Eventual product:** a **single, sport-agnostic, conversational** video model ?
upload a clip, then ask
18     contextual questions ("longest rally", "make a clip of all service faults", "rallies
over 20 shots").
19     - **On-device is LATER** (a model-compression goal so power users save upload
cost/hassle) ? definitely on the
20     roadmap, but **not** the immediate target. Don't over-index the strategy on it now.
21     - **The moat = the consented dataset + the eval harness, not the weights.**
22
23     ## Hard guardrails (non-negotiable ? "never corrupted")
24     - **Commercial-clean licensing for EVERY dependency ? code, data, AND model weights: MIT
/ Apache-2.0 / BSD only.**
25     (Ratifies the prior "Apache-2.0 only" to include MIT/BSD.) Reject viral/NC/custom-
restrictive: Gemma 3 (viral),
26     **Gemma 3n** (Gemma Terms follow derivatives ? even cleaner reject), Llama 4 (700M-MAU
cap), Commons-Clause repos.
27     - **Never train on paid-LLM (Gemini/OpenAI/Anthropic) outputs.** Also forbidden: public
grounding-CoT datasets that
28     were Gemini-generated (e.g. TVG-R1's 56K) ? we must originate our own CoT supervision
(a real, binding cost).
29     - **Exclude minors; per-user training data needs a separate explicit opt-in; split data
by match.**
30     - ? **Base-model pretraining provenance is unauditatable for *every* open VLM (incl.
Qwen3-VL).** Our "no paid-LLM
31     training" rule is enforceable at *our fine-tuning data* layer, not the base's
pretraining ? an **explicit owner
32     risk-acceptance** is required to use any open VLM. (Governed by the base's license,
not our pipeline.)
33
34     ## The model strategy (verified 2026-06-09)
35     **Framework ? base** (the Gemma-4-vs-ms-swift clarification): a *framework* is the forge
(runs SFT/GRPO, emits
36     weights); a *base* is the metal you forge. You pick one framework and feed it any clean
base.
37     - **Framework: ms-swift (Apache-2.0)** ? the one forge that does native **video + audio
+ timestamp-grounding +
38     GRPO/RFT with a custom video reward** in one stack. Keep **unsloth** (Apache-2.0 core;
*avoid its AGPL Studio UI*)
39     as the 8 GB-local SFT/compression tool for the deferred on-device goal. ? Point-in-
time snapshot ? re-verify at impl.
40     - **Base (when we reach the VLM): Qwen3-VL (Apache-2.0, uniform across sizes)** ? long-
video context, purpose-built
41     **timestamp emission** (second-level localization), multi-turn video chat. Re-confirm
the exact size's LICENSE
42     pre-release. **Gemma-4 stays AMBER and bench-only** ? its Apache grant is likely but

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its Prohibited-Use posture is

43 unconfirmed (counsel needed), *and* its video/timestamp capability is disputed; not the primary extractor.

44 **InternVL3 (MIT/Apache)** = clean fallback if a Qwen blocker appears. ? base swap is cheap *only* within

45 timestamp-capable Apache bases* (timestamp data-format + grounding-CoT are base-specific).

46 - **START WITH THE TAL HEAD, NOT THE VLM** (the decisive call): fine-tuned VLMs **miss strict temporal boundaries**

47 (best RL-post-trained video VLM ? R@0.7 50% on open-domain benchmarks; "coarse regions, not accurate boundaries").

48 **Boundary accuracy IS the product.** The head (`rally\_seq\_proto.py`) is already de-risked on our corpus (held-out

49 AUC 0.83?0.92), trains in minutes at ~4.5 GB on the 2070 Super (\$0), is ~1M params (trivially compressible later),

50 and **doubles as a pseudo-label teacher + the honest baseline the VLM must beat.** So: **head now ? VLM later**,

51 strictly sequenced. When the VLM comes, prefer **two-stage** (VLM coarse-proposes, head refines boundaries) over

52 wholesale replacement. *(Open: the VLM-ceiling number is open-domain; our amateur footage may differ ? re-measure.)*

53

54 ## Conversational video-QA ? build the bookkeeping NOW, punt the model

55 **Punt the conversational *feature*** until extraction quality is solid (current LOVO ~0.583 ? "very high"; a chat UI

56 over a weak extractor exposes the weakness). **Reject end-to-end-VLM-answers-each-turn** (expensive; bad at exact

57 counting; the strong teachers are license-forbidden). **But build the *seam* now** (the brief requires it):

58 - **Architecture = "extract once, query many":** EXTRACTION (TAL head ? later VLM) emits structured timestamped

59 spans ? a durable **per-video EVENT STORE** ? a **structured QUERY layer** (text-to-SQL + ffmpeg clip-cut). The

60 cited queries ("longest rally", "all service faults", "rallies > 20 shots") are **structured-numeric ? SQL, not a

61 VLM job** (SQL counts exactly where VLMs hallucinate).

62 - **Designed-now seam:** make the **per-video event-index schema a first-class data contract** alongside the M0.1

63 corpus spine: per rally `{video_id(MD5), rally_ordinal, sport, start, end, shot_count, ending_reason/fault_tags,

64 derived_clip_ref, provenance, confidence}`. \*\*Reserve the event/fault-tag taxonomy (e.g. `service_fault`) up front**

65 even if unpopulated. ? This is a real DB delta (a `PRAGMA user\_version` migration): `play\_segments` today has

66 `video\_id/start\_time/end\_time/duration/server/receiver/metadata` only; `highlights` isn't a table yet ? so it's an

67 *extension with new columns + a highlights table*, not a no-op. **Punt** the NL parser/agent until quality clears the bar.

68 - Paid Gemini may answer genuinely open-ended edge questions **at inference** over our structured index (allowed ?

69 it reasons over our data, it is not trained on its outputs).

70

71 ## Multisport (single sport-agnostic model)

72 - **Model/framework: UNCHANGED.** One shared TAL head + one Qwen3-VL base, fine-tuned across sports (sport = an input

73 feature / per-sport calibration). The collector + indexer are already sport-generic. ? **ACTION: rename the repo off

74 "badminton" ASAP** (owner emphatic) ? consistent with this.

75 - **What multisport changes = DATA + DETECTORS (scale up, the accepted "good problem"):** each new sport needs its own

76 labeled matches (split by match) + a **clean trajectory detector** to feed the head's features. **The binding blocker

77 is per-sport detectors:** WASB (badminton) has the unresolved **C3** checkpoint-provenance issue, and **clean MIT/

78 Apache/BSD shuttle/ball detectors for tennis/TT/pickleball/padel are not yet

```

identified.** Sequence by public-data
79     richness: **badminton ? tennis/TT ? pickleball/padel.**
80     - ? **Single-model-multisport-temporal-generalization is an OPEN HYPOTHESIS**, not
proven ? our corpus is badminton-only,
81     and the evidence once cited for it (RacketVision) was *refuted on verification* (it's
ball-tracking, not temporal). The
82     concept is sound on first principles (rally signal = a racquet-sport invariant) but
must be validated once a 2nd-sport
83     corpus exists. Don't cite RacketVision as evidence.
84
85     ## Language ? Python everywhere; on-device runtime is a deferred, layer-local export
shell
86     - **Training/eval = Python-locked, and that's fine** (PyTorch/ms-swift/ActionFormer);
offline/batch, no latency hot path.
87     - **Platform = no Python hot path to escape** ? the heavy bytes/GPU already run out-of-
process (ffmpeg, WASB-in-WSL, cloud
88     GPU per-job); the control plane is I/O-bound. **Reject a Go/Rust hot-path split** ?
single-digit-ms wins dwarfed by
89     ~40-min GPU jobs, and it would fracture the one codebase holding the QA-state + eval
harness + provider registry.
90     - **On-device (deferred) = the only place a 2nd language appears, and it doesn't dictate
our dev language:** train in
91     Python ? **export across the boundary** (ONNX/Core ML/ExecuTorch/GGUF) ? a thin device
shell (Swift/Kotlin/C++/Rust)
92     runs the artifact with zero Python. **The one concrete constraint now: train with
export-clean ops** so the compression
93     target stays reachable without re-architecting. (On-device base stays on the
Qwen3-VL-4B Apache lineage ? *not* Gemma-3-270M.)
94     - Escalation order if a real Python CPU hot path ever appears: free-threading ?
numpy/vectorize ? Cython/PyO3 ? only then a
95     separate service. **None exists today.**
96
97     ---
98
99     ## Phase 0 ? Foundation (start now, per-milestone greenlight; $0, local, no cloud/DPA)
100
101     ### M0.1 ? The labeled-corpus spine **+ the event-index contract** (the moat) .
*greenlight item*
102     Canonical JSONL row per rally (corpus) **and** the queryable per-video event-index
schema (the conversational seam)
103     ? same row, wired to a store. Fields: `{video_id(MD5), rally_ordinal, sport, start, end,
ending_reason/fault_tags,
104     shot_count, label_type, source(human|pseudo|vendor), consent/training_opt_in, split(BY
MATCH), trajectory_ref,
105     labeled_window, derived_clip_ref, confidence}`. 3 transcoders (heuristic/LogReg . TAL
ActivityNet-JSON . VLM clip?QA),
106     one scorer spine (`metrics.match_intervals`). Authored in `sports-data-collector`
(system-of-record; extends PR #13).
107     **Reserve fault/event taxonomy now.** Acceptance: round-trip test; split-by-match
enforced; **no Gemini-sourced rows in any training view**.
108
109     ### M0.2 ? Grow the corpus . **DONE (n=6), running** . *owner action + my scoring loop*
110     6 distinct matches labeled (3 singles incl. Mahadevpura, 1 doubles, 2 multi-court); per-
match + LOVO recorded; the
111     "contrast" data-quality metric validated. Owner adding 1 more varied multi-court-club
match ? margin. **Bottleneck =
112     data + calibration, not method.** Keep growing per new sport.
113
114     ### M0.3 ? Compliance cleanup (**hard blocker ? do before any training run**) .
*greenlight item*
115     - **(i) PURGE the Gemini-derived TRAINING labels** in `training.label_candidates` (over
Gemini CSVs) ? a **live
116     no-paid-LLM-training violation** that bakes in the moment a training run precedes it.
Retarget to human + open-teacher
117     (own-model/WASB) labels; Gemini = eval/reference/fallback only.

```

```

118     - **(ii) WASB C3 ? first-class action item (owner):** the academic-data checkpoint
provenance is a **commercial-ship
119     blocker** that a clean head does NOT launder. Resolve via **own-trained weights** or a
clean detector swap; **multiplies
120     per sport.** Carry as a tracked blocker that **gates SHIP** (not Gen-0 research).
121
122     ### M0.4 ? Gen-0 TAL harness (**FIRST-CLASS ACTION ITEM** per owner) · *greenlight item*
123     Productionize `rally_seq_proto.py` ? a **one-command train?eval?ship-gate** harness over
**ActionFormer (MIT, 1:1
124     rally=instance)** and/or **ASFormer (MIT, TAS)** (use `sj-li/MS-TCN2` if MS-TCN++, *not*
the Commons-Clause repo),
125     reusing `metrics.match_intervals`. **Define the ship-gate target up front** (open item):
a FLOOR of "beat the honest n=6
126     heuristic LOVO **0.480**" is necessary but not sufficient ? set an explicit **F1@tIoU**
target for "very high P/R" (e.g.
127     F1@tIoU=0.5 for highlights, tighter for precise rally cuts) before declaring victory.
Trains on the GPU/WSL box.
128
129     ---
130
131     ## Phase 1 ? Gen-0 ship decision (gated on n?5 + M0.4)
132     Run the A/B; if the learned Gen-0 clears the ship-gate with per-video calibration, it
becomes the rally/highlight
133     extractor (local, MIT). If not, the gap (more data/features) is the next signal ? no
overfitting to declare victory.
134
135     ## Phase 2 ? Gen-1 / Gen-2 (gated on platform scaffolding P0?P4 + healthy corpus +
explicit go)
136     - **Gen-1:** concatenate fully-local cue channels onto the head. ? **Audio is NOT a
promising cue here** ? our E1 probe
137     (PR #35) found it weak (~2.2× discrimination, AUC ~0.6) and the foreground-audio idea
also failed this session; keep it
138     parked, not "in reserve." Pose/court is the more credible #2 cue (license/cost-vet
first).
139     - **Gen-2 (VLM):** ms-swift fine-tune of **Qwen3-VL** (SFT cold-start ? GRPO/RFT with a
temporal-IoU-vs-golden reward,
140     our own CoT supervision). **Two-stage** (VLM proposes, head refines). Cloud train on
Modal (no-access) / RunPod-Secure.
141     - **Hard pre-serving gates (unmeasured, load-bearing):** the **$/call crossover** (self-
hosted VLM vs Gemini) *** a real
142     VRAM/token + cost benchmark** on rented A100/L40S (ms-swift multimodal-GRPO examples
use 6?8 GPUs ? **budget is the
143     largest open risk and is currently unquantified**). Don't stand up an owned serving
stack until these clear.
144
145     ---
146
147     ## Decisions / status
148     | # | item | status |
149     |---|---|---|
150     | 1 | Greenlight Phase-0 milestone(s) ? rec: **M0.1 corpus+event-index**, then **M0.4
harness** | awaiting owner |
151     | 2 | WASB C3 ? first-class action item (own-weights / clean swap) | **owner: yes** ? |
152     | 3 | Ratify licensing **MIT / Apache-2.0 / BSD** (code+data+weights) | **owner: yes** ?
|
153     | 4 | Collector = corpus + event-store system-of-record | **owner: confirm?** |
154     | 5 | Rename repo off "badminton" | **owner action** ? |
155     | 6 | Conversational-QA: build seam now, punt model | **rec accepted?** |
156
157     ## Open risks / unknowns (none block Phase-0 research)
158     Budget/cost to reach Gen-2 **unquantified** (largest risk) · ship-gate "very high P/R"
target undefined · per-sport clean
159     trajectory detectors not yet identified · base-model pretraining provenance unauditale
(risk-acceptance) · real VLM
160     VRAM/token cost · single-model-multisport-temporal generalization unproven · DPA + no-

```



```
train (minors-excluded) for cloud
161     training · on-device 4B latency on 8 GB unmeasured.
162
163     ---
164     *Design-only and reversible. Nothing starts until you greenlight a milestone. Sequencing
discipline holds: the committed
165     next platform slice remains the async job model (single-tenant); only the M0.1?M0.4
foundation is safe to start.*
166
```

6. user (2026-06-10T05:33:30.493Z)

```
1     # Roadmap ? from annotated data to a self-improving, offline rally segmenter
2
3     This document is the narrative thread: how the indexer evolves from an
4     AI-API-dependent prototype into a self-contained tool that learns rally detection
5     from its own commercially-licensed, annotated video corpus. It explains how we
6     got here, why each step matters, and how it shapes the final tool. Decisions
7     are recorded as ADRs in [DECISIONS.md](DECISIONS.md); the scoring mechanics live
8     in [EVALUATION.md](EVALUATION.md).
9
10    ## The thesis
11
12    The indexer detects badminton rallies in long recordings. Its strongest mode
13    (`yolo_hybrid`) is a two-stage pipeline: cheap permissive person-detection
14    player-motion filtering (Stage 1, local ? torchvision Faster R-CNN + ByteTrack
15    tracking) followed by Gemini semantic refinement (Stage 2, paid API). That
16    works, but it costs money, needs the network, and never improves on its own.
17
18    > Naming note: the `yolo_hybrid` registry key is retained for back-compat, but
19    > AGPL YOLOv8 was removed and replaced with a BSD/MIT-licensed torchvision +
20    > ByteTrack stack ? see ADR-005. "YOLO" in this doc means the local
21    > player-motion gating stage, not the Ultralytics library.
22
23    A sibling project, sports-data-collector, curates sports videos with
24    per-rally `[start, end)` annotations and a commercial-license flag. The thesis of
25    this roadmap: use that annotated, commercially-usable data to (1) measure the
26    pipeline objectively and (2) train an offline replacement for the paid Gemini
27    stage ? turning the tool into one that improves as more data is curated, runs
28    air-gapped, and generalizes across sports.
29
30    Why it matters: it removes the per-run API cost and network dependency, gives us
31    a defensible quality metric instead of eyeballing, and creates a flywheel ?
32    more curated matches ? a better offline segmenter ? better highlights.
33
34    ## The four phases
35
36    ### Phase 1 ? Bridge the repos (DONE)
37    A sport-agnostic `curate export` in the collector emits the indexer's `start,end`
38    ground-truth CSVs plus a `manifest.json` pairing each CSV with its video, gated on
39    the `is_commercial` flag. ? Why: one clean producer/consumer contract that works
40    for all five supported sports (badminton/tennis/cricket/pickleball/table-tennis) alike.
41    See ADR-002.
42
43    ### Phase 2 ? Acquire full-resolution video (DONE)
44    The corpus was 256x144 (the original crawl used `worstvideo`). The `fetch` command
45    re-acquires every video at best available resolution (1080p here), in place
46    (preserving annotations), via authenticated Firefox Premium cookies and yt-dlp's
47    `default` client. ? Why: YOLO/shuttle detection needs real pixels; and the
48    journey surfaced non-obvious traps (SABR streaming ? 144p; Chrome DPAPI; rate
49    limiting) now encoded in the tooling. 22/22 fetchable matches are now 1080p.
50    See ADR-003.
51
52    ### Phase 3 ? Evaluate + tune (IN PROGRESS)
53    `run_phase3_eval.py` walks the manifest, processes each video, and scores detected
```

rallies against the ground truth ? per video and corpus-aggregate (precision/recall/
F1, mean IoU, boundary error), for both the `candidates` (raw detector) and `final`
(refined) stages. ? *Why:* the candidates-vs-final split tells us *where* error
lives ? a **detection** problem (candidates miss rallies) vs. a **segmentation**
problem (candidates catch them, refinement is loose). That diagnosis decides where
effort goes, and establishes the baseline any trained model must beat.

Phase 4 ? Learn an offline, API-free segmenter (NEXT)
Replace the paid Gemini refinement with a **lightweight classifier trained on the
rally annotations**, consuming features the pipeline already computes and stores
(`candidate\_segments.stats`). No external API, no new labeling. ? *Why:* this is the
payoff ? a free, offline, self-improving rally mode. The **substrate was chosen from
data**: the ADR-004 update (2026-06-02) measured the YOLO/player-motion candidate
stage at **recall ~0.05 @ IoU 0.5** on `shuttleseat\_8` (median best coverage of a
rally only ~20%) ? too low a ceiling to build on. So the offline segmenter's
substrate is **WASB shuttle-motion** (ADR-001), not player-motion. See **ADR-004**.

Complementary local cues (beyond the segmenter swap)

Once the substrate is WASB shuttle-motion, the next lever is **fusing additional
fully-local cues** to raise rally recall/precision without reintroducing a paid
API. The first one explored is **audio "thwack" hit-detection** (**ADR-007**,
accepted as a direction 2026-06-07): a pure-numpy spectral-flux onset detector in
`backend/pipeline/audio/`. The **E1 GO/NO-GO probe** has run on real GoPro audio ?
verdict **AMBER / leaning-negative**: recall signal is real (hit-clusters recover
WASB-missed rallies) but the classical-onset precision ceiling (~2.2x) is too weak
to fuse cleanly, and a band-pass doesn't rescue it. The only path past that is a
learned timbre classifier ? which, like Phase 4, is **gated on golden labels**.
See `AUDIO\_E1\_FINDINGS.md`. Pose/court serve-gating is parked as the #2 cue.

The recurring blocker across Phases 3?4 and the audio work is the same: **more
golden labels** is the single highest-leverage move (see the cross-cut note below).

How this shapes the final tool

- **Offline & free by default.** The production rally mode becomes a local model;
Gemini is kept only as an upper-bound reference, not a dependency.
- **Self-improving.** Every newly curated, annotated match is more training signal;
re-training is a local, deterministic step.
- **Measurable.** Every change is scored against the golden set, so "better" is a
number, not a vibe.
- **Sport-generic.** The label ? feature ? classifier machinery and the export
bridge already span sports; adding a sport is data, not new architecture.
- **Commercially clean.** Only `is\_commercial` data flows through the bridge, so the
trained model and any derived product rest on a licensable foundation.

Status at a glance

Phase	What	Status
1	Export bridge (`curate export` ? manifest + CSVs)	? Done
2	1080p acquisition (`fetch`, in-place, cookies)	? Done (22/22)
3	Eval + tuning loop (`run_phase3_eval.py`)	? In progress
4	Offline learned segmenter (WASB shuttle-motion substrate)	? Next (substrate decided: WASB, ADR-004)
+	Audio hit-detection cue (`backend/pipeline/audio/`, ADR-007)	? E1 probe run ? AMBER, gated on golden labels

> **Cross-cutting:** golden-set ground truth (which powers Phases 3?4 and on-demand regression testing)
> is produced via a vendor process + pluggable annotation provider ? see
> GOLDEN_SET_VENDORING.md and **ADR-006**. As of 2026-06-02
the annotation
> guideline is authored (collector `docs/annotation/`) and **Roora** (Bengaluru) is
selected for an

```

114 > initial **3-sport pilot** (badminton, pickleball, table tennis). Input-capture quality
requirements
115 > that bound achievable precision/recall are tracked in
[VIDEO_RECORDING_GUIDELINES.md](VIDEO_RECORDING_GUIDELINES.md).
116 >
117 > **Observability:** every processed video now emits an always-on per-video diagnostics
118 > report (spec-driven warnings/errors with FAQ refs, soft-dependency on `obsreport`), so
119 > pipeline regressions and footguns surface without re-running eval ? see
120 > [CODE_MAP.md](CODE_MAP.md) §2 (reporting/observability).
121

```

7. user (2026-06-10T05:33:31.179Z)

```

1 # Deferred Hardening Plan ? Train/Eval Guardrails, Async Jobs, Packaging
2
3 **Status:** ? Design for review ? **no code yet**. Owner-requested follow-up to the
4 "config & identity consistency" tech-debt pass. Each item below becomes its own PR
5 only after this doc is approved.
6
7 **Why these three:** they were cataloged as out-of-scope during the consistency
8 pass because each is larger/design-heavy or needs the owner's machine. They are the
9 remaining high-value hygiene items not already done (config/structure/Result/logging/
10 reports) or in flight (detector abstraction, PR #49).
11
12 **Recommended sequence:** #7 Packaging (lowest risk, unblocks clean installs/CI) ?
13 #13 Train/eval guardrails (correctness of the data flywheel) ? #16 Async jobs
14 (largest, product-facing).
15
16 ---
17
18 ## 1. Packaging modernization (debt #7)
19
20 ### Problem
21 - No `pyproject.toml` on `master`. (PR #49's description lists packaging as "done",
22 but it isn't present ? reconcile with the owner.)
23 - The root `setup.py` is a **diagnostics/dependency-checker**, not a PEP 517 build
24 script ? its name collides with setuptools semantics and will confuse `pip`/`build`.
25 - Deps live only in `requirements.txt`; `obsreport` is a commented soft dep
26 (BACKLOG) and `supervision<0.30.0` is pinned (debt #14).
27
28 ### Approach
29 - Add `pyproject.toml` (PEP 621): project metadata + runtime deps mirroring
30 `requirements.txt`, with optional-dependency groups:
31 - `[project.optional-dependencies] dev` ? `pytest`, coverage tooling.
32 - `reporting` ? `obsreport` pinned to its git tag (`v0.1.3`, per BACKLOG).
33 - `gpu` ? torch/torchvision (note: the cu124 `--extra-index-url` is **not**
34 expressible in standard PEP 621 deps; document the install line or use a
35 `[tool.uv]/pip-conf` approach rather than faking it).
36 - Console entry points: `indexer = backend.main:main` (so `indexer --server` works),
37 optionally eval drivers.
38 - **Rename the diagnostic `setup.py`** (e.g. `scripts/doctor.py`) so it no longer
39 shadows the build system; update README/`run_all_tests.py`/CI references.
40 - Choose a build backend (hatchling recommended) and confirm the `backend/` import
41 root stays valid (keep current layout; no src/ move required).
42
43 ### Risks
44 - `setup.py` rename touches docs and any muscle-memory invocations ? grep first.
45 - Torch CUDA wheels can't be pinned cleanly in PEP 621; must document the
46 `--extra-index-url` install path (the current `setup.py` logic can move into docs).
47 - CI must switch to `pip install -e .[dev]`.
48
49 ### Verification
50 - `pip install -e .[dev]` in a clean venv ? `run_all_tests.py` green.
51 - `indexer --help` resolves via the entry point.
52

```

```

53 ---
54
55 ## 2. Train/eval separation guardrails (debt #13)
56
57 ### Problem
58 Separation is by convention only. `backend/eval/calibrate_local.py` and
59 `calibrate_wasb.py` do honest leave-one-video-out (LOVO), but nothing *enforces*
60 that a video used to train the learned segmenter
61 (`backend/eval/training.py::build_training_set` ? `distill_local`) is absent from the
62 eval/regression set. Leakage would silently inflate the very metrics the product
63 story rests on (? in BACKLOG).
64
65 ### Approach
66 - **Split unit = match, never clip** (per `GOLDEN_SET_PHASE1_VIDEOS.md`: split by
67 match, hold ~? of each sport as eval).
68 - Add a `split: "train" | "eval"` field to the collector's `curate export`
69 manifest (cross-repo contract, ADR-002 lineage) ? or a separate eval manifest.
70 - New `backend/eval/splits.py`:
71   - `load_split(manifest) -> (train_ids, eval_ids)` keyed by **match id** (not just
72     file MD5 ? re-encoded duplicates share a match but differ in
73     `compute_video_id`, so group at match level).
74   - `assert_disjoint(train_ids, eval_ids)` ? raise on any overlap.
75 - Wire the guard into the training entrypoint (`distill_local`/`training`) and into
76 `run_regression.py` so the baseline is computed strictly on the eval split.
77
78 ### Risks
79 - Requires a collector-side manifest change (coordinate the cross-repo contract).
80 - Small corpus ? tiny held-out set ? high metric variance; the guardrail is correct
81   but the numbers will be noisy until the golden set grows.
82 - MD5-only dedup is insufficient (re-encodes evade it) ? must track match identity.
83
84 ### Verification
85 - Unit test: overlapping train/eval ids ? `assert_disjoint` raises.
86 - Integration: a deliberately-leaked manifest fails the training entrypoint.
87
88 ---
89
90 ## 3. Async job model (debt #16)
91
92 ### Problem
93 `POST /api/process` runs validate ? segment ? report **inline**, blocking the
94 request for the full duration of a long video. Unworkable as a hosted/product API.
95
96 ### Approach (minimal first slice ? single self-hosted box)
97 - `POST /api/process` returns **202 + `job_id`**; add `GET /api/jobs/{job_id}`
98   returning `{status, progress, result, reports}`.
99 - Run the pipeline on a background worker ? start with a `ThreadPoolExecutor`
100   (segmenters already use threads internally for AI handoff, so this fits); leave a
101   real queue (RQ/Celery) for later if multi-process is needed.
102 - Persist job state in the existing SQLite DB via the migration ladder
103   (`backend/infrastructure/database.py::_init_db`, bump `PRAGMA user_version` 1 ? 2,
104   add a `jobs` table). The observability report becomes the job's artifact.
105 - Update the static UI to poll `GET /api/jobs/{id}` instead of blocking.
106
107 ### Risks
108 - **SQLite thread-safety:** verify `check_same_thread`/connection-per-thread;
109   `get_connection` already re-issues `PRAGMA foreign_keys=ON` per connection ?
110   confirm it's safe under the executor.
111 - Cancellation/timeout and crash recovery (a job in RUNNING after a restart).
112 - In-process store is not horizontally scalable ? acceptable for the current
113   single-box self-hosted target; call it out so it isn't mistaken for production-HA.
114
115 ### Verification
116 - Submit a job ? poll ? DONE with the same segments the sync path produced.
117 - Server restart mid-job leaves a recoverable/clearly-FAILED job, not a hang.

```

```

118
119 ---
120
121 ## Out of this doc
122 - **Detector/WSL abstraction (#6):** in flight (PR #49).
123 - **Unify hybrid segmenters (#7 in LOW_REGRET):** owner-deferred.
124 - **ByteTrack `supervision<0.30.0` migration + real-footage validation (#14/#15):**
125     need a GPU/WSL box and real video ? owner-machine work, tracked separately.
126

```

8. user (2026-06-10T05:33:37.512Z)

```

1      # Backlog ? deferred items to revisit
2
3      Single tracker for work we consciously deferred, so nothing gets lost between sessions.
Each item records **what**, **why it's deferred**, and **the trigger to revisit**. Every entry
traces to a real decision/PR/doc (no speculative work).
4
5      > Maintained as a version-controlled doc for now. Say the word and I'll mirror any/all
of these to **GitHub issues** instead (or in addition).
6      > Status legend: ? blocks something · ? do when triggered · ? nice-to-have/cleanup.
7
8      _Last updated: 2026-06-07._
9
10     ---
11
12     ## Monetization / licensing
13     *(source: `archives/MONETIZATION_AUDIT.md` §7, ADR-005)*
14
15     - ? **Ultralytics Enterprise pricing** ? get a quote **only if** the
torchvision/ByteTrack detector (PR #8) proves insufficient on real footage. Currently *not
needed* (swap done). Quote-only pricing. **Revisit:** if detector quality forces reconsidering
YOLOv8.
16     - ? **Codec patent thresholds (AVC/HEVC/AAC)** ? confirm exact Via-LA (AVC) / Access
Advance (HEVC) royalty rates & free thresholds **with legal counsel** before relying on the
"small SaaS is under threshold" reading. Mitigation already chosen: emit royalty-free AV1/Opus
output. **Revisit:** before commercial launch.
17     - ? **Per-checkpoint weight/dataset licenses** ? if we borrow pretrained weights (YOLOX,
ActionFormer, WASB, TrackNetV3), the *code* license ? the distributed *weights/dataset* terms.
Either train our own or confirm each checkpoint. **Revisit:** when adopting any borrowed
weights.
18     - ? **Real GPU $/video & latency** ? benchmark the three LLM-step options (offline
ActionFormer vs self-hosted Qwen2.5-VL-7B vs Gemini paid-tier) on the target VM before pricing
the product. **Revisit:** when choosing the production LLM step.
19
20     ## Detector swap follow-ups
21     *(source: PR #8, ADR-005)*
22
23     - ? **`supervision` ByteTrack deprecation** ? `sv.ByteTrack` is deprecated as of
supervision 0.28.0 and **removed in 0.30.0** ; `requirements.txt` is pinned `<0.30.0` as a
stopgap. Migrate to `ByteTrackTracker` from the separate `trackers` package before lifting the
cap. **Revisit:** before bumping supervision past 0.30.
24     - ? **Real-video validation of the swap** ? the torchvision + ByteTrack Stage-1 is unit-
tested with mocks but **not yet validated on actual footage** for detection/tracking quality.
(ADR-004 separately found player pixel-motion is a weak rally proxy on broadcast video, so
expect tuning.) **Revisit:** when a real test clip + the pipeline are run together.
25     - ? **`yolo_hybrid` naming** ? the registry key, `yolo_` config keys, and `yolo-
hybrid-` DB detector tags keep the "yolo" name for back-compat despite no YOLO remaining.
Optional rename. **Revisit:** if/when a clean break is worth the migration.
26
27     ## Golden-set subproject
28     *(source: `docs/GOLDEN_SET_IMPLEMENTATION.md`, `docs/GOLDEN_SET_VENDORING.md`)*
29
30     - ? **Annotation guideline** ? **DONE (2026-06-02):** authored as the Roora brief +
illustrated guide in the collector repo `docs/annotation/` ? rally CSV schema; ignore dead time

```

```

/ include every rally (incl. faults & lets) / shot = one contact; `ending_reason`
(forced/unforced-error taxonomy, shared across net-separated sports); singles & doubles for
badminton & pickleball.
31 - ? **Internal gold key** ? hand-annotate 5?10 licensed clips to near-perfect as the
vendor answer key. **Unblocked.** **Revisit:** anytime.
32 - ? **GS-3 ? `HumanVendorProvider`** ? concrete adapter for the chosen annotation vendor
(submit/poll/fetch + DPA/secure-link controls). **Vendor selected for the initial pilot: Roora**
(Bengaluru; iMerit / Taskmonk / TELUS remain the fallback shortlist). **Revisit:** once the
Roora pilot confirms quality, then build the adapter.
33 - ? **GS-4 oracle model** ? `AutomatedProvider` interface is stubbed (this work) with a
deterministic fake. Wire a *concrete* fast-tier auto-labeler later. **Oracle rule:** it MUST be
a different model lineage than production, or the regression test is blind to shared failures.
**Revisit:** when a production LLM step is chosen (pick the oracle's lineage to differ).
34 - ? **GS-5 ? regression triggers** ? wire `regress()` into CI (pre-merge gate), on-
demand CLI (the "surprise" case), and a scheduled run; include a deliberate re-baseline path.
**Revisit:** after GS-4.
35 - ? **Vendor pilot (Roora)** ? run the paid pilot with **Roora** as a **single-vendor
pilot first** (not a multi-vendor bake-off): assess quality-per-rupee against the $4 bar;
escalate to the iMerit/Taskmonk fallback only if it misses. **Revisit:** the pilot doubles as
calibration once the gold key exists.
36 - ? **Phase-1 video collection** ? gather the round-1 golden corpus per
`docs/GOLDEN_SET_PHASE1_VIDEOS.md`: **static self-recorded** footage (mirror deployment, not
broadcast), 3 sports incl. singles & doubles, across the diversity matrix; ~12 clips/sport with
**? held out by match**. **Unblocked.**
37 - ? **Train/eval separation guardrails** ? train & eval videos live in **separate
catalogs** today, kept apart by convention only. Add **hard guardrails** (permissions / explicit
approval / re-confirmation at ingest & export) so a video can **never** land in both train and
eval. **Revisit:** dedicated design session with the user.
38
39 ## Recording best practices
40 *(source: `docs/VIDEO_RECORDING_GUIDELINES.md`)*
41
42 - ? **Validate recording hypotheses** ? empirically test which capture conditions
(resolution, fps, camera angle/height, lighting) actually move precision/recall, then promote
the winners into user-facing "how to record" guidance. Only the static-camera rule is currently
evidence-backed (ADR-004). **Revisit:** once a controlled corpus + the eval loop can measure
per-condition deltas.
43
44 ## Code & test organization
45 *(source: owner review comments on PR #35, 2026-06-07)*
46
47 - ? **`tests/` reorganization to reduce fragmentation** ? owner wants to isolate test
components on a grouping
48 metric (by subsystem/component) so `tests/` is less fragmented, mirroring the new
`backend/pipeline/audio/` +
49 `backend/eval/audio/` layout (e.g. audio tests grouped under `tests/audio/`, eval
tests under `tests/eval/`).
50 The exact grouping **metric is the owner's design choice** (consolidate small files
vs. add per-area dirs ? the
51 "lesser fragmentation" goal slightly tensions with adding dirs), so this is left as a
**dedicated refactor PR**,
52 not guessed here. **Revisit:** when the owner specifies the grouping scheme. (The
`backend/eval/audio/` half of
53 the request ? comment 1 ? was done in PR #35.)
54
55 ## Commercial Readiness & Low-Regret Moves ? Approved Decisions Checkpoint (2026-06-07)
56
57 *(source: Commercial Readiness Review + structure/sequencing discussion with Grok as
second-in-command)*
58
59 - ? **Docs-only checkpoint PR** ? This update + the refreshed
`docs/LOW_REGRET_MOVES_PLAN.md` formally record the approved path before any code changes. The
initial review docs (PR #37) introduced the findings; this entry locks the execution decisions.
60 - ? **Approved directory structure** ? Adopt a layered approach under `backend/` for
professionalism and future growth (no flat dumping of new modules, no explosion of 15+ single-

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```

file dirs):
61     - `backend/config/` (package ? start with this)
62     - `backend/api/` (FastAPI app, routes, static UI)
63     - `backend/pipeline/` (validators + segmenters + stitcher + audio + detectors)
64     - `backend/infrastructure/` (database + external adapters)
65     - Keep `backend/eval/` prominent as a first-class concern.
66     - Logical grouping for providers, sports, annotations, utils, and a small `core/` only
when needed.
67     - ? **Sequencing (Config first, then atomic refactor)** ? Implement Deliverable 1
(Config) **first**, introducing the `backend/config/` package immediately. Perform the full
directory structure refactor as a **separate high-pri atomic PR** afterwards. This minimizes
conflicts, keeps PRs focused, and gives Claude a clean window with no concurrent moves.
68     - ? **Deliverable 1 (Config)** ? `backend/config/` package (`models.py` + `loader.py`)
instead of a single `config.py`. Typed `AppConfig`, central loader with built-in defaults + file
override, `extra="allow"` for migration. Update `setup.py` + `main.py`. Add tests. Land this
before the layout move.
69     - ? **Subsequent items** ? See the refreshed `LOW_REGRET_MOVES_PLAN.md` for the full
ordered list (Error contracts, logging, packaging, detector abstraction, etc.). The structure
refactor is now explicitly tracked as its own high-pri step right after Config.
70
71     **Status legend reminder:** ? = approved / in progress tracking, ? = do when triggered.
72
73     ## Observability reports ? shipped; intersections with the low-regret items (2026-06-07)
74     *(source: per-video observability feature ? indexer PR #41/#42/#43, collector PR
#11/#12; shared lib `obsreport`)*
75
76     - ? **Low-regret items 1 (Config pkg, PR #39) + 2 (directory restructure, PR #40) are
DONE.** Remaining = items 3?9 in `LOW_REGRET_MOVES_PLAN.md`.
77     - ? **Per-video observability reports shipped in BOTH tools** (a TECHNICAL report + a
CUSTOMER-friendly summary), via the **soft-dependency** lib `obsreport`
(github.com/avidullu/sports-obsreport, MIT, v0.1.3). Each run writes `**Item 8 (Input-Quality UX) is LARGELY SUBSUMED** by these reports (validation stage
+ footgun flags + verdict already surface quality; a clean customer summary already exists).
**Do NOT build a parallel mechanism** ? instead EXTEND the report: add an audio-readiness
stage/flag to `backend/reporting/` (harvest) + the obsreport spec. **Revisit:** when item 8 is
picked up.
79     - ? **Item 7 (unify hybrid segmenters) should land before the report's real AI-provider
timing.** The report records AI timing as `not_captured`; once `wasb_hybrid`/`tracknet_hybrid`
are unified there is ONE place to thread a recorder callback through `provider.analyze_video` ?
the report. **Revisit:** with item 7 (Claude to add the timing hook there).
80     - ? **Item 3 (Error/Result contract) synergizes with the report.** Today the report
INFERS failures (e.g. `silent_provider_noop` = 0 segments + no API key). A structured
Result/error from providers/segmenters would let the report read the real reason as a first-
class flag. **Revisit:** with item 3.
81     - ? **`obsreport` is a SOFT/commented dep** in `requirements.txt` (works locally via
editable install; pipeline no-ops if absent). **Pin the `git+ssh@v0.1.3` tag** in both repos
once SSH/CI access to the private lib repo exists.
82     - ? **`run_all_tests.py`** added at repo root ? run the full suite (clear PASS/FAIL,
pytest exit code) before sending PRs.
83     - ? **Config-migration follow-ups (DONE ? config & identity consistency pass)** ? the
two typed-config bugs from
84     #39/#40 are fixed: (1) **`/api/compile` now honors `--output-dir`** and (2)
**`output_dir` is a real field on
85     `OutputConfig`** (default `"output"`); the CLI overrides it on the typed model in
`main()`, retiring the
86     write-only `_runtime_overrides` hack. Same pass also unified `video_id` MD5 derivation
into
87     `backend/utils/hashing.py::compute_video_id` and migrated all four `run_*.py` scripts
off raw `json.load` onto
88     `backend.config.load_config`. See `CODE_MAP.md` §2.0/§6 (footguns retired).
89

```

```

1      # Commercialization Tracker
2
3      **Status:** ? LIVING ? track this in parallel with product/quality work.
4      **Owner:** avidullu . **Last refreshed:** 2026-06-07
5
6      **Purpose.** Single source of truth for the **commercial** dimension of the project:
7      licensing/legal posture, codec & output economics, hosting unit-economics, the
8      golden-set vendoring spend, packaging/distribution, and the product-market-fit
9      risks that gate monetization. Updated as decisions land ? see the
10     [Changelog](#changelog) at the bottom.
11
12     **Merged from / supersedes:**
13     - `archives/MONETIZATION_AUDIT.md` (2026-05-30 deep-research licensing & economics
14       audit) ? kept as the **research-of-record** (per-claim verification transcript,
15       primary sources). This doc carries its *current* status forward.
16     - `archives/COMMERCIAL_READINESS_REVIEW.md` (code-health + PMF assessment) ?
17       archived; its commercial findings + the quality-phase priorities live here now.
18
19     **Premise (unchanged, verified):** the product is a **closed-source hosted API** ?
20     users upload videos to a cloud NVIDIA GPU VM and get highlight reels back. We never
21     distribute code/binaries, so GPL/LGPL *distribution* copyleft is not triggered ? but
22     **AGPL §13 (network clause)**, **metered-API terms**, and **video-codec patent
23     royalties** all still apply.
24
25     ---
26
27     ## 1. Commercialization dashboard
28
29     Legend: ? done . ? in progress / watch . ? blocker / needs action . ?? needs legal
30     counsel . ? future
31     Priority: **P0** (do first) . **P1** . **P2** (lowest). Rows are grouped **Pending ? In-
32     flight ? Completed**, each sorted by priority. (C# IDs are stable identifiers ? they don't imply
33     order.)
34
35     | # | Workstream | Pri | Status | Where it stands | Next action / owner decision |
36     |---|---|---|---|---|---|
37     | | **? PENDING (needs action / not started)** | | | |
38     | C5 | **Output codec royalties** | **P0** | ? | Highlights are still encoded **H.264 +
39     AAC** (`libx264` in `backend/pipeline/stitcher/compiler.py`) ? encode-side AVC patent exposure
40     remains. | Switch output to **AV1 + Opus** (royalty-free) on an **Ada-class GPU (L4/L40)** for
41     HW encode. See §4. Decision: provision Ada-class GPU? |
42     | C9 | **Packaging / distribution** | **P0** | ? | No `pyproject.toml`; the diagnostic
43     `setup.py` shadows the build system. | Design in `DEFERRED_HARDENING_PLAN.md` (#7) ? **awaiting
44     your approval** before code. |
45     | C12 | **SaaS foundations (async/auth/tenancy)** | **P1** | ? | Sync `/api/process`
46     blocks on long videos; no auth/tenancy. | Multi-user/platform strategy in
47     `PLATFORM_ARCHITECTURE.md` (pluggable storage/compute, tenancy, orchestration); async slice =
48     `DEFERRED_HARDENING_PLAN.md` (#16). |
49     | | **? IN-FLIGHT (in progress / watch)** | | | |
50     | C4 | **Runtime AI strategy (A/B/C + owned model)** | **P0** | ? | Live = **(C) Gemini
51     2.5 Flash, paid tier**. Long-term = **own a self-hosted generative video-QA model** (fine-tuned
52     open VLM) as primary, **paid Gemini always-ready as fallback** via the provider registry. | Make
53     "provider registry + paid fallback" a core principle; build the owned model **after** the
54     scaffolding/flywheel are solid. See `PLATFORM_ARCHITECTURE.md` §5A. |
55     | C13 | **Beachhead validation** | **P0** | ? | Market research
56     (`PMF_MARKET_RESEARCH.md`) recommends **India badminton academies** as the #1 beachhead.
57     Unvalidated. | Cheapest tests: 10?15 academy/coach discovery interviews; rally-detection P/R
58     benchmark on 100+ real amateur clips; fake-door pricing (B2C ~$99/yr, B2B academy ~$600/yr). See
59     §6. |
60     | C8 | **Golden-set vendoring spend** | **P1** | ? | ADR-006: **Roora (Bengaluru)**
61     selected for an initial paid pilot; licensed/owned video only (no user uploads to vendors). |
62     Confirm quality-per-dollar from the pilot; then GS-3 `HumanVendorProvider`. See
63     `GOLDEN_SET_VENDORING.md`. |
64     | C3 | **Shuttle-tracking weights provenance** | **P1** | ? ?? | WASB-SBDT **code is MIT
65     (clear)**; distributed **checkpoints are academic-data-trained** ? not covered by the MIT code

```


grant. MonoTrack/YOLO-NAS banned (noncommercial). | Train-own weights (Phase-4 roadmap) ****or**** confirm per-checkpoint terms ?? before commercial deploy. |

43 | C14 | ****Owned-model base licensing + training-data policy**** | ****P1**** | ? ?? | Owned video-QA model (PLATFORM §5A) must build on ****Apache-2.0-clean, video-capable**** bases ? ****Qwen2.5-VL-7B/32B**** (Apache; 3B/72B are ***not***), ****Qwen3-VL-4B/8B****, or ****Gemma 4**** ? to keep weights private, gate, and monetize. ****Reject Gemma 3**** (viral derivative clause ? permanent Google PUP), Llama Vision (naming + 700M-MAU), ****gpt-oss**** (text-only). ****Hard rule: never train on paid-LLM (Gemini/OpenAI/Anthropic) outputs**** (competing-model clause). The moat is the ****consented dataset + eval harness****, not the model. | Lock to Apache-2.0; ****open-model teachers only****; separate explicit ****training opt-in****; ****exclude minors**** from the training pool; per-clip consent ledger. See PLATFORM §5A/§7. |

44 | C6 | ****Inbound HEVC decode**** | ****P2**** | ? ?? | Decoding users' GoPro H.264/****HEVC**** uploads is unavoidable; HEVC pools are fragmented + highest-uncertainty. | Low/uncertain residual; confirm decode-side exposure with counsel before launch. |

45 | C7 | ****Hosting unit-economics**** | ****P2**** | ? | Per-video cost is ***cheap*** for Gemini (~\$0.05 input for a 10-min clip); risk is volume × latency, not unit price. Self-host/offline costs unmeasured. | ****Benchmark \$/video & latency on the target VM**** for A/B/C before pricing. See §5. |

46 | C10 | ****Value layer (product surface)**** | ****P2**** | ? | Minimal exports shipped (EDL/JSON/CSV + ****basic_stats****, PR #51). | Deepen: per-rally tags/notes that feed golden, per-player/match stats, clip gallery. See §6. |

47 | C11 | ****Input-quality enforcement**** | ****P2**** | ? | Per-video observability reports surface a customer verdict + footgun flags incl. audio-present (PR #50). | Turn more ****VIDEO_RECORDING_GUIDELINES**** dimensions into validators/report rules. |

48 | C2 | ****supervision**** ByteTrack pin | ****P2**** | ? | Pinned ****<0.30.0**** (deprecation ahead). | Migrate before lifting the pin (BACKLOG). Not a legal risk, a maintenance one. |

49 | | ****? COMPLETED**** | | | |

50 | C1 | ****AGPL / copyleft removal**** | ? | ? | ADR-005 shipped: AGPL ****ultralitics**** YOLOv8 replaced with ****torchvision Faster R-CNN (BSD) + supervision**** ByteTrack (MIT). No AGPL code/weights in the stack. | Done. Keep new deps permissive. |

51

52 ---

53

54 ## 2. Licensing & dependency posture (refreshed risk table)

55

56 Verdicts updated against the current stack (****requirements.txt**** + code). Original

57 2026-05-30 verdicts and the 25-claim verification transcript are in

58 ****archives/MONETIZATION_AUDIT.md****.

59

60 | Dependency | License | SaaS obligation? | Current verdict | Notes |

61 |---|---|---|---|---|

62 | ****ultralitics**** + ****yolov8n.pt**** | AGPL-3.0 | Yes (§13) | ? ****REMOVED**** | Replaced per ADR-005 ? no longer in the stack. |

63 | ****torchvision**** detectors (Faster R-CNN/RetinaNet) | BSD-3 | No | ? LIVE | The chosen replacement; person = COCO class (see ****yolo_hybrid.py**** ****_PERSON_CLASS_ID****). |

64 | ****supervision**** (ByteTrack) | MIT | No | ? LIVE | Pinned ****<0.30.0**** ? migrate before bump (C2). |

65 | ****WASB-SBDT**** (shuttle) | MIT (code) | No | ? code clear | Weights provenance open (C3). |

66 | ****MonoTrack / YOLO-NAS**** | noncommercial | ? | ? BANNED | Do not use. |

67 | ****Gemini API**** (paid tier) | service terms | Paid: no training/review | ? LIVE WATCH | Must bill the paid tier to keep user video private (C4). |

68 | H.264 / HEVC / AAC codecs | patent pools | royalties on encode/decode | ? ?? | Separate from ffmpeg's license. Encode-side open (C5); decode-side residual (C6). |

69 | ffmpeg/ffprobe, google-genai, torch, opencv, numpy, fastapi/uvicorn/pydantic/jinja2/dotenv | LGPL/Apache/BSD/MIT | No | ? CLEAR | Invoked as subprocess / imported; all permissive. |

70 | ****obsreport**** (reports) | permissive (private) | No | ? soft dep | Pin the ****git+ssh@vx**** tag once the repo is published (BACKLOG). |

71

72 ****Bottom line:**** the two hard blockers from the audit (Ultralytics; banned weights)

73 are resolved/avoided. Remaining commercial-legal items are ****codec royalties (C5/C6)****

74 and ****weights provenance (C3)****, plus the metered-API privacy posture (C4).

75

76 ---

```

77
78 ## 3. Runtime AI strategy ? A/B/C
79
80 The semantic rally-boundary step is the one external/metered dependency. Three paths
81 (from the audit), with current standing:
82
83 - *(A) Eliminate ? offline learned segmenter.* ActionFormer (MIT) / our own learned
84 classifier on `candidate_segments.stats`. **Best steady-state margin** (no per-call
85 cost, no external dep) and matches the Phase-4 roadmap (substrate = WASB
86 shuttle-motion, decided in ADR-004). **Most engineering.** ? *target*
87 - *(B) Self-host VLM.* Qwen2.5-VL-7B/32B (Apache-2.0) or InternVL2.5-8B (MIT). No
88 per-call fees; pays in GPU VRAM (~16?24 GB for 7B). ?? Qwen 3B is noncommercial.
89 - *(C) Keep Gemini 2.5 Flash (paid).* Fastest, cheap per video, video-native;
90 ongoing metered cost + external dependency. **? current live path.**
91
92 Hosted alternatives for (C) comparison: Anthropic Claude (no training on
93 inputs/outputs; customer owns outputs; **no native video** ? weaker here);
94 OpenAI (no training by default; ZDR available); Gemini remains strongest video-native.
95
96 ---
97
98 ## 4. Codec / output economics (the live gap ? C5)
99
100 **The trap:** ffmpeg being free (LGPL) does **not** grant the *patent* rights to the
101 codecs it implements. Royalties attach to the act of encode/decode.
102
103 - **Encode side (we control it):** highlights currently emit **H.264/AAC**. AVC (Via LA)
104 is first 100k units/yr royalty-free with a permanent "free to end-users over the
105 internet" provision ? a small SaaS likely falls under thresholds, but ?? confirm.
106 - **Mitigation:** emit **AV1 (SVT-AV1, BSD + AOMedia royalty-free patent grant) + Opus
107 (royalty-free)**. NVENC AV1 HW-encode needs **Ada-class GPUs (L4/L40)**; Ampere/Turing
108 must use **SVT-AV1 (CPU)**.
109 - **Decode side (unavoidable, C6):** inbound GoPro **HEVC** decode ? fragmented pools,
110 highest uncertainty; low residual.
111
112 **Recommendation:** provision an **Ada-class GPU (L4/L40)** and standardize highlight
113 output on **AV1 + Opus** ? removes the encode-side royalty question almost entirely.
114 *Implementation lives in `backend/pipeline/stitcher/compiler.py` (codec args).*
115
116 ---
117
118 ## 5. Hosting unit-economics (C7)
119
120 Reference workload: ~30-min 1080p match on a cloud NVIDIA GPU VM.
121
122 - *(C) Gemini 2.5 Flash (paid):** input $0.30/1M tokens, output $2.50/1M; video ~263
123 tok/s ? 10-min clip ? 158k input tokens ? **~$0.05 input**. Cheap per unit; the
124 economic risk is **volume x latency**. Flash-Lite is cheaper; Pro ~10x.
125 - *(B) Self-host Qwen2.5-VL-7B:** $0 API; cost = GPU-hours / throughput; needs ?24 GB
126 VRAM (L4/A10). $/video vs Gemini is **open ? measure**.
127 - *(A) Offline ActionFormer/learned:** cheapest at inference; cost = one-time training
128 + light GPU. **Best steady-state margin.**
129
130 > The A/B/C $/video comparison depends on GPU instance + batch throughput ?
131 > **benchmark on the target VM before pricing the product.**
132
133 ---
134
135 ## 6. Product-market fit & the quality/data phase
136
137 ### Beachhead (from market research, 2026-06-07 ? see
138 [`PMF_MARKET_RESEARCH.md`](PMF_MARKET_RESEARCH.md))
139 The static-camera constraint is a **B2B filter**: the buyer must already record from a
140 fixed angle, which points away from the median casual player and toward coaches/clubs.
141 Ranked beachheads:

```

141 1. ****Badminton coaching academies & serious players, India**** ***(strongest)*** ? 20M+
 142 players,
 143 formalizing academy ecosystem (Khelo India), genuine fixed-angle recording, coaches
 144 who already monetize, badminton whitespace vs incumbents. Sell to the ****academy****,
 145 not
 146 the individual; needs INR-tuned pricing.
 147 2. ****US/global pickleball players & clubs**** ? fastest-growing sport (19.8M US, +45.8%
 148 YoY)
 149 but ****SwingVision** already owns single-camera tennis/pickleball AI** (USD 179.99/yr) ?
 150 head-to-head, not greenfield.
 151 3. ****Badminton academies/clubs in East/SE Asia + Denmark**** ? highest density; China
 152 needs
 153 local hosting/partners; Denmark/EU small but high-WTP, easy to pilot.
 154
 155 Pricing anchors ***(estimate)***: software-only BYO-camera ****B2C USD 60?180/yr****, ****B2B**
 156 academy/club USD 300?1,500/yr** per location (undercuts hardware-bundled Veo/Pixellot/
 157 Spiideo; comparable to SwingVision software-only).
 158
 159 **### PMF risks (from the readiness review)**
 160 - ****Capture quality is the gate.**** Accuracy is capture-conditional; broadcast/handheld
 161 footage is a failure mode. The wedge (static, tripod, full-court) is defensible but
 162 narrows the addressable market ? enforce/guide it (C11, `VIDEO\_RECORDING\_GUIDELINES`).
 163 - ****Value beyond raw segments is table stakes.**** Time-saved editing alone is weak;
 164 per-rally tags, stats, and a correction loop are what make it sticky (C10).
 165 - ****Golden labels are the recurring blocker**** across the offline segmenter, audio
 166 fusion, and measurable quality ? the highest-leverage investment (C8).
 167
 168 **### Refined priority order (quality/data phase ? "focus on quality once new videos
 169 land")**
 170 Carried from the readiness review; commercialization items above are tracked in
 171 parallel.
 172
 173 1. ****Golden-set expansion + data-flywheel UX**** (highest with new labeled videos):
 174 ingest the new batch, expand calibration/CV/regression baselines, make label
 175 contribution + safe retraining easy and non-contaminating (eval/training separation).
 176 2. ****Hybrid unification + observability hooks**** (deferred item 7, now higher): extract
 177 shared logic from `tracknet\_hybrid`/`wasb\_hybrid` before heavy AI timing/metrics
 178 land.
 179 3. ****Detector/windowing quality iteration**** with new data: re-tune velocity/merge/
 180 min-duration + `rally\_gate`; validate WASB on real video; trial audio fusion
 181 (ADR-007).
 182 4. ****Deeper value layer**** (beyond minimal exports, C10): per-rally tagging/notes ?
 183 `candidate\_segments`, per-player/match stats, clip gallery, human-correction loop.
 184 5. ****Input-quality enforcement & guidance polish**** (C11): more guideline dimensions ?
 185 validators/report rules; "your video is good enough" UX.
 186 6. ****Multi-sport maturation + recording best practices.****
 187 7. ****SaaS / multi-user / async foundations**** (C12) ? lower until monetization is firm.
 188 8. ****Polish & distribution**** (C9): installer story, first-run path, codec/output,
 189 economics.
 190
 191 ****Recommendation:**** with the mid-week labeled-video batch, spend ~80% on #1 + #3
 192 (flywheel + quality iteration on real data), then unification (#2) and deeper value
 193 (#4).
 194 ---
 195
 196 **## 7. Open questions needing owner / counsel sign-off ??**
 197
 198 1. ****Ultralytics Enterprise price**** ? only relevant if a permissive detector ever proves
 199 insufficient (fallback). Quote-only.
 200 2. ****Per-checkpoint / dataset licenses**** for WASB / TrackNetV3 / any VLM ? train-own or
 201 confirm each (C3).
 202 3. ****Exact Via LA (AVC) / Access Advance (HEVC) rates & free thresholds**** ? confirm live
 203 figures with counsel before relying on the "small SaaS under threshold" reading
 204 (C5/C6).

```

195 4. **Real per-video GPU cost & latency** of self-hosted VLM vs Gemini paid at expected
196 throughput (C7).
197 5. **Provision Ada-class GPU (L4/L40)** for HW AV1 encode? (C5 ? gates the codec
switch.)
198
199 ---
200
201 ## Changelog
202
203 - **2026-06-07** ? Added the market-research-of-record (`PMF_MARKET_RESEARCH.md`):
204 ranked beachhead (India badminton academies #1), pricing benchmarks, and a C13
205 "beachhead validation" workstream. Wired the summary into §6.
206 - **2026-06-07** ? Created this living tracker. Merged the 2026-05-30 monetization
207 audit + the commercial-readiness review. **Refreshed since the audit:** C1 AGPL
208 removal now ? shipped (was the #1 blocker); added C5 as a ? (output is still H.264,
209 the audit's AV1 recommendation is not yet implemented); C9 packaging tied to
210 `DEFERRED_HARDENING_PLAN.md`; reflected PR #50/#51 (input-quality reports, value-layer
211 exports). Archived `COMMERCIAL_READINESS_REVIEW.md`.
212
213 ---
214
215 ## See also
216 - `PMF_MARKET_RESEARCH.md` ? market-study-of-record (sizing, segments, competitors,
beachhead; cited).
217 - `archives/MONETIZATION_AUDIT.md` ? licensing/economics research-of-record (sources +
verification transcript).
218 - `archives/COMMERCIAL_READINESS_REVIEW.md` ? superseded assessment (history).
219 - `archives/decisions/DECISIONS.md` ? ADR-005 (permissive stack), ADR-006 (vendor
process).
220 - `ROADMAP.md` ? Phase-4 offline segmenter (the C4 "(A) eliminate" path).
221 - `DEFERRED_HARDENING_PLAN.md` ? packaging (C9), async (C12) designs (pending approval).
222 - `GOLDEN_SET_VENDORING.md` ? vendor process + Roora pilot (C8).
223 - `BACKLOG.md` ? `supervision` migration (C2), `obsreport` pin, codec follow-ups.
224

```

10. user (2026-06-10T05:33:42.883Z)

```

1  # Platform Architecture ? Multi-User "Assistant Coach"
2
3  **Status:** ? Strategy / research ? for review. No code yet.
4  **Date:** 2026-06-07 . **Owner:** avidullu
5
6  **Purpose.** Map the path from today's **single-user, local** tool to a **multi-tenant
7  "assistant coach" platform**: users upload videos, get highlights **stored and
8  browsable by history**, and choose **where their data lives** (Google Drive / cloud
9  bucket / local) and **where the compute runs** (our GPU / their VM). The organizing
10 principle is to **decouple the layers** ? identity, ingestion, storage, orchestration,
11 execution/compute, the detection+highlight pipeline, and data-serving ? behind small
12 **pluggable interfaces**, reusing the registry pattern the codebase already uses for
13 segmenters/providers/validators/sports/annotations.
14
15 **Product framing (the north star *and* the v1 line).** This is an **assistant coach**,
16 not just a clipper ? but the strategy is **phased and unambiguous about scope**:
17 - **v1 (concrete, near-term):** **rally detection ? highlight-reel compilation ? per-
user
18 history ? a coach/user correction loop that feeds the golden set** ("assistant-coach
19 *lite**"). Essentially today's pipeline made multi-user ? this is the real v1
deliverable.
20 - **Mid-term:** **rich rally/shot semantic metadata** (shot type, rally outcome) and the
21 **owned video-QA model** (§5A; §6 P4?P5).
22 - **North-star vision (strictly future ? *not* v1):** **gait / biomechanics, per-athlete
23 identifying templates, and deep coaching feedback under a supervising coach** (§6 P6).
24 Inspiring, but explicitly deferred ? and the part that raises the **biometric**
privacy
25 bar (§7).

```

[illegible]

```

84         ? Storage (local upload | Google Drive | S3 | ?
85         ? GCS | Azure) ? raw video + highlight outputs ?
86         ?????????????????????????????????????????????
87     ...
88
89     The two new pluggable seams are Storage and Compute/Execution; everything
90     else either exists (pipeline) or is net-new infra (identity, orchestration, serving). A
91     model/training layer ? the owned generative video-QA model ($5A) ? sits atop these,
92     fed by Storage (training corpus), Data (labels), and Compute (training + inference), and
93     is delivered through the existing AI-provider registry with a paid fallback.
94
95     ## 3. Pluggable abstractions (extend the existing registry pattern)
96
97     The codebase already proves the model: an `ABC` + a `Registry` singleton with
98     `register/get/list_available` (e.g. `SegmenterRegistry`). Add two more registries with
99     the same shape so they're plug-and-play and testable in isolation.
100
101     ### 3.1 `StorageBackend` registry
102     Replace the bare `video_path: str` that flows through the pipeline with a
103     StorageRef (a backend id + a key/URI), resolved by a backend.
104
105     ```python
106     class StorageBackend(ABC):
107         name: str
108         def open_read(self, ref: StorageRef) -> BinaryIO: ... # stream large video
109         def download_to_local(self, ref: StorageRef, dst: str): ... # for ffmpeg/cv2/WSL
110         def upload(self, local_path: str, ref: StorageRef): ... # persist
111         def signed_url(self, ref: StorageRef, ttl: int) -> str: ... # hand to LLM / browser
112         def stat(self, ref: StorageRef) -> StorageStat: ...
113         def list(self, prefix: StorageRef) -> list[StorageRef]: ...
114     ...
115
116     - Recommended implementation: build the backend over fsspec (one matrix ?
117     local / S3 (`s3fs`) / GCS (`gcsfs`) / Azure (`adlfs`) + Google Drive via `gdrivefs`,
118     Dropbox via `dropboxdrivefs`), reaching for smart_open as the robust large-MP4
119     byte-streaming primitive. Storage becomes one more registry. (Treat Drive/Dropbox as
120     add-ons ? less battle-tested than `s3fs`/`gcsfs`; cover with integration tests.)
121     - Backends: `local` (upload dir), `gdrive`, `s3`, `gcs`, `azure`. BYO-storage:
122     the tenant registers their own bucket/Drive; raw video never leaves their account.
123     - Auth ? never persist users' long-lived keys. For buckets, have the user create an
124     IAM role you AssumeRole into, scoped to one bucket/prefix, and mint presigned
125     URLs so raw video moves browser ? the user's bucket directly, never through our
126     servers. For Drive, OAuth with the narrow drive.file scope; for academics, a
127     service account with domain-wide delegation. Store any per-tenant creds encrypted.
128     - Why it matters for the pipeline: ffmpeg/OpenCV/WSL need a local file, while
129     the LLM provider can take a signed URL. The backend exposes both `download_to_local`
130     (for compute) and `signed_url` (for the provider/browser), so the pipeline stops
131     caring where the bytes live.
132     - Migration path: introduce StorageRef alongside the current local path; a local
133     backend wraps today's behavior so nothing breaks; adapters land incrementally.
134
135     ### 3.2 `ComputeBackend` / `ExecutionBackend` registry
136     Where the pipeline actually runs for a given job.
137
138     - local ? in-process / local subprocess (today's behavior; dev + self-host).
139     - hosted ? our GPU VM pool (the SaaS default).
140     - byo_worker ? the tenant runs a pull-based worker (a small agent) on their
141     own VM/GPU. Critical security property: the worker pulls jobs scoped to its tenant
142     from the control plane, pulls input from the tenant's own storage, runs the
143     pipeline, and writes artifacts + results back ? so the orchestrator never holds the
144     user's cloud credentials and needs no inbound access to their network. This

```

```

145 mirrors the **self-hosted CI runner** pattern (GitHub Actions runners use a 50s
146 outbound-only HTTPS long-poll). This pull-based runner is the **cheapest, most
147 credential-safe v1** for BYO-compute.
148 - For users who want the platform to **provision GPUs in their *own* cloud account on
149 demand** (an academy with an AWS account but no standing GPU box), add **SkyPilot**
150 (BYOC: everything launches in the user's accounts/VPCs; creds stay with them).
151 **Modal** is the managed fallback for users wanting zero infra (accepting that raw
152 video then flows through Modal).
153 - **?? Operational / support burden (a real adoption risk).** For the primary beachhead
154 ?
155 Indian badminton **academies/coaches** ? asking them to *run and maintain a worker on
156 their own VM/GPU* is a meaningful ongoing load (updates, monitoring, firewall/NAT, the
157 "it stopped after our IT change" ticket). So: **default academy users to `hosted`**,
158 and
159 treat `byo_worker`/SkyPilot as a **power-user / enterprise / privacy-sensitive tier**,
160 not the academy default. If/when BYO ships, plan for **auto-updating, self-healing
161 agents + health heartbeats + clear worker status in the UI**, and budget the support
162 cost.
163 - **Middle tier to consider ? "zero-infra BYO":** a managed-but-isolated path where the
164 **storage** stays in the user's bucket (signed URLs) while compute runs on **our**
165 GPUs
166 in a **strictly isolated, per-tenant, ephemeral sandbox** (nothing persists our side).
167 This gives the BYO *privacy* benefit (raw video never persists on our infra)
168 **without**
169 the user running a worker ? a pragmatic default between full-hosted and full-BYO.
170
171 The pipeline code is unchanged; only *where it's invoked* and *how it gets bytes* vary.
172
173 ### 3.3 Reuse, don't reinvent
174 The existing `segmenter` / `provider` / `validator` / `sport` / `annotation` registries
175 are the proven precedent and stay as the **pipeline** layer. The `annotation` registry +
176 `candidate_segments` table are also the natural seam for the **coach-correction / golden
177 -set** loop in the assistant-coach phase ($6, P4).
178
179 ## 4. Multi-tenant data model & serving layer
180
181 ### 4.1 Entities ? generic core, with the coach model as an optional overlay
182 The foundational schema is deliberately **domain-neutral** so we don't overfit to the
183 coach mindset ? the coach/athlete framing is a **market-fit hypothesis that will
184 iterate**
185 (see C13 / PMF), not a load-bearing assumption. **Generic core:**
186 ```
187 User ??< VideoCollection ??< Video ??< Highlight ??< HighlightMetadata
188             ???< PlaySegment ??< Event
189 Video / Highlight ??< Job (processing record: status, compute backend, timings)
190 ```
191 - **Generic & re-targetable:** a `VideoCollection` groups a user's videos (a season, an
192 event, a "folder") with no assumption of a coach or academy ? a power user, a coach,
193 or a
194 parent are all just `User`s with collections.
195 - **Highlights are first-class** (today compile only writes a file): a `highlights`
196 table
197 (segment ids + output `StorageRef` + params + created_at) powers "see past videos and
198 highlights by history," and **`HighlightMetadata`** carries the **rich semantic tags**
199 (shot type, rally outcome, etc. ? see $6 metadata stage) plus titles / ratings /
200 notes.
201 - **`video_id` stays the content MD5** (dedup within a tenant), scoped by `tenant_id`.
202 - **Coach overlay (optional, additive):** for the academy/coach GTM, layer
203 `Org/Academy ? Coach ? Athlete` *on top* (e.g. a `Membership`/role table + an
204 `athlete_id`
205 tag on collections/videos) **without** baking it into the core tables ? so the schema
206 stays usable for solo power users and the coach model can evolve, or be dropped, as
207 PMF
208 dictates.
209 - **History/serving API:** list collections/videos/highlights per user (and, with the

```

```

199 overlay, by athlete/coach/date); **query highlights by `HighlightMetadata`** (e.g.
shot
200 type / outcome); re-stitch from stored segments without reprocessing.
201
202 ### 4.2 Tenancy isolation ? options & recommendation
203 | Model | Isolation | Ops cost | Best for |
204 |---|---|---|---|
205 | **Shared DB + `tenant_id`** (row-level, ideally Postgres **RLS**) | logical | low |
**Hosted SaaS, many small tenants** ? *recommended default* |
206 | Schema-per-tenant | medium | medium | tens?hundreds of mid tenants |
207 | **DB-per-tenant** (incl. **SQLite-per-tenant**) | strong/physical | higher at scale |
**self-hosted single-academy**, or high-isolation enterprise |
208
209 **Recommendation:** for the hosted product, **migrate SQLite ? Postgres with shared-DB +
210 `tenant_id` + Row-Level Security**; keep **SQLite-per-tenant** as the **self-hosted /
211 single-academy** deployment (it reuses today's code almost verbatim). Either way,
212 **abstract DB access behind a repository layer now** so the engine is swappable, and
213 graduate the `PRAGMA user_version` migration discipline to **Alembic** under Postgres.
214
215 ## 5. Orchestration / async jobs
216
217 This is the serving layer that makes multi-user real. It is the **multi-user superset of
218 [`DEFERRED_HARDENING_PLAN.md`](DEFERRED_HARDENING_PLAN.md) #16** (async job model) ?
build
219 that first, single-tenant, then generalize.
220
221 - **Job record** (`jobs` table): `id, tenant_id, video_id, type (index|compile|analyze),
222 status (queued|running|done|failed), progress, compute_backend, result_ref, error,
223 timestamps`. `/api/process` returns **202 + job_id** ; `/api/jobs/{id}` polls.
224 - **Idempotency** keyed on `(tenant_id, video_id, pipeline_version)` ? reuse the
225 content-hash discipline already in `compute_video_id`.
226 - **Dispatch:** the orchestrator enqueues; a worker (hosted or `byo_worker`) claims jobs
227 for its tenant, resolves input via the tenant's `StorageBackend`, runs the pipeline,
228 writes artifacts + DB rows. **Observability reports** (already per-video) become the
229 job artifact surfaced in history.
230 - **Queue/worker backend ? a decision to prototype in #16, *not* a default here.** The
231 *requirements* are firm ? **idempotency**, **resumability** (a 40-min GPU job mustn't
232 restart on a worker blip), **progress states** ? which rule out fire-and-forget
Celery/RQ
233 for the long jobs. The *choice* should be made by prototyping in the
234 [DEFERRED_HARDENING_PLAN.md](DEFERRED_HARDENING_PLAN.md) #16 slice, weighing:
235 - **Start simplest:** **Arq** (async-native, natural FastAPI fit) or even a **plain
236 Postgres-backed queue + worker** ? likely sufficient for the *first* single-tenant
237 async slice, with the lowest operational complexity and the easiest debugging.
238 - **Durable-execution upgrade if/when needed:** **Hatchet** (Postgres-backed, per-step
239 retries/timeouts/concurrency; pairs neatly with the pull-based worker) ? attractive
240 because it adds **no new datastore** once we're on Postgres, but it *is* a new
241 dependency + operational/debugging surface; adopt only when long-job resumability/
242 observability demands it.
243 - **Heaviest:** **Temporal** ? most mature durable workflows, operationally heavy;
244 reserve for genuinely complex multi-stage pipelines at scale.
245
246 Record the decision criteria in #16: operational complexity, maturity, **debugging
247 experience for long GPU jobs**, and whether durability is worth a new dependency.
248 **Bias: start with the simplest thing that gives retries + status; graduate only on
249 evidence.**
250
251 ## 5A. Owned generative model (the video-QA assistant) & the data flywheel
252
253 The north-star capability: an **owned, self-hostable generative video model** ?
254 fine-tuned on sports video (including **consented, user-contributed** footage) ? that
255 **answers questions about rallies, shots, and technique**. This is the conversational
256 brain of the "assistant coach." Crucially, it does **not** replace the known stack; it
257 is introduced **behind the same abstraction, with the paid API as an always-ready
258 fallback**.

```



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259
260 > **The moat is *not* the model.** Anyone can download Qwen-VL/Gemma. The defensible
261 > assets are **(a) the consented, badminton-specific video?QA dataset** and **(b) the
262 > eval harness**. Build and message *those* ? not "we own an LLM."
263
264 > **? Committed guardrail (legal, non-negotiable).** We respect the licensing findings
in
265 > full. **Paid frontier models (Gemini / OpenAI / Anthropic) are used for *inference /
266 > serving / fallback* only ? never as a *training* data source.** Harvesting their
outputs
267 > to train our owned model would breach their "no competing model" terms (a
terms/copyright
268 > violation), so it is prohibited in both engineering and legal process. **Training
teachers
269 > are open, Apache-2.0 models only** (Qwen-VL / InternVL / Gemma 4). Detail in the
270 > distillation constraint below.
271
272 ### Core principle: AI is a pluggable provider with a paid fallback
273 The codebase already abstracts the cloud AI behind `provider_registry` /
274 `AIVideoProvider` (`backend/providers/`). Make that a **first-class platform
principle**:
275 - Every generative capability routes through the **provider registry**.
276 - The **owned model** plugs in as a new provider (`owned_vlm`); **Gemini (paid, via
277 Vertex AI)** stays registered as the **fallback** (failover on error/low-confidence,
278 or as the quality ceiling/teacher reference).
279 - This lets us **ship today on the known paid stack** and **swap in the owned model when
280 its quality clears a measured bar** ? with zero architectural churn and the paid path
281 always one config flip away. (Mirrors the COMMERCIALIZATION C4 A/B/C strategy: the
282 owned model is the long-run "(A) eliminate the per-call dependency"; paid is "(C)".)
283
284 ### The model / training layer (new sub-layer, fed by the scaffolding)
285 ```
286 Data curation      Training      Eval      Serving
287 (consented user vid (fine-tune an open, (extend the existing (self-host inference
288 + golden set +      commercially-usable  eval harness to QA   as a provider /
289 rally/segment/event base VLM via      quality + the        ComputeBackend;
290 annotations ?      LoRA/QLoRA)          regression gate)     paid fallback)
291 video-QA pairs)
292 ```
293 This layer **reuses every other layer**: **Storage** holds the training corpus;
294 **Data** supplies labels (the `annotation` registry + `candidate_segments` +
295 golden set are already the labeling seam); **Compute/Orchestration** runs training and
296 inference jobs (even **BYO-compute** can serve inference on the tenant's GPU); the
297 **eval harness** already in `backend/eval/` extends to grade QA quality. **In other
298 words, the rock-solid scaffolding is exactly what makes the model flywheel possible** ?
299 which is why scaffolding comes first (see §6).
300
301 ### Base model & training (Apache-2.0 only ? licensing is decisive)
302 **Recommended near-term base: `Qwen2.5-VL-7B-Instruct`** ? Apache-2.0, native long-video
303 (dynamic-FPS sampling, temporal mRoPE), mature LoRA tooling. Upgrade path
304 **Qwen3-VL-4B/8B (Apache-2.0)**; **Gemma 4 multimodal (Apache-2.0, 2026)** is a clean
305 Google-stack alternative to evaluate.
306 - ?? **Lock to Apache-2.0-clean bases** so you can fine-tune, **keep weights private,
307 gate behind subscriptions, and never owe open-sourcing or branding**. (Note Qwen2.5-VL
308 is Apache **only at 7B/32B** ? 3B/72B use the custom Qwen license.)
309 - ? **Reject for the flagship:** **Gemma 3** ? custom terms with a **viral derivative
310 clause** (anything fine-tuned on it, *or even trained on its synthetic outputs*, stays
311 permanently bound to Google's *updatable* Prohibited-Use Policy); **Gemma 4's switch
to
312 Apache-2.0 removes this**. **Llama 3.2 Vision** ? custom license forces a "Llama-?"
313 model name + "Built with Llama" badge and a 700M-MAU ceiling. **`gpt-oss`** ?
Apache-2.0
314 but **text-only**, so usable only as an optional text-reasoning layer, never the video
315 encoder.
316 - **Training method:** **QLoRA** (? r=64, ?=128) on the 7B ? fits one prosumer GPU and

```

```

317     avoids catastrophic forgetting of general video skills. Build the dataset by
converting
318     the existing rally/shot/segment annotations + golden set into templated **clip?QA
319     pairs**, expand synthetically with **open-model teachers only** (see constraint
below),
320     and hold out a **golden QA eval** that gates every fine-tune. A few hundred gold pairs
?
321     **5k?20k** curated is a realistic band for this narrow domain.
322 - **Serving economics:** a 7B VLM ? 14 GB VRAM (FP16) / ~3.5 GB (INT4); a ?16 GB GPU is
a
323     comfortable serve target (vLLM for throughput). Self-hosting wins **only** once you
have
324     sustained video-QA volume **** a defensible consented dataset **** real per-call API
325     cost pain ? below that, the paid API is cheaper than a standing GPU and the second
stack
326     is overhead. Hence: **paid-primary now ? owned-primary / paid-fallback later.**
327
328     Base-model licensing + training-data policy are tracked as **COMMERCIALIZATION C14**.
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329

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330     ### Training backend ? managed service vs self-managed (the fine-tune "forge")
331     *Where* you run the LoRA/RL fine-tune. Weigh on **video support, weight
export/ownership,
332     smallest servable model, RL support, no-train DPA, and infra burden**.
333
334     **Decisive finding (researched 2026-06):** managed fine-tuning services are
335     **image-based** (JSONL + base64 images) ? for video you pre-sample frames and feed them
336     as image sequences. **Only the self-managed `ms-swift` explicitly advertises *native*
337     video (+audio) training.** So for true video-QA, a self-managed framework is the
338     strongest technical fit ? and it also lets you fine-tune the **small 7B** (cheap to
339     serve) and keep full ownership. Managed services trade some of that for zero infra.
340
341     | Option | Type | Video / VLM support | Smallest VL | RL | Export weights | Best for |
342     |---|---|---|---|---|---|---|
343     | **ms-swift** (ModelScope) | Self-managed (rented/own GPU) | **Native video+audio**;
```

Qwen3-VL/Omni, 300+ MLLMs | any incl **7B** | ? DPO/GRPO | ? full ownership | **? video-QA;

slots into BYO-compute (\$3.2)** |

```

344     | **LLaMA-Factory** | Self-managed | Qwen3-VL + 100+ VLMs; video datasets | any incl 7B
| ? | ? | popular, well-documented |
345     | **Unsloth** | Self-managed (1-GPU optimized) | Qwen3-VL LoRA (vision+lang; frames) |
any incl 7B | ? GRPO | ? | cheapest first runs |
346     | **Tinker** (Thinking Machines) | Managed API (low-level) | Qwen3-VL **30B-A3B+ only**;
```

image input (video **unverified**) | 30B-A3B MoE | ? PPO/GRPO, custom loss | ? adapters + ckpt;

no-train ToS | zero-infra RL/research at MoE scale |

```

347     | **Together AI** | Managed (tune + serve) | Qwen3-VL / Gemma-3 / Llama-4; image data |
verify | partial | ? adapters downloadable | managed Qwen3-VL + serving |
348     | **Fireworks AI** | Managed (tune + serve) | Full Qwen2.5-VL (3B?72B SFT; LoRA@32B +
LoRA-upload) | **7B** | partial | ?? verify export (lock-in risk) | managed *small*-VL + serving
|
349     | **Predibase** (Rubrik) | Managed *** VPC self-host** | VLM instruct-tune; **LoRAX**
multi-adapter | verify | ? (VPC) | ? **RFT/GRPO** flagship | RL "creativity" + multi-adapter +
privacy |
350     | ~~OpenAI GPT-4o / Gemini tuning~~ | Closed | ? | ? | ? | ? no export | **non-fit ?
violates "own the model"*** |
351
352     *(Enterprise clouds ? Databricks Mosaic / SageMaker / Vertex on open models ? give full
353     control but are only worth it if you're already on that stack.)*
354
355     **Recommendation:**
356     - **Early iteration / true video-QA ?** self-managed **`ms-swift`** (or LLaMA-Factory /
357     Unsloth) on a **rented GPU** (RunPod / Lambda / Modal / SkyPilot-to-your-cloud):
native
358     video, any size incl. **7B** (cheap to serve), full ownership, and it drops into the
359     **ComputeBackend / BYO-compute** seam ($3.2). **Likely a better fit than Tinker** for
360     the video requirement *and* serving economics.
361     - **Managed, no GPU ops ?** **Together AI** (Qwen3-VL + downloadable adapters + serving)
```

```

362 or **Fireworks** (full Qwen2.5-VL incl. 7B + serving) ? confirm frame/video support
and
363 adapter export.
364 - **If RL "coaching creativity" + per-sport multi-adapter is central ?** **Predibase**
365 (RFT + LoRAX, VPC self-host for the biometric/privacy posture) or **Tinker** (lowest-
366 level RL primitives).
367 - **Tinker** remains a strong zero-infra RL-research option with clean weight export ?
just
368 not the cheapest path to a *small video* model (its VL floor is 30B-A3B).
369 - **Portability is the hedge:** you own the **data + recipe + exported weights**, so the
370 backend is swappable ? same ethos as the provider/registry pattern. Don't get locked
in.
371
372 **Verify before committing (any backend):** (1) **native video / frame-sequence**
373 ingestion + max frames per example; (2) **weight export** + a **commercial DPA**
covering
374 consented, minors-excluded video (biometric, §7); (3) the **inference plan** for the
size
375 you train (a self-managed **7B** is far cheaper to serve than a 30B-A3B MoE).
376
377 ### Training harness ? make it agent-runnable (nanochat-style operability)
378 The goal: **an agent (or CI) can run a fine-tune on a fresh batch of golden-labeled
video
379 and get a trustworthy ship / no-ship verdict, unattended.** Karpathy's **nanochat** is
the
380 reference for that *operability* ? one MIT repo, one `speedrun.sh` that runs
381 tokenizer?train?eval?serve, a single complexity dial, and an automatic **report-card**
382 score. It is **not** a usable engine here (text-only, from-scratch GPT-2-class), but its
383 five properties are exactly what make training agent-runnable, and we copy them:
384 ** (1) ** one command, end-to-end; ** (2) ** one complexity dial (sane auto-derived
defaults);
385 ** (3) ** an automatic **report card** ? a deterministic pass/fail vs a target (the crux);
386 ** (4) ** reproducible + deterministic + cheap; ** (5) ** minimal/readable.
387
388 **Engine (don't reinvent):** **`ms-swift`** already delivers this for VLMs ? CLI-driven,
389 **native video**, Qwen3-VL support, an eval backend (EvalScope), and reproducibility via
390 `args.json` + `full_determinism`. (LLaMA-Factory is the close second; HF **`nanoVLM`** ?
391 ~750 lines ? is the minimal reference to *learn* VLM training, not the production path.)
392
393 **The harness = thin glue over components we already have** ? one declarative,
394 agent-invokable job (call it `train_speedrun`):
395 1. **Build dataset** from golden labels ? clip?QA pairs (reuse the `annotation` registry
+
396 `candidate_segments` + golden set ? the labeling seam in `backend/`).
397 2. **Fine-tune** via ms-swift on a **ComputeBackend** GPU (§3.2), run as an
orchestration
398 job (§5 / DEFERRED #16): one config, idempotent, logged.
399 3. **Auto-eval + report card** ? extend the existing `backend/eval/` regression harness
to
400 **QA quality**; emit a `report.md` pass/fail vs the **held-out golden QA eval**.
401 4. **Register** the result as the `owned_vlm` provider (Gemini paid fallback stays)
**only
402 if it beats the bar**.
403
404 So: **golden labels ? ms-swift train (on a compute backend) ? eval gate ? register
405 **provider**, behind one command an agent or CI runs. nanochat is the inspiration for the
406 wrapper; ms-swift is the engine; the **golden QA eval is the report card**.
407
408 > **Prerequisite (the crux):** the harness is only as trustworthy as the **golden QA
eval
409 > set**. nanochat's speedrun is meaningful *because* its eval (CORE score) is automatic
and
410 > trustworthy; ours is the golden QA eval ? **build that first**, or an agent-run loop
411 > optimizes blind. (Same "moat = dataset + eval", scaffolding-first principle.)
412

```

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413     ### ?? Teacher / distillation constraint (the single biggest legal landmine)
414     You **cannot** train the owned model on outputs from the paid frontier APIs ? a general
415     video-QA assistant is squarely a "competing model": **Gemini** ("may not use the
416     Services to develop models that compete"), **OpenAI** ("may not use Output to develop
417     models that compete"), **Anthropic** (bars using Outputs to train competing models). Two
418     safe lanes: ** (1) ** generate synthetic QA with **open-model teachers only** (Qwen-VL /
419     InternVL / Gemma 4 ? Apache-clean); ** (2) ** keep the paid API strictly as a **runtime
420     fallback** ? serving answers is fine, *harvesting its outputs to train the model that
421     replaces it is not*. Write a hard ***"no training on competitor-API outputs"*** rule into
422     eng + legal process. (This makes the ROADMAP's "Gemini as offline teacher only" precise:
423     the teacher is *open models*, not the paid APIs.)
424
425     ### Data flywheel, personalization & consent tiers
426     Two complementary loops, each with its **own** opt-in (training consent is separate from
427     processing consent ? §7):
428
429     - **Global model (shared ? raises the floor).** Train the base owned model on an
430     **explicitly-contributed, licensed** corpus ? *not* on whatever sits in a tenant's
431     bucket
432     (preserves "your video, your storage"). Flywheel: *more consented, labeled matches ? a
433     better shared model ? better highlights/answers ? more reason to contribute.* Higher
434     consent bar (pooled data, biometric/minor care ? §7); **minors excluded by default**.
435
436     - **Per-user adapters (personalized ? raises the ceiling) ? owner hunch, under
437     exploration.**
438     A small **per-user (or per-tenant) LoRA adapter, fine-tuned *only on that user's own
439     consented videos*** on top of the shared base, served via **multi-LoRA** (LoRAX /
440     Together / Predibase). Why it's compelling:
441     - **Precision:** adapts to *that user's* court, camera angle, lighting, and players'
442     styles ? materially better detection/highlights for them.
443     - **Cleanest consent story:** *"improve **your** detection using **your** data"* is
444     far
445     lower-friction and lower-risk than contributing to a global model ? the data
446     improves
447     **only the user's own adapter**, with **no cross-user mixing**, sidestepping much of
448     the
449     pooled-corpus biometric/minor entanglement. Reinforces the differentiator: *your
450     video,
451     your storage, your compute ? and now **your model**.* With BYO-compute the adapter
452     can be
453     trained/served on the user's own GPU and **never leave their boundary**; **deletion
454     =
455     drop the adapter** (clean erasure).
456
457     - **Open questions (still a hunch):** cold-start (need enough of the user's labeled
458     data
459     before a personal adapter beats the base ? fall back to the shared model meanwhile);
460     a
461     **per-user held-out eval** to prove the adapter helps (don't ship a regression);
462     base +
463     optional global-adapter + per-user-adapter stacking; per-user training cost at
464     scale.
465     Worth a spike once the harness + golden eval exist.
466
467     **Two consent tiers, then:** ** (a) ** "improve **my own** detection with **my** data"
468     (per-user, low friction) and ** (b) ** "contribute to the **shared** model" (global,
469     higher
470     bar). **Default for both is off.**
471
472     ## 6. Phased roadmap
473
474     | Phase | Deliverable | Build / reuse |
475     |---|---|---|
476     | **P0** | **Async job model, single-tenant** (DEFERRED #16) | `jobs` table via
477     `user_version` ladder; worker; 202+status endpoints |
478     | **P1** | **Identity + multi-tenant data model + upload + highlight persistence +

```

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history** | auth (build vs Auth0/Clerk/Supabase ? decide); Postgres + `tenant_id`/RLS;
`highlights` table; repository layer; history API + UI |
464 | **P2** | **`StorageBackend` registry + BYO-storage** | new ABC+registry;
`local`?`gdrive`/`s3`/`gcs`; replace `video_path` with `StorageRef`; encrypted per-tenant creds
|
465 | **P3** | **`ComputeBackend` registry + BYO-compute** | `local`/`hosted`/`byo_worker`;
pull-based worker agent; control/data-plane split |
466 | **P4** | **Rich rally/shot semantic metadata & tagging** *(the next value stage after
solid rally detection + highlights)* | shot type (smash / backhand drive / net), rally
**outcome** (winner / forced / **unforced error**), context (e.g. **open court but error**) ?
query-API filters + `HighlightMetadata` + a **human correction/review loop** feeding the golden
set via the `annotation` registry + `candidate_segments` |
467 | **P5** | **Owned generative video-QA model** ($5A) | global flywheel **per-user
adapters** ; P4 metadata is training signal ? fine-tune an open VLM (Qwen-VL) via LoRA ? serve as
`owned_vlm` **provider with Gemini paid fallback** ; extend the eval harness to QA quality |
468 | **P6** *(visionary)* | **Gait / biomechanics under a supervising coach** |
deliberately **back-burner** ; technique feedback; highest Art. 9 bar ? gate behind explicit
biometric + parental consent ($7) |
469
470 P0?P1 deliver the headline ask (per-user upload + stored highlights + history). P2?P3
471 deliver the "plug-and-play storage/compute" ask. **P4 (rich semantic metadata) is the
472 immediate next value stage once rally detection + highlights are solid** ? e.g.
**"rallies
473 that ended in a smash or backhand drive,"* **rallies where the court was open but the
474 player made an unforced error."* **P5** owns the model; **P6 (gait) is the visionary
475 horizon, intentionally deferred.**
476 **Sequencing principle (owner directive):** make the scaffolding (P0?P4) **rock-solid
477 first**, keep the **known paid stack as an always-ready fallback**, and only then shift
478 focus to **owned-model quality** (P5). The provider registry keeps the paid path one
479 config flip away.
480
481 ## 7. Privacy & legal (gait raises the bar ? design for it now)
482
483 - **The load-bearing nuance:** biometric data is special-category **only when used to
484 uniquely identify* a person** (GDPR Art. 9). **Highlight/shot detection on players a
485 coach already knows is general analytics ? generally *not* Art. 9.** But
486 **gait/biomechanics that builds a per-athlete identifying template tips into Art. 9**
487 and needs **explicit, granular consent** referencing biometric data. Engineer gait
488 with that boundary (consent gate, purpose limitation, no cross-account
identification);
489 decide whether it runs **locally/on-tenant only** (strong posture) vs our cloud. (EDPB
490 Guidelines 3/2019 on video devices apply.)
491 - **Minors (e.g. academies).** Parental/guardian consent and safeguarding are first-
order,
492 not an afterthought. Model consent explicitly per relationship (user?subject,
coach?athlete
493 where the overlay applies, parent?minor).
494 - **BYO-storage is the central privacy/liability lever:** if raw video stays in the
495 tenant's own bucket/Drive and we process via signed URL or their compute, we minimize
496 what we hold ? shrinking breach surface, residency, and erasure obligations.
497 - **Paid-LLM no-train (C4):** use **Gemini via Vertex AI** (contractual no-training),
498 **not the consumer/free tier** ; set short/zero retention, document it in the DPA, and
499 surface "your footage is never used to train AI" as a trust feature for academies. For
500 the gait phase, prefer keeping biometric inference off third-party APIs entirely.
501 - **Training consent is a distinct, higher bar (for the owned model, $5A).** Training on
502 user video needs a **separate, explicit, opt-in training licence** ? not bundled with
503 processing consent ? and it **compounds** the biometric (Art. 9) and minor-consent
504 issues above. Train **only on explicitly contributed/licensed video** ; never train on
505 what merely passes through a tenant's bucket. **Exclude minors' video from the
training
506 pool by default** , keep the training corpus a **separate, firewalled, opt-in sub-
pool** ,
507 and log **per-clip consent/provenance** for revocation + audit. Incentives must not
make
508 consent non-free (the product stays fully usable without contributing). And the

```

509 ****teacher-distillation rule**** (§5A): never train on paid-LLM outputs. ****Two consent**
510 tiers
511 (§5A):**** (a)* "improve **my own** detection with **my** data" ? a per-user adapter**
512 trained
513 only on the user's own video (lowest friction; no cross-user mixing; deletion = drop
514 the
515 adapter); ***(b)* "contribute to the **shared** model" ? pooled corpus, higher bar.**
Default
516 for both is ****off****.
517 - ****Right to erasure / retention:**** per-tenant, per-athlete delete must cascade across
DB +
518 storage artifacts (the `ON DELETE CASCADE` discipline already in `database.py` extends
here).
519 **## 8. Key decisions for the owner (the forks)**
520 1. ****Hosted-first vs BYO-first.**** Recommended: ****hosted control-plane first**** (P0?P1)
with
521 ****BYO-storage soon after**** (P2); ****BYO-compute later**** (P3) as a power-
user/enterprise
522 option ? it's the highest-complexity, highest-security item.
523 2. ****Database now or later.**** Move to ****Postgres at P1**** for the hosted path, or stay
524 SQLite-per-tenant if self-hosted-single-academy is the first GTM. (Abstract DB access
525 regardless.)
526 3. ****Auth: build vs buy**** (Auth0 / Clerk / Supabase Auth / Cognito).
527 4. ****Orchestration tech**** (see research recommendation) ? bias to the simplest that
gives
528 retries + status.
529 5. ****Gait processing location**** ? local/on-tenant (privacy-max) vs cloud (capability-
max).
530 6. ****How far to push BYO-compute**** given its security/ops complexity.
531 7. ****Owned-model base + licensing (§5A).**** Which commercial-OK video VLM to build on
532 (Qwen-VL Apache-2.0 vs Gemma 3 custom license vs ?) and confirm a gated commercial
533 derivative is permitted ? see COMMERCIALIZATION C14.
534 8. ****Training-data sourcing & consent.**** Opt-in contribution terms, incentives, and the
535 teacher-distillation constraint (whether paid-LLM outputs may seed the corpus).
536 9. ****When to start the owned model**** ? strictly after the scaffolding + flywheel are
537 solid (the explicit directive), with the paid provider as the standing fallback.
538 ****Risks:**** scope explosion (sequence strictly P0?P5); multi-tenant data-leak bugs (RLS +
539 tests); BYO-compute security; ****biometric/minor liability**** (get this right before any
540 gait feature ships); ****training on user video without proper opt-in/licence****, and
541 ****base-model / teacher-distillation licence violations****; cost of hosted GPU (training
542 *and* inference); ****premature model-building**** before data/eval are ready (mitigated by
543 keeping the paid fallback and gating P5 behind the flywheel).
544 **## 9. Sources**
545 External best-practices research (2026-06-07) backing the recommendations above. Each
546 underlying claim in the research was tagged fact-vs-synthesis; figures are as-of and
547 should be re-validated before committing budget.
548 ****Storage abstraction:**** [fsspec](https://fsspec.com/) ·
549 [cloudpathlib](https://github.com/drivendataorg/cloudpathlib) ·
550 [smart_open](https://github.com/piskvorky/smart_open) · [Apache
551 Libcloud](https://libcloud.apache.org/) · [gdrive-fsspec](https://github.com/fsspec/gdrive-
fsspec) · [dropboxdrivefs](https://github.com/fsspec/dropboxdrivefs) · [AWS S3 presigned
URLs](https://docs.aws.amazon.com/AmazonS3/latest/userguide/using-presigned-url.html) · [token-
based S3 with STS](https://oneuptime.com/blog/post/2026-02-12-token-based-access-s3-sts/view) ·
[signed URLs across clouds](https://medium.com/@shaikhsarwan49/signed-urls-in-aws-gcp-and-azure-
a-beginner-friendly-guide-e2e549425865)
552 ****Multi-tenancy:**** [Bytebase: multi-tenant DB
553 patterns](https://www.bytebase.com/blog/multi-tenant-database-architecture-patterns-explained/)
· [Hunchbite: RLS vs schema-per-tenant](https://hunchbite.com/guides/multi-tenant-saas-

architecture) · [PlanetScale: tenancy in Postgres](https://planetscale.com/blog/approaches-to-tenancy-in-postgres) · [DEV: DB-per-tenant vs shared schema](https://dev.to/young_gao/multi-tenant-architecture-database-per-tenant-vs-shared-schema-1n2e)

554

555 ****BYO-compute / orchestration:**** [Pinecone BYOC](https://www.pinecone.io/blog/byoc/) · [Nuon: control/data-plane](https://nuon.co/blog/byoc-control-plane-data-plane-architectures) · [GitHub self-hosted runners (ARC)](https://docs.github.com/en/actions/hosting-your-own-runners/managing-self-hosted-runners-with-actions-runner-controller/about-actions-runner-controller) · [SkyPilot](https://github.com/skypilot-org/skypilot) · [SkyPilot AI control plane](https://blog.skypilot.co/ai-job-orchestration-pt2-ai-control-plane/) · [Modal (AWS Marketplace)](https://aws.amazon.com/marketplace/pp/prodview-j727623xqhh2k) · [Northflank: Anyscale alternatives](https://northflank.com/blog/anyscale-alternatives-for-ai-ml-model-deployment) · [DevPro: Python task queues 2025](https://devproportal.com/languages/python/python-background-tasks-celery-rq-dramatiq-comparison-2025/) · [Judoscale: choosing a task queue](https://judoscale.com/blog/choose-python-task-queue) · [Hatchet vs Temporal](https://hatchet.run/versus/hatchet-vs-temporal) · [Hatchet](https://github.com/hatchet-dev/hatchet)

556

557 ****Privacy / legal:**** [GDPR Art. 9](https://gdpr-info.eu/art-9-gdpr/) · [GDPR Local: biometric compliance](https://gdprlocal.com/biometric-data-gdpr-compliance-made-simple/) · [Legiscope: Art. 9](https://www.legiscope.com/blog/gdpr-article-9-special-categories.html) · [EDPB Guidelines 3/2019 (video)](https://www.edpb.europa.eu/sites/default/files/files/file1/edpb_guidelines_201903_video_devices_en_0.pdf) · [Fox Rothschild: EDPB video guidance](https://www.foxrothschild.com/publications/video-surveillance-under-gdpr-edpb-issues-final-guidance) · [ICO: children & UK GDPR](https://ico.org.uk/for-organisations/uk-gdpr-guidance-and-resources/childrens-information/children-and-the-uk-gdpr-old/what-are-the-rules-about-an-iss-and-consent/) · [EDPB Guidelines 05/2020 (consent)](https://www.edpb.europa.eu/sites/default/files/files/file1/edpb_guidelines_202005_consent_en.pdf) · [AI data-retention compared](https://ax-sentinel.com/blog/ai-data-retention-policies-compared) · [Xenoss: enterprise LLM platforms](https://xenoss.io/blog/openai-vs-anthropic-vs-google-gemini-enterprise-llm-platform-guide)

558

559 ****Comparable products:**** [Veo](https://www.veo.com/) · [Veo Help Center](https://support.veo.com/hc/en-us/articles/4459159526929-Overview-of-cam-veo-co-Record-manage-and-upload-footage-with-your-Veo-Cam-1) · [Latterly: Hudl competitors](https://www.latterly.org/hudl-competitors/) · [Apple: Behind the Design ? SwingVision](https://developer.apple.com/news/?id=0pg4dthn) · [SwingVision FAQ](https://swing.vision/faq) · [Sportsvisio: coach video-analysis comparison](https://www.sportsvisio.com/stories/video-analysis-software-for-coaches-complete-comparison)

560

561 ****Training backends & harness (\$5A):**** [karpathy/nanochat (harness reference)](https://github.com/karpathy/nanochat) · [HF nanoVLM (minimal VLM)](https://github.com/huggingface/nanoVLM) · [nanoVLM blog](https://huggingface.co/blog/nanovlm) · [Tinker (GA + vision)](https://thinkingmachines.ai/blog/tinker-general-availability/) · [Tinker model lineup](https://tinker-docs.thinkingmachines.ai/model-lineup) · [Tinker ToS (weights/no-train)](https://thinkingmachines.ai/legal/tos/) · [Fireworks VLM fine-tuning](https://fireworks.ai/blog/vlm-tuning) · [Together AI fine-tuning](https://www.together.ai/fine-tuning) · [Predibase: fine-tune & serve VLMs](https://predibase.com/blog/how-to-fine-tune-and-serve-vlms-in-predibase) · [Predibase RFT/GRPO](https://docs.predibase.com/user-guide/fine-tuning/grpo) · [ms-swift (native video/audio)](https://github.com/modelscope/ms-swift) · [LLaMA-Factory (VLMs)](https://github.com/hiyouga/LlamaFactory) · [Unsloth (Qwen3-VL)](https://qwen.readthedocs.io/en/latest/training/unsloth.html) · [Fine-tuning tools 2026 (LoRA/QLoRA)](https://www.index.dev/blog/top-ai-fine-tuning-tools-lora-vs-qlora-vs-full)

562

563 ****Owned model & licensing (\$5A):**** [Qwen2.5-VL](https://qwenlm.github.io/blog/qwen2.5-vl/) · [Qwen2.5 license split](https://qwenlm.github.io/blog/qwen2.5/) · [Qwen3-VL](https://github.com/QwenLM/Qwen3-VL) · [Gemma 3 (multimodal)](https://huggingface.co/blog/gemma3) · [Gemma Terms of Use](https://ai.google.dev/gemma/terms) · [Gemma license-risk analysis](https://wcr.legal/google-gemma-license-risks/) · [Gemma 4 / Apache-2.0](https://www.mindstudio.ai/blog/what-is-gemma-4-apache-2-license-commercial-ai-deployment) · [Llama 3.2 license (700M MAU + naming)](https://www.llama.com/llama3_2/license/) ·

[gpt-oss (text-only, Apache-2.0)](<https://openai.com/index/introducing-gpt-oss/>) · [Gemini API terms (compete clause)](<https://ai.google.dev/gemini-api/terms>) · [OpenAI terms](<https://openai.com/policies/row-terms-of-use/>) · [Anthropic: outputs & training](<https://support.claude.com/en/articles/12326764-can-i-use-my-outputs-to-train-an-ai-model>) · [QLoRA practical](<https://apxml.com/courses/fine-tuning-adapting-large-language-models/chapter-4-parameter-efficient-fine-tuning/practice-qlora-fine-tuning>) · [VLM LoRA cookbook](https://huggingface.co/learn/cookbook/en/fine_tuning_vlm_trl) · [GDPR Art. 9](<https://gdpr-info.eu/art-9-gdpr/>) · [athletes & biometric data](<https://www.architectureandgovernance.com/applications-technology/biometric-data-and-athletes-privacy-law-and-compliance-implications/>) · [federated learning + DP](<https://research.google/blog/federated-learning-with-formal-differential-privacy-guarantees/>) · [self-hosted LLM VRAM/cost](<https://www.gpu-mart.com/guides/self-hosted-llm>) · [Gemini API pricing](<https://ai.google.dev/gemini-api/docs/pricing>)

564

565 ## See also

566 - [``COMMERCIALIZATION.md``](COMMERCIALIZATION.md) ? C12 (SaaS foundations), C4 (paid-LLM), economics.

567 - [``DEFERRED_HARDENING_PLAN.md``](DEFERRED_HARDENING_PLAN.md) ? #16 async job model (the P0 infra slice), #7 packaging.

568 - [``PMF_MARKET_RESEARCH.md``](PMF_MARKET_RESEARCH.md) ? academy/coach beachhead (who this serves).

569 - [``CODE_MAP.md``](CODE_MAP.md) ? current architecture + the registry extension points.

570 - [``ROADMAP.md``](ROADMAP.md) ? offline self-improving rally detection (the pipeline's own trajectory).

571

11. user (2026-06-10T05:33:50.771Z)

1 # PMF & Market Study ? AI Highlight-Reel Tool for Self-Recorded Racket Sports

2

3 > **Research snapshot ? 2026-06-07.** Figures are cited as-of with source + year;

4 > validate live before relying on them for investment/pricing decisions. This is the

5 > market-research-of-record behind the PMF section of the living

6 > [``COMMERCIALIZATION.md``](COMMERCIALIZATION.md) tracker (§6). Product framing:

7 > a closed-source hosted API/SaaS (plus local mode) that ingests long-form,

8 > self-recorded, **static-camera, full-court amateur** racket-sports video and

9 > auto-detects rallies and compiles highlight reels. Primary sport: badminton;

10 > sport-generic architecture roadmapped to tennis, pickleball, table tennis, cricket.

11

12 > **Confidence convention.** Figures from named industry reports, federations, or

13 > company pricing pages are tagged with source + year. Inferred items are tagged

14 > ****(estimate)**** / ****(low confidence)****. Emerging-category market sizes vary widely

15 > between vendors; ranges are shown rather than false precision.

16

17 ---

18

19 ## 1. Executive summary

20

21 The product sits at the intersection of three favorable trends: (a) a fast-growing

22 **sports analytics market** (~USD 5?6B in 2025, ~18?22% CAGR to USD 24?36B by ~2034

23 depending on vendor); (b) a structural shift to **amateur self-recording** on action

24 cameras and phones (action-camera market ~USD 6.5B in 2024, sports the largest

25 segment); and (c) **strong racket-sport participation** ? 220M+ badminton players,

26 106M tennis players (2024), and a near-vertical pickleball curve (~19.8M US players in

27 2024, +45.8% YoY).

28

29 The central tension is the **static-camera constraint**: the tool works best on fixed,

30 full-court amateur footage and **fails** on broadcast/panning footage. This is both a

31 moat (most competitors target team sports / broadcast) and a filter on the addressable

32 user ? the customer must already set a camera on a tripod behind the court. That

33 behavior is concentrated in **coaches/academies, clubs, and serious enthusiasts**, not

34 the median casual player.

35

36 **Recommended beachhead (strongest single):** **badminton coaching academies and**

37 **serious club/competitive players in India** ? a large, rapidly formalizing academy

ecosystem (Khelo India tailwind), genuine fixed-angle recording behavior, existing willingness to pay for coaching, an underserved tooling gap (incumbents are team-sports/broadcast), and a direct-but-immature competitor field. The product is *native badminton-first* ? a genuine edge here. Secondary: ***(2) US/global pickleball players & clubs*** (capital-rich, fastest-growing, but SwingVision already strong); ***(3) badminton academies/clubs in East/SE Asia + Denmark*** (highest density).

****Pricing**** clusters into hardware-bundled auto-filming (Veo, Trace, Pixellot, Spiideo) and software-only phone AI (SwingVision USD 179.99/yr; BadPro+ pay-per-use). A software-only BYO-camera model undercuts the hardware players; realistic anchors ***(estimate): B2C USD 60?180/yr; B2B academy/club USD 300?1,500/yr per location.***

****Top risks:**** (1) amateurs won't reliably set up a tripod ? TAM collapses to coaches/clubs; (2) free/near-free alternatives cap B2C WTP; (3) capture-quality variance is the core failure mode; (4) badminton's strongest markets (India, Indonesia, China) are the most price-sensitive and hardest to bill at Western SaaS prices.

2. Market sizing & trends

2.1 Sports analytics / sports-video market

- **2025 size:**** ~USD 5.47B (Precedence Research) to ~USD 5.79B (Fortune Business Insights); other vendors USD 5.6?6.0B. ***(2025)***
- **Growth:**** CAGR 15.7%?22.5%. Precedence ? ~USD 29.75B by 2034 (20.63%); Fortune BI ? USD 24.03B by 2032 (22.5%); MarketsandMarkets (narrower) USD 4.75B by 2030 (15.7%); Future Market Insights ? ~USD 36.2B by 2035 (22.1%). ***(reports 2024?2025)***
- **Video analytics is the leading segment**** within sports analytics (Grand View / Fortune BI) ? the directly relevant sub-market.
- **Interpretation (estimate):**** the *amateur/club auto-highlight* slice is a small but fast-growing fraction; most analytics revenue today is pro/broadcast/team-sports. The product targets the long tail the big numbers under-serve.

2.2 Action-camera / amateur recording adoption

- **2024 size:**** ~USD 6.5B (GMIInsights) ? ~USD 16.9B by 2034 (~10% CAGR). ***(2024?2025)***
- **Sports is the largest end-use segment**** of action cameras (GMIInsights, 2024).
- **GoPro ~47% share, DJI ~20%**** (electroIQ/industry, 2024); phones + cheap tripods are the de-facto amateur rig.
- **Implication:**** the capture-side behavior the product depends on is already widespread and growing ? *for the segments that record* ? validating "BYO-camera, software-only."

2.3 Participation & popularity by sport and geography

Sport	Participation signal	Source (year)
Badminton	220M+ play regularly; 160+ countries BWF est. via BadmintonBites (2023)	
Badminton ? China	~80M registered, up to ~100M (largest base) industry compilations (2023)	
Badminton ? India	20M+ recreational by 2023; academy boom + Khelo India industry compilations (2023)	
Badminton ? Indonesia	National sport; ~1M club players industry (est.)	
Badminton ? Denmark	100,000+ registered club members (Europe #1) industry (2023)	
Badminton ? Japan/Korea	Major school-club + workplace sport industry (2023)	
Tennis	106M players (2024)**; +25.6% over 5 yrs (87M in 2021) ITF Global Tennis Report (2021, 2024)	
Tennis ? Asia	~1/3 of players (33M of 87M) ITF (2021)	
Pickleball ? US	19.8M players (2024)**; +45.8% YoY, +311% over 3 yrs; ~6.2M core SFIA / Pickleheads / The Dink (2024)	
Pickleball ? US courts	~70,000 courts, +55% places to play YoY SFIA / Pickleheads (2024)	
Pickleball ? global	IFP 78 member countries; UK +73% in 2024; ~282M played monthly per PPA claim *(low confidence)* various (2024?2025)	
Table tennis	30M+ players; 227 member associations ITTF (current)	

96 | **Cricket** | 2.5B+ fans; 500M+ in India (mostly **fans**, not self-recorders) | Cricket
 Rise / Nielsen (2024?2025) |

97

98 - **Pickleball market value:** ~USD 2.2B (2024) ? USD 9.1B by 2034 at 15.3% CAGR
 99 (Market.us/Custom Market Insights, 2024?2025).

100 - **Badminton equipment market:** ~USD 7.23B global (2023), Asia-Pacific dominant.

101

102 **Geographic read:** badminton density is overwhelmingly Asian + Denmark; tennis is
 103 broad/high-income; pickleball is US-dominant, capital-rich, globalizing fast; cricket
 104 is huge as **fandom** but its amateur self-recording + static-full-court fit is weak
 105 (large field, not a fixed-frame court). For **this product's static-camera fit**,
 106 badminton/pickleball/tennis/table-tennis are the core; cricket is a long-tail roadmap
 107 item, not a beachhead.

108

109 ---

110

111 **## 3. Customer segment evaluation**

112

113 | Segment | Market-size signal | Willingness to pay | Accessibility | Static-camera fit
 | B2C/B2B | Verdict |

114 |---|---|---|---|---|---|

115 | **Individual amateur/enthusiast** | Largest raw TAM (100s of millions) | **Low** ?
 competes with free; ~USD 5?15/mo ceiling | High via app stores; high CAC/churn | **Mixed** ?
 most won't set a tripod | B2C | Volume play, weak economics; use as funnel |

116 | **Coaches & coaching academies** | India academy boom; thousands across Asia |
High ? already sell coaching; USD 300?1,500/yr plausible | Reachable via
 federations/distributors/social | **Strong** ? already film fixed-angle | B2B/B2B2C | **Best**
 fit ? buyer, recorder, payer aligned |

117 | **Clubs & leagues** | Denmark 100k+; thousands of clubs Asia/EU | Medium-high; USD
 500?3,000/yr | Medium ? committee sales cycle | **Strong** | B2B/B2B2C | Strong secondary;
 longer cycle |

118 | **Tournament/event organizers** | Many small/regional events | Medium ? per-
 event/seasonal | Medium ? episodic | **Strong** | B2B | Good per-event upsell; lumpy |

119 | **Schools & universities** | Large (esp. Asia/Japan clubs) | Low-medium ? budget-
 constrained | Low ? slow procurement | Strong | B2B | Slow; grant/Khelo-India dependent |

120 | **National/regional federations** | Few, high-value | Medium-high; procurement-
 heavy/political | Low ? long cycles | Strong | B2B | Lighthouse/credibility, not beachhead |

12. user (2026-06-10T05:33:51.695Z)

1 # Architecture Decision Records

2

3 **## ADR-001: Precise shuttle tracking via self-hosted TrackNetV4/WASB on WSL2**

4

5 **Date:** 2026-05-25

6 **Status:** Accepted

7

8 **### Context**

9

10 The pipeline uses an AI multimodal provider (Gemini) for **semantic** rally segmentation.
 11 We need ***precise frame-level detection of the moving shuttlecock*** (and, generically,
 12 balls for other sports) to produce higher-quality candidate rallies and a fine-tuning
 13 signal.

14

15 We evaluated whether a commercially available model/API could do this. Findings:

16

17 - **Multimodal LLM APIs** (Gemini, Claude, GPT-class) are the wrong tool for precise
 18 localization. They reason well about **what** is happening (rally vs. dead time) but
 19 cannot reliably track a tiny, fast (~up to 400 km/h), motion-blurred shuttle frame by
 20 frame.

21 - The precision leaders are ***purpose-built trajectory models***, which are open source
 and
 22 self-hosted, not big-vendor APIs:

23 - **TrackNetV4** (TensorFlow) ? badminton-specific, heatmap + motion-attention,
 24 highest reported shuttle accuracy; ships `src/predict.py` producing per-frame (X, Y)

```

25     coordinate CSV. https://github.com/AR4152/TrackNetV4
26 - **WASB** (PyTorch) ? multi-sport strong baseline (badminton/tennis/volleyball?),
27   fits our `SportProfile` abstraction; needs a custom-video inference wrapper.
28   https://github.com/nttcom/WASB-SBDT
29 - True turnkey commercial APIs (Roboflow hosted models) exist but cap below a tuned
30   TrackNet; enterprise systems (Hawk-Eye, SkillCorner, TRACAB) are not API-accessible.
31
32   ### Decision
33
34   1. Adopt **self-hosted TrackNetV4** as the primary precise shuttle detector, with
**WASB**
35     as the multi-sport fallback.
36   2. Run these models under **WSL2 (Ubuntu)**, **not** native Ubuntu. The entire indexer
37     already runs on Windows; WSL2 keeps a single dev environment and quarantines the
38     TensorFlow/PyTorch dependencies. Native Ubuntu would require dual-boot and migrating
39     the whole project for marginal gain.
40   3. Integrate via a **subprocess adapter**: a detector module that shells out to the WSL
41     `predict.py`, reads the coordinate CSV, and converts trajectory ? candidate rallies
42     feeding the existing stitching flow. This keeps the TF dependency fully isolated from
43     the main Python environment.
44
45   ### Consequences
46
47   - **WSL I/O caveat:** `/mnt/c` access is slow. Stage match videos inside the WSL
48     filesystem (e.g. `~/clips/`) before inference rather than reading large mp4s across
the
49     Windows?WSL boundary.
50   - TrackNetV4 (TensorFlow) and WASB (PyTorch) live in **separate conda envs**.
51   - Next step after integration: manual video annotation ? fine-tune the detector.
52
53   ---
54
55   ## ADR-002: Use the sports-data-collector's rally annotations as a golden eval +
temporal-segmenter training set
56
57   **Date:** 2026-05-26
58   **Status:** Accepted
59
60   ### Context
61
62   A sibling project, **sports-data-collector**, curates sports videos and stores per-rally
63   `[start, end)` timestamps (plus stroke counts, scores) in an SQLite DB, with a
64   commercial-license flag per source. We asked whether this data could improve the
65   indexer's rally detection, and for which task.
66
67   Key findings:
68
69   - The annotations are **temporal-only** (when rallies start/end). They do **not**
contain
70     per-frame shuttle `(x, y)` coordinates, so they **cannot train the shuttle detector**
71     (TrackNetV4/WASB ? see ADR-001), which needs coordinate labels.
72   - They are an excellent **golden evaluation set** and a **training signal for the
73     **temporal* rally segmenter** (active vs. dead play over time) ? the stage that is
today
74     a hand-tuned heuristic, not a learned model.
75   - The collector's five sports tables (`badminton_rallies`, `tennis_points`,
76     `cricket_deliveries`, `pickleball_rallies`, `tabletennis_rallies`) all share a
77     `(video_id, ordinal, start_time, end_time)` shape, and the indexer already has
78     `sport_registry` / `segmenter_registry`. So a **generic** `(start, end)` bridge serves
79     every sport.
80   - The two repos key videos differently: the collector by a human `video_id`, the indexer
81     by **file MD5**. A re-encode/re-download changes the bytes, hence the MD5.
82
83   ### Decision
84

```

```

85 1. Treat the collector's rally timestamps as (a) the golden eval set for rally
86 segmentation and (b) the labels for training a temporal segmenter ? explicitly
87 not as shuttle-detector training data.
88 2. Bridge the repos with a sport-agnostic export (`curate export` in the collector):
89 per-video `start,end` CSVs (the indexer's eval format) + a `manifest.json` that pairs
90 each CSV with its video file. Gate on `is_commercial` so only commercially usable
data
91 flows through.
92 3. Honour the MD5 keying contract: evaluation/processing must run on the exact video
93 file recorded in the manifest; acquisition (ADR-003) therefore precedes processing.
94
95 ### Consequences
96
97 - The collector becomes the producer of labels; the indexer the consumer (processing
+
98 scoring + training). Neither takes a hard dependency on the other beyond the manifest.
99 - Scoring is free and deterministic (reads the DB; no API). See `docs/EVALUATION.md`.
100 - Only invest in frame-level coordinate labels (for the detector) if evaluation
shows
101 detection, not segmentation, is the bottleneck.
102
103 ---
104
105 ## ADR-003: Video acquisition ? always best resolution, upgrade in place, authenticated
cookies
106
107 **Date:** 2026-05-26
108 **Status:** Accepted
109
110 ### Context
111
112 The collector's initial crawl downloaded videos with `yt-dlp -f worstvideo` (256x144),
113 mislabelled as 1080p. The indexer needs 720p MP4 for meaningful YOLO/shuttle detection.
114 Re-acquiring full-resolution sources surfaced several hard-won realities:
115
116 - The ShuttleSet corpus tops out at 1080p (no 4K); "best available" resolves to
1080p.
117 - Authenticating with a YouTube Premium account is required for reliable,
unthrottled
118 downloads ? anonymous downloads are throttled (~1.3 MiB/s) and non-deterministic
119 (identical commands variously yielded 1080p DASH, 1080p HLS, or a 144p fallback).
120 - Chrome cookies can't be read on Windows (app-bound encryption / DPAPI, yt-dlp
121 #10927); Firefox cookies read cleanly even while open.
122 - The `web`/`web_safari` player clients are forced into SABR streaming, which strips
123 download URLs and silently drops to 144p (yt-dlp #12482). yt-dlp's `default` client
124 serves real downloadable 1080p DASH with auth cookies.
125 - Bulk bursts trip YouTube rate-limiting (a 22-video burst failed all at once,
though
126 each worked in isolation).
127
128 ### Decision
129
130 1. Always fetch the best resolution available (`bv*+ba/b -S res,...`, merged to
MP4);
131 ffprobe the result and store the actual width/height/fps ? never assume.
132 2. Upgrade in place, never clean-slate: the `fetch` command re-downloads a video and
133 updates only `local_path`/`resolution`/`fps`, preserving every rally annotation and
134 curation status (deleting would cascade-delete the valuable labels).
135 3. Use authenticated cookies (`--cookies-from-browser firefox` / `--cookies file`)
and
136 the default player client; enable the EJS challenge solver (deno).
137 4. Throttle bulk runs (`--sleep` + per-video `--retries`/`backoff`) and treat a
138 below-min-height download as a failure (discard, don't commit) so a flaky low-res
139 result never overwrites a good entry.
140

```

```

141     ### Consequences
142
143     - All 22 fetchable ShuttleSet matches are now true 1080p (one match, shuffleset_12, has
no
144         source URL in the catalog and remains video-less but keeps its annotations).
145     - Acquisition is reproducible and resumable; the ~26 GB corpus is kept outside git.
146     - Future non-YouTube or 4K sources are supported by the same code (optional
`max_height`).
147
148     ---
149
150     ## ADR-004: Add an offline, API-free *learned* rally segmenter as a first-class mode
151
152     **Date:** 2026-05-26
153     **Status:** Proposed (substrate pending data)
154
155     ### Context
156
157     The default high-quality path (`yolo_hybrid`) refines YOLO candidate windows with
**Gemini**,
158     which costs money and requires network access. With a now-1080p, commercially-licensed,
159     rally-annotated corpus (ADR-002/003), we can learn the refinement step offline instead.
160
161     Crucially, the **API-free signal already exists and is already persisted**:
`yolo_hybrid`
162     Stage 1 is pure YOLO (player tracking ? per-window motion features: peak/mean velocity,
163     active-frame count, track-id count) stored in `candidate_segments.stats`. Gemini is only
164     the Stage-2 refiner. So the rally annotations can **train a classifier to replace
Gemini**,
165     with no external API and **no new labeling**.
166
167     Three candidate substrates were weighed:
168
169     - **Tune `motion` thresholds** on the annotations ? trivial, fully offline, but a low
170       ceiling (pixel motion can't separate rally from crowd/pan/replay). Use as a *floor*.
171     - **YOLO features ? trained classifier** ? reuses already-stored, semantically
meaningful
172       player-motion features; medium-high ceiling; moderate effort. *Recommended primary.*
173     - **WASB shuttle-motion ? rally detector** ? highest ceiling (the shuttle defines play),
174       fully offline but heaviest (needs the WSL2 WASB runner, ADR-001); can't train WASB
175       itself (no xy labels), only a rally layer on its output.
176
177     ### Decision
178
179     1. Build an **offline, API-free learned segmenter** as a new registry mode (e.g.
180       `yolo_learned`): label detector windows by overlap with GT rallies, train a
lightweight
181       classifier on the persisted features, emit final segments with **zero API calls**,
and
182       score it with the existing harness against `motion` and a Gemini reference.
183     2. **Choose the substrate from data, not assumption.** A learned refiner can only
sharpen
184       the windows the detector found, so the deciding metric is the detector's **candidate-
stage
185       recall** vs. the annotations: high recall ? build on YOLO features; low recall ?
bring in
186       WASB's shuttle signal to raise the ceiling.
187     3. Keep the Gemini-backed `yolo_hybrid` as an **upper-bound reference**, not a
production
188       dependency.
189
190     ### Consequences
191
192     - The final tool gains a **self-contained, offline, free** rally mode trained on its own
193       curated data ? runnable air-gapped, improvable as more matches are annotated.

```

194 - The approach is **sport-generic** : the same label?feature?classifier machinery extends
to
195 tennis/cricket once those have annotations.
196 - The full paid 22-video Gemini evaluation is **not** run; spend is limited to a single
197 reference video.
198
199 **### Update (2026-05-26): Phase 3 evidence ? substrate is WASB, not YOLO**
200
201 The deciding candidate-recall measurement is in. On shuttleset_8 (64 GT rallies), pure
202 YOLO Stage-1 produced 181 candidate windows with:
203
204 - **Recall @ IoU 0.5 = 0.05** , but recall @ IoU 0.1 = 0.53 and @ IoU 0.01 = 0.72.
205 - **72%** of rallies have *some* candidate overlap, yet the **median best coverage of a**
206 rally by any candidate is only **~20%** .
207 - Post-hoc **merging adjacent candidates does not help** (recall stays ~0.05 and
degrades
208 as the merge gap grows ? windows overshoot rather than align).
209
210 Interpretation: YOLO isn't *blind* to rallies, but its windows are **badly misaligned**
?
211 because the **broadcast camera tracks the players** , their on-screen (pixel) velocity
dips
212 *during* rallies and spikes on pans/cuts/replays. Player-pixel-motion is therefore a
poor
213 rally proxy on broadcast footage (it was designed for *static* GoPro recordings). A
214 classifier that only *filters* these windows is capped at ~0.25 recall ? below any
useful
215 bar ? and no simple boundary post-processing recovers it.
216
217 **Decision update:** per this ADR's data-driven rule (low candidate recall ? bring in
the
218 shuttle signal), the offline segmenter's substrate is **WASB shuttle-motion** (ADR-001),
219 not YOLO player-motion. The shuttle's motion/presence is camera-invariant and directly
220 defines active play. The *motion* segmenter is likewise unusable here (0 TP ? moving-
camera
221 pixel motion is everywhere).
222
223 **Caveats / next:** this is one video; the mechanism (camera tracking) generalizes but
224 should be confirmed on 2?3 more via a *free, local, candidates-only* YOLO pass (no
Gemini).
225 Processing is also slow (~50 min per 1080p video) ? the detection pass likely needs
reduced
226 resolution and/or cached detector outputs to be practical across 22 matches. The
227 substrate-agnostic training-data + classifier layers already built still apply: they
will
228 consume **WASB-derived** windows/features instead of YOLO's.
229
230 ---
231
232 **## ADR-005: Permissive Dependency Stack for the Hosted Monetization Path**
233
234 **Date:** 2026-05-30
235 **Status:** Accepted
236
237 **### Context**
238
239 The project will be monetized as a **closed-source hosted API** (users upload videos to
a cloud GPU VM and get highlights). Because we never distribute code/binaries, GPL/LGPL
distribution copyleft is not triggered ? but **AGPL §13** (network clause) and **metered-API**
terms are, and **video-codec patent royalties** are a separate concern from software licenses.
A deep-research audit (2026-05-30, 25 adversarially-verified claims against primary sources ?
see *docs/MONETIZATION_AUDIT.md*) identified the blockers and clean alternatives. This ADR also
 resolves the open licensing question in ADR-001/ADR-004 about WASB.
240
241 **### Decision**

242

243 1. ****Drop `ultralitics`.** Both the library and the `yolov8n.pt` weights are ****AGPL-3.0****; §13 makes them a blocker for a closed-source hosted service. Swapping the codebase while keeping the `.pt` does ****not**** escape ? the weights are AGPL too. Replace player/motion-gating detection with a permissive detector: ****torchvision Faster R-CNN / RetinaNet (BSD, default ? zero new deps, person = COCO class 0)****, escalating to ****YOLOX**** or ****RT-DETRv2 (Baidu repo, Apache-2.0)**** if more speed/accuracy is needed. ****Ultralytics Enterprise License**** is a paid fallback only (quote-only pricing).

244 2. ****Shuttle tracking stays permissive.**** ****WASB-SBDT and TrackNetV3 are MIT (CLEAR for commercial use)**** ? this resolves the licensing unknown flagged in ADR-001/ADR-004. ****MonoTrack (Adobe Research, noncommercial) and YOLO-NAS weights (Deci, noncommercial) are banned.**** Train our own checkpoints (or confirm per-checkpoint terms), since distributed research weights/datasets aren't covered by the MIT *code* grants.

245 3. ****LLM step ? keep all three strategies open:**** (A) eliminate via an offline temporal segmenter (****ActionFormer, MIT**** ? the Phase-4 direction), (B) self-host ****Qwen2.5-VL-7B/32B (Apache-2.0)**** or ****InternVL2.5 (MIT)****, or (C) keep ****Gemini 2.5 Flash on the paid tier**** (required ? the free tier trains on and human-reviews uploaded video). The `google-genai` SDK itself is Apache-2.0 (clear).

246 4. ****Emit royalty-free output.**** Standardize highlight output on ****AV1 (+ Opus)**** to avoid AVC/HEVC encode-side patent exposure; provision an ****Ada-class GPU (L4/L40)**** for hardware AV1 (NVENC), falling back to ****SVT-AV1 (CPU, BSD-3-Clause-Clear)**** on T4/A10/A100.

247

248 **### Consequences**

249

250 - Eliminates the only two hard legal blockers; the rest of the stack (torch, opencv, fastapi, numpy, lapx, ffmpeg-as-subprocess, google-genai SDK) is already permissive.

251 - Detector swap is contained to `backend/indexer/yolo\_hybrid.py` (the `from ultralytics import YOLO` load + the `model.track(...)` call) plus `config.json` knobs and `tests/test\_yolo\_hybrid.py` mocks. Stage-1 feature outputs (velocity/active-frame/track-id stats) and Stages 2?4 are detector-agnostic and unchanged.

252 - We now own the frame loop + tracking (ultralitics did both internally). Use a permissive ByteTrack (e.g. `supervision`, MIT, or the MIT ByteTrack repo) on top of `lapx` (already a BSD dep).

253 - Residual items needing legal sign-off: exact codec-pool royalty thresholds (AVC/HEVC decode of uploads), Ultralytics Enterprise pricing, and per-checkpoint weight licenses ? tracked in `docs/MONETIZATION\_AUDIT.md` §7.

254 - The `yolo\_hybrid` registry key is now a slight misnomer (no YOLO); keep it for back-compat or rename in a follow-up.

255

256 ---

257

258 **## ADR-006: Golden-Set Generation via Vendor Bake-off + a Pluggable Annotation Provider**

259

260 ****Date:**** 2026-05-30

261 ****Status:**** Accepted (process v1; vendor shortlist pending research)

262

263 **### Context**

264

265 The product needs ****golden-set ground truth**** (hand-verified rally `[start,end)` boundaries) to (a) measure pipeline quality and (b) run ****on-demand / surprise regression tests**** against the production tool. Two forces shape the design: the hosted-API monetization goal raises a "who views the user's video" concern (ADR-005 / MONETIZATION_AUDIT.md), and the founder is ****Bangalore-based**** and prefers ****local/India vendors****. We already own a deterministic interval scorer (`backend/eval/metrics.py`) and a growing collection of ****licensed/owned**** match videos.

266

267 Decisions locked with the owner: ****licensed/owned video only**** (no user uploads to vendors) . ****rally `[start,end)` annotation depth**** . ****tiered turnaround**** (fast automated + slow human) . ****vendor bake-off**** to choose . ****Bangalore/India-local preference****.

268

269 **### Decision**

270

271 1. ****One scorer, three jobs.**** A vendor's `[start,end)` output is structurally identical to a prediction, so `metrics.py` grades product-vs-truth, vendor-vs-key, and annotator-vs-annotator (IAA) with no new scoring code.

272 2. ****Select vendors by measured quality-per-dollar****, via a bake-off: write a precise rally-definition guideline, build a small internal gold key (5?10 licensed clips), send the same clips to 2?3 vendors (favoring Bangalore/India), and grade every submission with our own harness. The numbers are the decision.

273 3. ****Pluggable `AnnotationProvider` registry**** mirroring the existing registry pattern: `FileProvider` (today's manual/ShuttleSet flow), `HumanVendorProvider` (slow tier ? authoritative truth), `AutomatedProvider` (fast tier ? independent model oracle).

274 4. ****Regression = existing harness + one new component.**** `regress(video)` calls a provider for ground truth, runs the production pipeline, and scores with the unchanged `metrics.py` against a per-video `baseline.json`.

275 5. ****Privacy by construction.**** Golden sets use only licensed/owned footage; all third-party video egress flows through the single provider boundary where DPA / no-reuse / delete-on-delivery / access-logging controls live.

276

277 **### Consequences**

278

279 - Eliminates the "human views user video" concern for golden-set/regression work, because user uploads never enter the pipeline.

280 - ****The oracle problem is a first-class caveat:**** the fast-tier `AutomatedProvider` must be a **different model lineage** than production (else shared blind spots make the test falsely green); fast-tier alarms escalate to the slow human tier before blocking a release.

281 - New docs: `docs/GOLDEN\_SET\_VENDORING.md` (the full process v1) and `docs/VIDEO\_RECORDING\_GUIDELINES.md` (a tracked draft: capture-quality thresholds to be derived empirically from precision/recall, since input quality bounds output quality ? cf. ADR-004's broadcast-camera finding).

282 - The vendor shortlist itself is ****pending a primary-source research pass**** (favoring India/Bangalore) and is intentionally left blank until verified ? no un-sourced vendor claims.

283 - ****Honest limitation:**** a licensed/owned corpus may not fully represent real user capture, so regression greens are strong evidence, not guarantees ? the recording-guidelines work aims to narrow that gap.

284

285 **### Update (2026-06-02): shortlist researched; Roora selected for the initial pilot**

286

287 - The ****vendor shortlist**** is filled in (GOLDEN_SET_VENDORING.md §6, primary-source pass 2026-05-30).

288 - The founder selected ****Roora ML Pvt Ltd**** (HSR Layout, Bengaluru) for the ****initial paid pilot****, diverging from the doc's iMerit/Taskmonk default toward a Bangalore-local, startup-friendly shop. Roora's existence, image/video/temporal capability, and location are verified (rooragr.com); ****pilot quality is still to be measured**** ? run as a ****single-vendor pilot first**** (the pilot **is** the calibration), not a multi-vendor bake-off.

289 - ****Step 0 ? the annotation guideline ? is authored**** (the highest-leverage artifact): collector repo `docs/annotation/` ? `ROORA\_BRIEF.md` (rally CSV schema), `ANNOTATION\_GUIDE.md` (illustrated guide), plus intake-manifest and NDA templates.

290 - ****Pilot scope is 3 sports**** (badminton, pickleball, table tennis); the collector's `import-local` ingestion for all three is implemented (collector ADR-001).

291 - ****Still pending:**** the internal ****gold key**** (5?10 expert-labeled clips) and the vendor-dependent ****GS-3 `HumanVendorProvider`**** ? GS-1/GS-2/GS-4 (FileProvider, regression runner, automated-oracle scaffold) have already landed (see GOLDEN_SET_IMPLEMENTATION.md).

292

293 ---

294

295 **## ADR-007: Audio "thwack" hit-detection (fused with WASB) is the next rally-detection cue; pose/stance parked**

296

297 ****Date:**** 2026-06-07

298 ****Status:**** Accepted (direction); experiments gated on golden eval labels

299

300 **### Context**

301

302 After building the WASB shuttle-trajectory substrate (ADR-001), a Gemini-teacher ? local-student distillation

303 (ADR-004 lineage), and measuring both, two honest findings emerged: **** (a) **** threshold-tuning the local

304 windowing ceilings around F1 ? 0.44 and the learned student **underperforms** the threshold baseline at n=2

305 labeled videos; **the dominant failure is candidate RECALL ? WASB-derived windows simply don't align**
 306 with (or miss) true rallies (IoU-0.5 recall ceiling ~0.43 on one clip). To decide whether teacher?student was
 307 even the right backbone, we ran a deep-research survey
 (`docs/RALLY\_DETECTION\_RESEARCH.md` ? 23 sources, 25
 308 adversarially-verified claims). Verdict: **teacher?student is sound but NOT uniquely best**; the strongest
 309 direction is a **complementary, fully-local multi-cue stack**, and the **#1 cheapest lever is audio hit**
 310 detection fused with the trajectory (audio+visual fusion measurably lifts rally *recall* ? tennis HMM 83%
 311 fused vs 54% visual / 63% audio). Pose/court serve-gating is the validated #2 but is engineering-heavy.
 312
 313 **### Decision**
 314
 315 1. **Adopt a fully-local AUDIO hit-detection cue** (shuttle "thwack" impacts) and **fuse it with the WASB**
 316 **trajectory** ? primarily to recover rallies WASB misses (recall) and secondarily as features for the
 317 distilled student. Detailed design: `docs/AUDIO\_RALLY\_DETECTION\_PLAN.md`.
 318 2. **Gate on an empirical GO/NO-GO probe (E1)**. Shuttle impacts are quiet vs tennis/TT, so before building
 319 anything heavy, measure on *real GoPro audio* whether hit-clusters align with golden rallies and recover
 320 WASB misses. Build E2/E3 only if E1 passes.
 321 3. **Keep Gemini as an OFFLINE teacher only** (never runtime) ? research confirms this is the correct role for
 322 a cloud VLM; it is not the deployed localizer.
 323 4. **Park pose/court-geometry stance detection as the #2 cue** (serve-ready START + flight-gradient END). It is
 324 expected to be substantially harder to get right; revisit after audio (and a TBD owner hunch).
 325 5. **Make "audio always enabled" a capture requirement** (`docs/VIDEO\_RECORDING\_GUIDELINES.md`), since the cue
 326 is only as good as the recording.
 327 6. Stay permissive (ADR-005): classical DSP (numpy + stdlib `wave`, BSD), no restrictive weights; any later model
 328 trained in-house.
 329
 330 **### Consequences**
 331
 332 - New `backend/pipeline/audio/` module (extraction + onset/peak + confidence), feeding `distill\_local` and scored
 333 by the existing `metrics.py` / golden loaders. (As built: the detector is **pure numpy + stdlib `wave`**, no
 334 **scipy** ? the anticipated scipy dependency proved unnecessary.)
 335 - **All audio measurement is blocked on golden eval labels** ? which the VLC rally-annotator (now a standalone
 336 MIT, multi-sport open-source repo: <https://github.com/avidullu/rally-annotator>) lets the owner self-produce,
 337 and the in-repo `--labels` loaders / collector `import-local` ingest.
 338 - Sport-generic: tennis/table-tennis hits are *louder*, so the cue should transfer at least as well there.
 339 - **Honest risk**: if E1 shows shuttle SNR is too poor on real amateur audio, audio is deferred ? this ADR
 340 commits to the *experiment and direction*, not to a guaranteed win.
 341
 342 **### Integration architecture: modular fusion now (Option A), an owned end-to-end model later (Option B)**
 343
 344 A recurring design question: do we **feed audio into WASB** so it refines its own output, or **keep audio as a**
 345 **separate signal fused in our code**? The answer today is the latter ? and understanding

```

*why* sets up the
346     long-term route.
347
348     **Why fusion lives in our rally layer, not inside WASB (Option A ? now).** WASB is a
**frozen, vision-only
349     shuttle *detector** (image frames ? per-frame shuttle `(x, y, visibility)`; MIT weights
we *run*, not train).
350     It has no audio pathway and it does not decide rallies at all ? **rally detection is our
own layer** on top of
351     its trajectory (`trajectory_to_action_windows` ? gate ? the distilled classifier).
Making WASB consume audio
352     would mean bolting on an audio encoder and **retraining the network**, which needs per-
frame
353     shuttle-coordinate-*plus*-audio labels we don't have (we only have rally `[start,end]`
labels) and would fork
354     the clean MIT model. It also conflates two different questions: WASB answers ***where*
the shuttle is**; audio
355     answers ***when* a hit happened**. Audio doesn't make WASB localize better ? it's an
independent rally-activity
356     cue. So fusion belongs where cues are *combined into rallies*: our layer.
357
358     ``
359     frames ?? WASB (frozen, MIT) ?? shuttle trajectory ??
360     audio ?? hit detector (ours) ?? hit events ?????????? RALLY LAYER (ours) ?? rallies
361     [pose ?? stance heuristic (ours; the parked #2) ???? (feature + decision fusion)]
362     ``
363
364     Two complementary fusion modes (both in `docs/AUDIO_RALLY_DETECTION_PLAN.md`):
365     1. **Feature-level (primary):** audio features (hit_rate, inter-hit cadence/regularity,
time-since-last-hit)
366         join WASB-derived features (velocity, net-crossings) in the **single distilled
classifier** that emits
367         rallies ? "use audio to sharpen our rally metrics" (experiment E3).
368     2. **Decision-level recall-recovery:** where WASB proposed nothing but a dense hit-
cluster exists, propose a
369         rally there ? directly correcting WASB's recall misses.
370
371     Each cue (WASB, later the person/pose detector) stays a **swappable black box** ; every
signal is independently
372     testable ? which matters while we're label-starved.
373
374     **The long-term route (Option B ? the intended destination, once a healthy golden corpus
exists).** As we
375     accumulate enough golden eval/training videos (owner self-rating via the annotator +
vendor labels), the right
376     end-state is to **collapse the modular cues into a single OWNED multimodal rally
model**, trained/fine-tuned
377     end-to-end on (frames + audio + pose/stance + ?) ? rally boundaries ? the HitNet /
temporal-action-localization
378     direction surfaced in `docs/RALLY_DETECTION_RESEARCH.md`. Crucially, that lets us
**fine-tune from our own code
379     and data instead of depending on WASB's pretrained weights** ? removing both the *recall
ceiling* imposed by
380     WASB's candidate windows and the *licensing question* on its academic-trained weights.
381
382     **Why we don't jump to Option B now ? and why Option A is the path to it.** Option B
needs data we don't yet
383     have, and it's far harder to debug as one black box. So we deliberately build **modular
cues first** ? audio
384     (this ADR), then a stance heuristic (the parked #2), then other signals ? each cheap,
debuggable, label-
385     efficient, and *each one generating the very golden labels an end-to-end model will
need*. When the corpus is
386     healthy, we train the unified model. **Option A is not a detour from Option B; it is how
we earn it.**

```

13. user (2026-06-10T05:33:56.141Z)

```

1      # Checkpoint: 2026-06 Strategy Foundation (post-scaffolding)
2
3      **Date:** June 2026
4      **STATUS: COMPLETED**
5      **Trigger:** Owner merged the post-scaffolding strategy docs ? #52 (config/identity
cleanup), #53 (deferred-hardening design), #55 (commercialization tracker), #57 (PMF market
study), #58 (platform architecture).
6
7      ## Purpose
8      Captures the state once the **strategic / direction-setting foundation** for the
9      post-scaffolding phase was laid: the commercial dimension, market/PMF, the multi-user
10     platform architecture (pluggable storage/compute + the owned-model layer), and the
11     deferred-hardening designs. **Strategy only ? no implementation started.** Provides a
12     regression-detection baseline for priorities/decisions as quality + implementation work
begins.
13
14     ## Key living documents at checkpoint time
15     - **COMMERCIALIZATION.md** ? commercial tracker (dashboard C1?C14, grouped Pending ? In-
flight ? Completed with a P0/P1/P2 column).
16     - **PMF_MARKET_RESEARCH.md** ? cited market study; beachhead = **India badminton
academies**.
17     - **PLATFORM_ARCHITECTURE.md** ? multi-user "assistant coach" platform: pluggable
`StorageBackend`/`ComputeBackend` (BYO), generic schema (coach overlay optional), owned video-QA
model layer ($5A), training-backend survey + nanochat-style harness, committed legal guardrail,
per-user-adaptor direction, v1-vs-vision scoping.
18     - **DEFERRED_HARDENING_PLAN.md** ? packaging, train/eval guardrails, async job model
(#16) ? *pending approval*.
19     - **CODE_MAP.md / BACKLOG.md / ROADMAP.md** ? updated for the config/identity cleanup.
20
21     See git history around #52/#53/#55/#57/#58 for exact snapshots.
22
23     ## Committed decisions / non-negotiables recorded here
24     - **Paid frontier models (Gemini/OpenAI/Anthropic): inference / serving / fallback only
? never a training data source** (competing-model terms). Teachers/bases = open Apache-2.0 only.
25     - **Owned-model base = Apache-2.0-clean, video-capable** (Qwen2.5-VL-7B/32B, Qwen3-VL,
Gemma 4); **reject Gemma 3** (viral license clause); **minors excluded** from the training pool;
per-user training needs a separate opt-in.
26     - **v1 = rally detection + highlight compilation + per-user history + coach/user
correction loop** ("assistant-coach lite"); **gait/biomechanics strictly future** (Art. 9).
27     - **Generic data schema** (User ? VideoCollection ? Video ? Highlight ?
HighlightMetadata); coach ? athlete is an **optional overlay** (a market-fit hypothesis).
28
29     ## What comes next (per owner)
30     - First implementation slice = **async job model (DEFERRED #16)**, single-tenant ?
**pending explicit go-ahead** (no implementation started without approval).
31     - Quality iterations with new labeled videos; golden-set expansion + the **golden QA
eval set** (the prerequisite for the owned-model training harness).
32
33     When the first implementation slice or a quality milestone completes, add a new dated
34     checkpoint here (or under `quality-iterations/`).
35

```

14. user (2026-06-10T05:33:56.695Z)

```

1      # Owned-model training study (rally segmentation ? end-to-end multimodal)
2
3      **Date:** 2026-06-08 . **STATUS: DESIGN BLUEPRINT (no implementation until PLATFORM
P0?P4 + healthy corpus; owner sign-off required).**
4
5      > How to train our **OWNED** rally/highlight model **reusing the videos we label**,
aligned to the

```

```

6      > Apache-2.0 / no-paid-LLM-training / privacy-local guardrails. Produced by an 11-agent
research
7      > workflow (ground ? 5 web-research dimensions ? adversarial verification of
costs/licenses/privacy ?
8      > synthesis + completeness critic; 35 sources). Empirical results from THIS session
(LOVO + a learned
9      > prototype) are folded in and **resolve the study's biggest open unknown** ? see $"What
we proved today".
10     > Numbers from web sources are 2026 snapshots in a  $\pm 15\% \pm 25\%$  band; **re-quote live before
budgeting.**
11
12     ## Update (2026-06-09, post-PR-#61-review) ? authoritative roadmap is now
`docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md`
13     A verified strategy workflow + owner review refined this study. Key changes (the plan
supersedes where they differ):
14     - **VISION (foregrounded):** the goal is **rally/shot/player analytics + highlights for
ALL net-separated racquet
15     sports** (badminton/tennis/TT/pickleball/padel) ? eventually all sports, as **ONE
sport-agnostic, conversational
16     model**. This does **not** change the model/framework picks; it scales **data + per-
sport trajectory detectors** (the
17     accepted "good problem") and adds a licensing front (clean per-sport detectors).
Single-model temporal generalization
18     is an **open hypothesis** (corpus is badminton-only; the once-cited RacketVision
evidence was **refuted** ? it's
19     ball-tracking, not temporal).
20     - **Framework = ms-swift (Apache-2.0)** (only forge with native video+audio+timestamp-
grounding+video-GRPO).
21     **Base = Qwen3-VL (Apache-2.0, uniform sizes, timestamp emission)** ? primary,
**replacing the Qwen2.5-VL-7B framing**.
22     - ? **Gemma 4 stays AMBER and bench-only** (NOT "confirmed clean" as the table below
says): Apache grant likely but
23     Prohibited-Use posture unconfirmed (counsel) + video/timestamp capability disputed ?
not the primary extractor.
24     - **Start with the TAL head, not the VLM** (VLMs miss strict boundaries ? boundary
accuracy is the product). Head now ? VLM later.
25     - **Immediate goal = a high-P/R extraction *harness*, not on-device** (on-device
deferred). **Reframe supersedes the on-device-now framing.**
26
27     ## Thesis
28
29     Train the owned model in **three staged generations off ONE model-agnostic labeled
corpus** ? a label
30     collected once feeds every generation, so labels are never re-collected. **The moat is
the consented
31     dataset + the eval harness, never the weights.**
32
33     | Gen | When | Approach | Base / model | Compute |
34     |---|---|---|---|---|
35     | **Gen-0** | now ? unblocks at n $\geq 5$  matches | tiny **temporal action segmentation
(TAS)** head over a hand-built per-frame feature stack from the WASB trajectory we already
produce; per-frame rally(1)/non-rally(0) ? merge into [start,end] | **ASFormer (MIT, ~1.1M
params)** primary; **ActionFormer (MIT)** A/B challenger (overtakes once corpus = hundreds of
matches) | **local RTX 2070 Super**, trains in minutes, $0 |
36     | **Gen-1 / Option A** | 6 $\pm$ 12 mo | concatenate more local cue channels onto the SAME
head: trajectory + audio-onset + pose/court serve-gating; fusion in our rally layer, cues
swappable | same ASFormer/ActionFormer, wider input | local + occasional cloud burst (pseudo-
labeling) |
37     | **Gen-2 / Option B** | 12 $\pm$ 24+ mo, after scaffolding + healthy corpus | end-to-end
multimodal VLM: (frames+audio+pose+trajectory) ? rally [start,end] (+semantics); SFT cold-start
then **GRPO/RFT with a temporal-IoU-vs-golden reward** | **Qwen3-VL-4B-Instruct** (Apache-2.0,
on-device) primary; **Qwen2.5-VL-7B** fallback; **Gemma 4 E4B/12B-Unified** (Apache-2.0 + native
video+audio) end-state | rented 24 $\times$ 48 GB A100/L40S (does NOT fit 8 GB box) |
38
39     ## What we proved today (de-risks the blueprint)

```

40

41 The study's **"dominant empirical unknown"** was: **can a low-dim WASB trajectory (?3?4*

42 raw dims) feed a

43 TAS head, given ASFormer's small-data result used 2048-d I3D features?* This session answered it.

44 - ****Cross-video LOVO of the current heuristic**** (``calibrate_local``, Adarsh-96 + Kushagra-32, human labels):

45 mean held-out F1 **0.443 ± 0.279** ; Adarsh 0.722, ****Kushagra 0.163**** (its own ceiling ? **no* velocity-windowing*

46 config does better). ****The ± 0.279 spread is method variance, not config.**** ? This also ****corrects the study's**

47 `"~0.73 bar"`: that was a fit-on-self single-match number; the honest cross-video heuristic bar is **~ 0.44** ,

48 which **lowers** the Gen-0 ship gate.

49 - ****A learned per-frame prototype**** (``backend/eval/rally_seq_proto.py``: 6 hand-built features ? visibility rate,

50 mean/max shuttle speed, ****net-crossing rate****, x/y spread ? into the repo's dependency-free LogisticRegression,

51 honest LOVO): ****per-frame AUC 0.878 (Adarsh) / 0.841 (Kushagra), held-out.**** On Kushagra ? the match the

52 heuristic **cannot** segment (0.163) ? a calibrated threshold reaches ****segment F1 0.578 (3.5 $\times$)****. The naive

53 cross-video threshold gives only 0.049, so ****the remaining gap is threshold/operating-point calibration at n=2,**

54 not representation.**

55

56 - ****n=3 update (doubles match added ? `LargeTestVideo`, 49 rallies):**** the learned prototype ****overtakes the**

57 heuristic. ****Mean held-out LOVO F1 ? learned 0.377 ± 0.615 , heuristic 0.443 ± 0.544 ****. Kushagra's learned

58 held-out F1 jumped **0.049 ± 0.417** (one added match closed most of the calibration gap); the ****doubles**** match,

59 held out and trained only on the two singles, scored AUC **0.914** / F1 0.729 (generalizes across play type +

60 frame rate). The heuristic still cannot rescue Kushagra at any n (0.151, capped). ****The honest "bar to beat" is**

61 thus ~ 0.54 (n=3), not 0.73 ? and the learned path already clears it on the mean and keeps rising with n.**

62

63 ****Conclusion:**** the rally signal IS learnable from the WASB trajectory features, cross-footage (AUC $\sim 0.85 \pm 0.91$) ?

64 the heuristic just can't exploit it. This ****validates the Gen-0 ASFormer-over-WASB-features path****, shows the

65 hand-built feature stack the study calls for is the right lever, and confirms the bottleneck is ****data + per-video**

66 calibration (n $\geq$ 5 matches), not method.** The prototype is the seed of Gen-0, and the n=2?n=3 trend (learned mean

67 +0.24, overtaking the heuristic) is the empirical green light.

68

69 **## Platform ranking (privacy + cost lens; the fine-tune is a sub-24h, \$10?50 job ? privacy/ergonomics dominate)**

70

71 1. ****Modal ? PRIMARY.**** Strongest **written** no-access contract (verbatim: "Modal will never access? the inputs/outputs to your Functions, or any data you store"), ?7-day TTL, SOC2-II, HIPAA-BAA; per-second scale-to-zero matches retrain-on-each-batch. A100 80 GB ~\$2.50/hr, H100 ~\$3.95/hr (corrected down from cited). Privacy is the load-bearing reason, not price.

72 2. ****RunPod (Secure Cloud tier ONLY) ? FALLBACK / Option-B scale-up.**** Cheapest vetted-DC self-serve; A100 80 GB ~\$1.39?1.49/hr. Community/P2P tier is UNVETTED ? ****never**** touches consented video. Confirm AES-256/DPA at booking.

73 3. ****Vertex AI / GCP ? ON BENCH.**** Only strong-privacy option with an India region, but it's ****capacity-gated**** (asia-south1 high-end GPU is invitation-only, H100-class A3 not A100) and ****tuning Gemini yields NO portable on-device weights**** ? only tuning an Apache-2.0 base in Model Garden does. Use only if India residency-at-rest is mandated.

```

74      4. **Self-hosted RTX 2070 Super 8 GB** ? Gen-0 training + ALL inference + CI only.
**Cannot** QLoRA a 7B video VLM (~20724 GB). Best privacy (nothing leaves the box).
75      5. **Fireworks** (managed, contractual no-train + ZDR) usable; **Together BLOCKED**
until a signed DPA + explicit no-train clause. AWS SageMaker / Azure ML = heavy for a solo
founder; India-residency only.
76
77      ## Label reuse ? one corpus, three transcoders (the actual moat)
78
79      Promote the rally CSVs from eval-only files to **one model-agnostic corpus**: JSONL, one
row per
80      `(video_id, rally_ordinal)` = `{start, end, ending_reason, sport, source
(human|pseudo|vendor),
81      annotator/provenance, consent/opt-in, label_type, split (BY MATCH not clip),
trajectory_ref}`. The
82      provenance/consent fields are the **enforcement point** for the legal guardrails. Three
deterministic
83      transcoders read it, zero re-labeling:
84      1. **Heuristic/LogReg view** ? already exists
(`backend/eval/training.py::label_candidates` via `metrics.match_intervals`).
85      2. **TAL view** ? ActivityNet JSON (`segment:[start,end]`, `label:'rally'`); our seconds
map 1:1.
86      3. **VLM clip?QA view** ? "list every rally [start,end] in this clip"; `ending_reason` ?
a second QA task.
87
88      **One scorer spine:** reuse `metrics.match_intervals` for heuristic-labeling, TAL eval,
AND the Gen-2 RFT
89      reward ? no metric drift. **Leakage guard:** split **by match, never by clip**.
**Amplification (all
90      guardrail-clean):** open-teacher pseudo-labels (WASB/own-model, confidence-gated ? never
paid-LLM),
91      boundary-preserving temporal augmentation, active learning to spend the Roora budget on
highest-uncertainty
92      matches. ? **The new partial-label windowing**
(`eval_partial_labels.restrict_to_window`) must flow into the
93      TAL/VLM views so prefix-labeled videos aren't mis-scored (a corpus that's partially
tagged is the norm).
94
95      ## Licensing & privacy guardrails (all respected; verify per-checkpoint)
96
97      - **Bases: Apache-2.0/MIT only.** APPROVED: Qwen2.5-VL-7B/32B (Apache-2.0),
Qwen3-VL-4B/8B (Apache-2.0),
98      Gemma 4 E2B/E4B/12B-Unified (Apache-2.0 + native video/audio), InternVL3 ?38B
(MIT/Apache), ASFormer/
99      ActionFormer/TriDet (MIT). **REJECTED:** Gemma 3 (viral Gemma Terms + PUP),
Qwen2.5-VL-3B (NC research),
100     Qwen-72B/InternVL3-78B (100M-MAU cap), Llama Vision (700M-MAU), DAMO VideoLLaMA3
weights (NC + paid-LLM-trained),
101     TimeChat/VTimeLLM/Momentor (NC/no-license = method-reference only), VideoMAE weights
(CC-BY-NC), `yabufarha/ms-tcn`
102     (Commons Clause ? use `sj-li/MS-TCN2` MIT). ? **Gemma 4 size names =
E2B/E4B/26B-A4B/31B/12B-Unified ? there is no plain "4B".**
103     - **No paid-LLM training data** ? Gemini/OpenAI/Anthropic = inference/eval/teacher-
reference only, **never** a
104     training signal. Gemini silver labels stay out of every transcoder.
105     - **Exclude minors by default + separate training opt-in;** per-user adapters trained
only on that user's
106     consented video (deletion = drop the adapter). Enforced at the corpus-membership layer
before transcoding.
107     - **Privacy lever = local inference** (Gen-0 ~1.1M params; Gen-2 4B quantized). Any
cloud *training* needs a
108     signed DPA + no-train assurance; route only owned/consented footage, never user
uploads.
109
110     ## Cost (spend on labels, not GPUs)
111

```

```

112 - **Gen-0:** ~$0 (local 2070 Super; trains in minutes; ~4.5 GB VRAM).
113 - **Gen-2 LoRA:** ~$5?40/run on a rented 24?48 GB GPU (RunPod cheapest; **VRAM/token
cost on real 1080p/60fps
114 clips is UNVERIFIED ? 24 GB is a floor, re-benchmark**). Managed (Fireworks/Together)
~$2?30 *if* text-priced
115 (video tokens can balloon). RFT = compute-heavy (multiple rollouts/sample).
116 - **Inference crossover (the unanchored number):** self-hosted 7B beats Gemini
(~$0.05/10-min clip) **only**
117 above sustained QA volume; below it, **paid Gemini is cheaper** ? paid-primary now,
owned-primary later.
118 **Benchmark $/call before committing** (COMMERCIALIZATION C7).
119
120 ## Open risks & critic caveats (read before acting)

```

15. user (2026-06-10T05:34:01.817Z)

```

120 ## Open risks & critic caveats (read before acting)
121
122 - **?? WASB checkpoint provenance (C3) propagates.** The study calls WASB "MIT," but
that's the *code*; the
123 distributed `wasb_badminton_best.pth.tar` is **academic-data-trained, provenance
unresolved** (COMMERCIALIZATION
124 C3, still open). Every Gen-0/1 plan uses WASB features as input AND as a pseudo-label
teacher ? a "clean" owned
125 model trained on provenance-unclear features **does not launder the provenance.**
ROADMAP Phase-4 "train own WASB
126 weights" is a **prerequisite, not an aside.**
127 - **Gemma 4 "clean + native-video" rests on post-cutoff secondary sources** ? keep AMBER
until a raw HF LICENSE
128 revision is verified, given the Gemma-3 rejection. Its timestamp-emission recipe is
less documented than Qwen's.
129 - **Existing distilled model was trained on Gemini silver labels** ? the (correct) eval-
only policy now **forbids
130 this; that pipeline must be purged of Gemini-derived training labels** to comply.
131 - **Policy ratifications needed:** guardrail literally says "Apache-2.0 only" but the
approved list includes MIT ?
132 ratify "Apache-2.0/MIT". The MAU-cap rejections (72B/78B) may be over-conservative at
our scale.
133 - **On-device privacy claim is unbenchmarked** ? no tokens/sec for a quantized 4B VLM on
8 GB Turing at
134 1080p/60fps. **Measure before relying on it.**
135 - **Amplification is load-bearing for Gen-2** ? 128 raw rallies can't reach the ~5k?20k
clip?QA band alone; if
136 augmentation + QA-multiplexing + pseudo-labels underdeliver, Option B is under-fed.
Amateur 60fps may need more
137 labels than broadcast-trained literature implies.
138 - **Dropped option worth keeping:** InternVideo2 encoder (MIT) + light grounding head ?
the *only* data-realistic
139 owned-video experiment available at n?128 today. And: a **one-time ~$1?2k 24 GB local
card** (e.g. used 3090)
140 would preserve the privacy lever for *training* and dissolve most of the cloud-DPA
analysis ? uncoded here.
141
142 ## Open questions (need the GPU/WSL machine-side suite ? not runnable in this dev
container)
143
144 1. Does ASFormer/ActionFormer over the WASB feature stack beat the **honest ~0.44 LOVO**
bar at n?5 matches?
145 (Today's prototype says the per-frame AUC is there ? 0.84 ? so this is now *likely*;
confirm at segment level.)
146 2. Real VRAM + token cost of fine-tuning Qwen3-VL-4B / Qwen2.5-VL-7B on our actual
clip?QA frame counts?
147 3. Working Gemma-4 timestamp-emission recipe + exact checkpoint (E4B vs 12B-Unified)?
148 4. Measured self-hosted-7B $/call + latency vs Gemini on the target workload (the owned-
model ROI crossover)?

```

```

149     5. Will Together/Lambda/Azure sign a DPA with explicit no-train for consented video,
India-configurable?
150     6. Realistic clip?QA yield from 128 rallies via augmentation + QA-multiplexing + pseudo-
labels?
151
152     ## Sequencing (owner directive, non-negotiable)
153
154     This is a **design blueprint only.** It unblocks PLATFORM **P5** but starts **no
implementation** until P0?P4
155     scaffolding (async jobs, storage/compute backends) is rock-solid AND the golden corpus
is healthy. Paid Gemini
156     stays the always-ready primary/fallback. The only items safe to act on now ? **and even
these need owner
157     sign-off before code** ? are (a) the model-agnostic corpus format and (b) growing n?2 ?
n?5 matches (a small
158     Roora batch), which unblocks honest tuning immediately and is the **highest-ROI spend
(labels, not GPUs).**
159
160     *Full agent transcript + 35 sources: workflow `wf_49aab888-e10` (this session).*
161

```

16. user (2026-06-10T05:34:13.032Z)

Found 2 files

```

C:\Users\avidu\Projects\badminton-highlight-indexer\backend\eval\distill_local.py
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\eval\training.py

```

17. user (2026-06-10T05:34:14.505Z)

```

C:\Users\avidu\Projects\badminton-highlight-indexer\backend\main.py:105: default_seg =
getattr(getattr(global_config, "indexing", None), "default_segmenter", "motion")
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\main.py:371: seg_name =
args.segmenter or getattr(getattr(global_config, "indexing", None), "default_segmenter",
"motion")
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\config\models.py:47:
default_segmenter: str = "yolo_hybrid"
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\config\models.py:69:
@field_validator("default_segmenter")
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\reporting\harvest.py:210:
config_default = (config.get("indexing") or {}).get("default_segmenter")
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\reporting\spec.yaml:112: The
segmenter that ran differs from config.json's default_segmenter, and no
C:\Users\avidu\Projects\badminton-highlight-indexer\README.md:74:This outputs a validation
summary and indexes all rallies directly into the local SQLite database, using whichever
segmenter is set as `default_segmenter` in `config.json` (`yolo_hybrid` by default).
C:\Users\avidu\Projects\badminton-highlight-indexer\README.md:124: "default_segmenter":
"yolo_hybrid",
C:\Users\avidu\Projects\badminton-highlight-indexer\README.md:155:*
`indexing.default_segmenter` ? `yolo_hybrid` | `gemini` | `motion`.
C:\Users\avidu\Projects\badminton-highlight-indexer\README.md:215: *Or set
`"default_segmenter": "yolo_hybrid"` (or `"gemini"`/`"motion"`) under `"indexing"` in
`config.json`.*
C:\Users\avidu\Projects\badminton-highlight-indexer\README.md:219:[Omitted long matching line]
C:\Users\avidu\Projects\badminton-highlight-indexer\README.md:237: "default_segmenter":
"gemini",
C:\Users\avidu\Projects\badminton-highlight-indexer\config.json:26: "default_segmenter":
"gemini",
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\CODE_MAP.md:29: -> seg_name =
req.segmenter or global_config.indexing.default_segmenter (fallback 'motion')
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\CODE_MAP.md:86: - ? `--segmenter`
argparse `choices` frozen at build time from `segmenter_registry.list_available()` ? a lazily-
registered segmenter wouldn't appear. CLI default_segmenter fallback `motion`.
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\CODE_MAP.md:96:[Omitted long matching
line]
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\CODE_MAP.md:102:- `@config.json` ? on-

```



```

disk config; mirrors `AppConfig`. ? **`indexing.default_segmenter='gemini'`** diverges from
model default `yolo_hybrid` ? file wins at runtime.
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\CODE_MAP.md:177:[Omitted long matching
line]
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\CODE_MAP.md:238:- **Segmenter
default:** single source of truth = model `IndexingConfig.default_segmenter='yolo_hybrid'`;
`config.json` may override at runtime (ships `gemini`). run_*.py read the model value (no
literals); `main.py` keeps the defensive `motion` getattr fallback noted above.
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\CODE_MAP.md:264:| Change a config
default / add a config field | `@backend/config/models.py` (single source of truth) ->
regenerate via `setup.py`/`save_default_config`; ? config.json may override (default_segmenter)
|
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\TRACKNET_WSL_SETUP.md:71:Then select it
as the segmenter (e.g. set `indexing.default_segmenter` to
C:\Users\avidu\Projects\badminton-highlight-indexer\run_process_and_stitch.py:72:
seg_name = cfg.indexing.default_segmenter
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_config.py:24:    assert
cfg.indexing.default_segmenter == "yolo_hybrid"
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_config.py:34:
"default_segmenter": "gemini",
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_config.py:45:    assert
cfg.indexing.default_segmenter == "gemini"
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_config.py:58:
"indexing": {"default_segmenter": "yolo_hybrid"},
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_config.py:66:    assert
cfg.indexing.default_segmenter == "yolo_hybrid"
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_config.py:78:    assert
cfg.indexing.default_segmenter == "yolo_hybrid"
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_config.py:122:    assert
cfg.get("indexing", {}).get("default_segmenter") == "yolo_hybrid"
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_reporting_harvest.py:46:
config={"ai_provider": "gemini", "ai_model": "gemini-flash", "indexing": {"default_segmenter":
"wasb_hybrid"}},
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_reporting_harvest.py:98:
config={"indexing": {"default_segmenter": "motion"}})
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_reporting_harvest.py:107:    cfg
= {"ai_provider": "gemini", "indexing": {"default_segmenter": "wasb_hybrid"}}
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_reporting_harvest.py:150:
config={"ai_provider": "gemini", "indexing": {"default_segmenter": "wasb_hybrid"}},

```

18. user (2026-06-10T05:34:21.768Z)

```

1      """Build a training set for the offline, API-free learned rally segmenter.
2
3      Phase 4 of the collector->indexer bridge (see docs/ROADMAP.md, ADR-004). The idea:
4      a detector (YOLO player-motion today, optionally WASB shuttle-motion later)
5      produces candidate rally windows with features; the collector's ground-truth
6      rally intervals tell us which windows are real. A lightweight classifier learns
7      that mapping, replacing the paid Gemini refinement with a local model.
8
9      This module is the *data layer* ? pure and unit-tested:
10
11      - `label_candidates` ? generic: label windows positive/negative by whether they
12        match a ground-truth rally (same IoU matching the evaluator uses, so training
13        labels are consistent with how we score).
14      - `extract_yolo_features` ? YOLO-specific feature vector from a candidate row's
15        persisted `stats`. Pluggable: a WASB substrate supplies its own feature_fn.
16      - `build_training_set` ? glue: rows + GT -> (X, y, intervals).
17
18      No model training or I/O here; that lives in the trainer/segmenter once the
19      substrate is chosen from the Phase 3 candidate-recall result.
20      """
21
22      from typing import Callable, Dict, List, Tuple
23

```

```

24     from backend.eval.metrics import Interval, match_intervals
25
26     # YOLO Stage-1 persists these per candidate window in `candidate_segments.stats`
27     # (see HybridYoloSegmenter). Duration is derived from the window bounds.
28     YOLO_FEATURE_NAMES = [
29         "duration", "peak_velocity", "mean_velocity", "active_frame_count",
30         "track_id_count",
31     ]
32
33     def extract_yolo_features(row: Dict) -> List[float]:
34         """Feature vector for one YOLO candidate row (dict from query_candidate_segments).
35
36         `stats` is already JSON-deserialized by the DB layer. Missing fields default
37         to 0.0 so a sparse/old row still yields a well-formed vector.
38         """
39         stats = row.get("stats") or {}
40         duration = float(row["end_time"]) - float(row["start_time"])
41         return [
42             duration,
43             float(stats.get("peak_velocity", 0.0)),
44             float(stats.get("mean_velocity", 0.0)),
45             float(stats.get("active_frame_count", 0)),
46             float(stats.get("track_id_count", 0)),
47         ]
48
49     def label_candidates(candidate_intervals: List[Interval], gt_intervals: List[Interval],
50                          iou_threshold: float = 0.5) -> List[int]:
51         """Label each candidate window 1 (real rally) or 0 (spurious).
52
53         A window is positive iff it is matched one-to-one to a ground-truth rally at
54         `iou_threshold` ? the *same* matching the evaluator uses for true positives,
55         so the classifier is trained on exactly what we score. Substrate-agnostic:
56         works for YOLO or WASB-derived windows alike.
57         """
58         mr = match_intervals(candidate_intervals, gt_intervals, iou_threshold)
59         positive = {m.pred_index for m in mr.matches}
60         return [1 if i in positive else 0 for i in range(len(candidate_intervals))]
61
62     def build_training_set(
63         candidate_rows: List[Dict],
64         gt_intervals: List[Interval],
65         iou_threshold: float = 0.5,
66         feature_fn: Callable[[Dict], List[float]] = extract_yolo_features,
67     ) -> Tuple[List[List[float]], List[int], List[Interval]]:
68         """Turn one video's candidate rows + ground truth into (X, y, intervals).
69
70         `feature_fn` is pluggable so a WASB substrate can supply shuttle-motion
71         features without changing the labeling logic. Returns the feature matrix `X`,
72         binary labels `y`, and the candidate intervals (handy for downstream merging).
73         """
74         intervals: List[Interval] = [
75             (float(r["start_time"]), float(r["end_time"])) for r in candidate_rows
76         ]
77         y = label_candidates(intervals, gt_intervals, iou_threshold)
78         X = [feature_fn(r) for r in candidate_rows]
79         return X, y, intervals
80
81
82

```

19. user (2026-06-10T05:34:30.473Z)

```

1     """Distill Gemini (teacher) into a LEARNED local rally classifier (student).
2

```

```

3     Threshold-tuning the local knobs hit a low ceiling (see calibrate_local). This goes
4     further: learn "is this WASB candidate a real rally?" from Gemini silver labels,
5     using a few resolution/fps-INVARIANT local features, so the model transfers across
6     videos and runs 100% local at inference (no Gemini, privacy lever intact).
7
8     Flow (all local at runtime):
9     WASB trajectory -> recall-oriented candidate windows (propose many; classifier
10    filters) -> per-candidate features [duration, peak_velocity, mean_velocity,
11    active_ratio, net_crossings] -> LogisticRegression(prob >= threshold) -> merge
12    -> rally windows.
13
14    Training labels come from Gemini full-context labels via the SAME IoU matching the
15    evaluator uses (training.label_candidates). Honest leave-one-video-out: train on the
16    other videos' candidates, predict the held-out one, score vs its Gemini labels, and
17    compare against the threshold baseline. Reuses classifier.py, training.label_candidates,
18    rally_gate, metrics, calibrate_wasb.load_golden_gts, gemini_refine.merge_overlaps.
19
20    Run:
21    python -m backend.eval.distill_local --manifest videos.json --model-out
output/local_rally_model.json
22    """
23    from __future__ import annotations
24
25    import json
26    import argparse
27    from typing import Dict, List, Tuple
28
29    import numpy as np
30
31    from backend.pipeline.detectors.tracknet_runner import TrackNetRunner
32    from backend.pipeline.detectors import rally_gate
33    from backend.eval.calibrate_wasb import load_golden_gts
34    from backend.eval.training import label_candidates
35    from backend.eval.classifier import LogisticRegression
36    from backend.eval.metrics import score
37    from backend.eval.gemini_refine import merge_overlaps
38
39    Interval = Tuple[float, float]
40
41    FEATURE_NAMES = ["duration", "peak_velocity", "mean_velocity", "active_ratio",
"net_crossings"]
42    # Recall-oriented proposal: deliberately permissive so real rallies are among the
43    # candidates (the classifier removes the junk). (velocity_thresh, merge_gap, min_dur)
44    PROPOSAL = (0.005, 1.0, 0.5)
45    BASELINE_LOCAL = (0.02, 2.0, 2.0) # today's production-ish windowing, no learning
46
47
48    def build_candidates(trajectory_csv: str, fps: float, frame_width: float,
49    proposal: Tuple[float, float, float] = PROPOSAL) ->
Tuple[List[Interval], List[List[float]]]:
50    """WASB trajectory -> (candidate intervals, fps/resolution-invariant feature
rows)."""
51    points = TrackNetRunner.parse_trajectory_csv(trajectory_csv)
52    vt, mg, mwd = proposal
53    wins = TrackNetRunner.trajectory_to_action_windows(
54    points, fps=fps, frame_width=frame_width,
55    velocity_thresh=vt, merge_gap=mg, min_window_duration=mwd)
56    intervals = [(w["start"], w["end"]) for w in wins]
57    gated = rally_gate.gate_windows(points, intervals, fps, min_crossings=0) # for net-
crossings
58    X: List[List[float]] = []
59    for w, g in zip(wins, gated):
60        dur = max(1e-6, w["end"] - w["start"])
61        win_frames = max(1.0, dur * fps)
62        X.append([

```

```

63         dur,                                     # rally length (sec) ?
fps-invariant
64         float(w["peak_velocity"]),               # already normalized by
frame_width
65         float(w["mean_velocity"]),
66         float(w["active_frame_count"]) / win_frames, # active-shuttle ratio ?
fps-invariant
67         float(g.crossings),                       # net crossings (count) ?
the rally signal
68     ])
69     return intervals, X
70
71
72     def baseline_windows(trajecory_csv: str, fps: float, frame_width: float) ->
List[Interval]:
73         points = TrackNetRunner.parse_trajectory_csv(trajecory_csv)
74         vt, mg, mwd = BASELINE_LOCAL
75         wins = TrackNetRunner.trajectory_to_action_windows(
76             points, fps=fps, frame_width=frame_width,
77             velocity_thresh=vt, merge_gap=mg, min_window_duration=mwd)
78         return [(w["start"], w["end"]) for w in wins]
79
80
81     def load_video(spec: Dict, iou: float = 0.5) -> Dict:
82         fps, fw = float(spec["fps"]), float(spec["frame_width"])
83         intervals, X = build_candidates(spec["trajectory"], fps, fw)
84         gts = load_golden_gts(spec["labels"])
85         y = label_candidates(intervals, gts, iou)
86         return {"name": spec.get("name", spec["trajectory"]), "intervals": intervals, "X":
X,
87                 "y": y, "gts": gts, "baseline": baseline_windows(spec["trajectory"], fps,
fw)}
88
89
90     def predict_rallies(model: LogisticRegression, X: List[List[float]], intervals:
List[Interval],
91                       threshold: float = 0.5) -> List[Interval]:
92         if not intervals:
93             return []
94         proba = model.predict_proba(X)
95         pos = [iv for iv, p in zip(intervals, proba) if p >= threshold]
96         return merge_overlaps(pos, gap_tol=1.0)
97
98
99     def run(specs: List[Dict], iou: float = 0.5, threshold: float = 0.5, model_out: str =
None) -> dict:
100         videos = [load_video(s, iou) for s in specs]
101         print(f"=== Distilled local rally classifier vs Gemini silver-GT ")
102         f"({len(videos)} videos, IoU>={iou}, prob>={threshold}) ===")
103         for v in videos:
104             ceil = score(v["intervals"], v["gts"], iou).recall
105             print(f"[{v['name']}] {len(v['intervals'])} candidates ({sum(v['y'])} pos) | "
106                   f"proposal recall ceiling {ceil:.3f} | {len(v['gts'])} Gemini rallies")
107
108         result: dict = {"videos": [v["name"] for v in videos]}
109         if len(videos) >= 2:
110             print("\n--- HONEST leave-one-video-out: learned vs threshold baseline (held-
out) ---")
111             clf, base = [], []
112             for i, held in enumerate(videos):
113                 others = [v for j, v in enumerate(videos) if j != i]
114                 X = [row for v in others for row in v["X"]]
115                 y = [t for v in others for t in v["y"]]
116                 model = LogisticRegression().fit(X, y, FEATURE_NAMES)
117                 preds = predict_rallies(model, held["X"], held["intervals"], threshold)

```

```

118         sc = score(preds, held["gts"], iou)
119         bsc = score(held["baseline"], held["gts"], iou)
120         clf.append(sc)
121         base.append(bsc)
122         print(f"    held-out {held['name']}: LEARNED P={sc.precision:.3f}
R={sc.recall:.3f} F1={sc.f1:.3f}"
123             f"    | baseline P={bsc.precision:.3f} R={bsc.recall:.3f}
F1={bsc.f1:.3f}")
124         mean = lambda xs, a: sum(getattr(s, a) for s in xs) / len(xs)
125         print(f"    MEAN held-out: LEARNED P={mean(clf, 'precision'):.3f}
R={mean(clf, 'recall'):.3f} "
126             f"F1={mean(clf, 'f1'):.3f} | baseline P={mean(base, 'precision'):.3f} "
127             f"R={mean(base, 'recall'):.3f} F1={mean(base, 'f1'):.3f}")
128         result["mean_learned_f1"] = mean(clf, "f1")
129         result["mean_baseline_f1"] = mean(base, "f1")
130
131         # Final model on ALL videos (for production / running on a fresh held-out video).
132         Xall = [row for v in videos for row in v["X"]]
133         yall = [t for v in videos for t in v["y"]]
134         final = LogisticRegression().fit(Xall, yall, FEATURE_NAMES)
135         weights = dict(zip(FEATURE_NAMES, [round(float(w), 3) for w in final.w]))
136         print(f"\nfinal model standardized weights (sign/size = influence): {weights}")
137         if model_out:
138             final.save(model_out)
139             print(f"final model (all videos) -> {model_out}")
140             result["model_out"] = model_out
141         return result
142
143
144     def main() -> None:
145         ap = argparse.ArgumentParser(description="Distill Gemini labels into a local rally
classifier.")
146         ap.add_argument("--manifest", required=True,
147                         help="JSON: [{name, trajectory, fps, frame_width, labels}, ...]
(labels = Gemini CSV)")
148         ap.add_argument("--iou", type=float, default=0.5)
149         ap.add_argument("--threshold", type=float, default=0.5, help="classifier probability
cutoff")
150         ap.add_argument("--model-out", default="output/local_rally_model.json")
151         args = ap.parse_args()
152         with open(args.manifest, encoding="utf-8") as f:
153             specs = json.load(f)
154         run(specs, iou=args.iou, threshold=args.threshold, model_out=args.model_out)
155
156
157     if __name__ == "__main__":
158         main()
159

```

20. user (2026-06-10T05:34:45.341Z)

No files found

21. user (2026-06-10T05:34:46.446Z)

C:\Users\avidu\Projects\badminton-highlight-indexer\docs\archives\quality-iterations\2026-06-multivideo-lovo\README.md:25:GPU never double-booked, race-free via a `kushagra\_traj.csv`-exists gate). Manifest: `output/human\_lovo\_manifest.json`.

C:\Users\avidu\Projects\badminton-highlight-indexer\docs\archives\quality-iterations\2026-06-multivideo-lovo\README.md:76: `output/targettestvideo\_vs\_human.csv`. Manifest now n=3 (`output/human\_lovo\_manifest.json`).

C:\Users\avidu\Projects\badminton-highlight-indexer\docs\archives\quality-iterations\2026-06-multivideo-lovo\README.md:241:python -m backend.eval.calibrate_local --manifest output/human_lovo_manifest.json --metric f1 # heuristic LOVO

C:\Users\avidu\Projects\badminton-highlight-indexer\docs\archives\quality-

```
iterations\2026-06-multivideo-lovo\README.md:242:python -m backend.eval.rally_seq_proto
--manifest output/human_lovo_manifest.json # learned prototype
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\CODE_MAP.md:138:[Omitted long matching
line]
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\eval\rally_seq_proto.py:26:
python -m backend.eval.rally_seq_proto --manifest output/human_lovo_manifest.json
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\NEXT_STEPS.md:44:- Eval corpus:
`output/human_lovo_manifest.json` (badminton n=6). Roora CSVs:
`%TEMP%\roora\{pickleball,tabletennis}_rallies.csv`;
```

22. user (2026-06-10T05:34:52.548Z)

```
C:/Users/avidu/Projects/badminton-highlight-indexer/backend/eval/__init__.py
C:/Users/avidu/Projects/badminton-highlight-indexer/backend/eval/annotations.py
C:/Users/avidu/Projects/badminton-highlight-indexer/backend/eval/classifier.py
C:/Users/avidu/Projects/badminton-highlight-indexer/backend/eval/metrics.py
C:/Users/avidu/Projects/badminton-highlight-indexer/backend/eval/training.py
C:/Users/avidu/Projects/badminton-highlight-indexer/backend/eval/calibration.py
C:/Users/avidu/Projects/badminton-highlight-indexer/backend/eval/gemini_refine.py
C:/Users/avidu/Projects/badminton-highlight-indexer/backend/eval/audio/__init__.py
C:/Users/avidu/Projects/badminton-highlight-indexer/backend/eval/audio/e1_probe.py
C:/Users/avidu/Projects/badminton-highlight-indexer/backend/eval/calibrate_local.py
C:/Users/avidu/Projects/badminton-highlight-indexer/backend/eval/calibrate_wasb.py
C:/Users/avidu/Projects/badminton-highlight-indexer/backend/eval/distill_local.py
C:/Users/avidu/Projects/badminton-highlight-indexer/backend/eval/export_wasb_segments.py
C:/Users/avidu/Projects/badminton-highlight-indexer/backend/eval/harness.py
C:/Users/avidu/Projects/badminton-highlight-indexer/backend/eval/regression.py
C:/Users/avidu/Projects/badminton-highlight-indexer/backend/eval/batch.py
C:/Users/avidu/Projects/badminton-highlight-indexer/backend/eval/eval_partial_labels.py
C:/Users/avidu/Projects/badminton-highlight-indexer/backend/eval/rally_seq_proto.py
```

23. user (2026-06-10T05:34:55.316Z)

```
EXISTS on disk
.gitignore:39:output/    output/human_lovo_manifest.json
```

24. user (2026-06-10T05:36:17.534Z)

```
docs/NEXT_STEPS.md | 46 ++++++
1 file changed, 46 insertions(+)
26ee2de docs: next-steps + multisport findings (WASB transfers to table tennis, not pickleball)
```

25. user (2026-06-10T05:36:18.138Z)

```
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\BACKLOG.md:25:- ? **`yolo_hybrid`
naming** ? the registry key, `yolo_*` config keys, and `yolo-hybrid-*` DB detector tags keep the
"yolo" name for back-compat despite no YOLO remaining. Optional rename. **Revisit:** if/when a
clean break is worth the migration.
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\CODE_MAP.md:243:- **spec.yaml <->
harvest.py coupling:** rule `path` strings dotted-navigate harvest's stage/metric/detail names;
rename a detail in harvest.py and the matching rule silently stops firing. Add rules in
spec.yaml, not code.
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\DEFERRED_HARDENING_PLAN.md:38:-
**Rename the diagnostic `setup.py`** (e.g. `scripts/doctor.py`) so it no longer
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\DEFERRED_HARDENING_PLAN.md:44:-
`setup.py` rename touches docs and any muscle-memory invocations ? grep first.
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\NEXT_STEPS.md:39:3. **Housekeeping:**
**rename repo off "badminton" (multisport); confirm collector = corpus/event-store system-of-
record.
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\OWNED_MODEL_IMPLEMENTATION_PLAN.md:73:
feature / per-sport calibration). The collector + indexer are already sport-generic. ? **ACTION:
rename the repo off
C:\Users\avidu\Projects\badminton-highlight-
indexer\docs\OWNED_MODEL_IMPLEMENTATION_PLAN.md:154:| 5 | Rename repo off "badminton" | **owner
```

```
action** ? |
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\archives\decisions\DECISIONS.md:254:-
The `yolo_hybrid` registry key is now a slight misnomer (no YOLO); keep it for back-compat or
rename in a follow-up.
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\archives\past_projects\low-regret-
moves\LOW_REGRET_MOVES_PLAN.md:75:??? segmenters/ # (renamed/moved from current
indexer/)
```

26. user (2026-06-10T05:36:28.630Z)

```
28:         CREATE TABLE IF NOT EXISTS videos (
29:             id TEXT PRIMARY KEY,
30:             filepath TEXT NOT NULL,
31:             processed_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
32:             status TEXT NOT NULL,
33:             validation_results TEXT
34:         )
35:     """)
36-
37-     # Play Segments Table (Extensible metadata JSON)
38-     cursor.execute("""
39:         CREATE TABLE IF NOT EXISTS play_segments (
40:             id INTEGER PRIMARY KEY AUTOINCREMENT,
41:             video_id TEXT NOT NULL,
42:             start_time REAL NOT NULL,
43:             end_time REAL NOT NULL,
44:             duration REAL NOT NULL,
45:             server TEXT,
46:             receiver TEXT,
47:             metadata TEXT,
48:             FOREIGN KEY (video_id) REFERENCES videos(id) ON DELETE CASCADE
49:         )
50:     """)
51-
52-     # Events Table
53-     cursor.execute("""
54:         CREATE TABLE IF NOT EXISTS events (
55:             id INTEGER PRIMARY KEY AUTOINCREMENT,
56:             segment_id INTEGER NOT NULL,
57:             timestamp REAL NOT NULL,
58:             type TEXT NOT NULL,
59:             player TEXT,
60:             details TEXT,
61:             FOREIGN KEY (segment_id) REFERENCES play_segments(id) ON DELETE CASCADE
62:         )
63:     """)
64-
65-     # Versioned migrations live here. PRAGMA user_version is the schema
66-     # contract ? bump it for every change and add an ordered `if version < N`
67-     # block below. Tests assert the expected version, so accidental drift fails CI.
68-     version = cursor.execute("PRAGMA user_version").fetchone()[0]
69-
70-
71-     CREATE TABLE IF NOT EXISTS candidate_segments (
72:         id INTEGER PRIMARY KEY AUTOINCREMENT,
73:         video_id TEXT NOT NULL,
74:         detector TEXT NOT NULL,
75:         chunk_index INTEGER,
76:         start_time REAL NOT NULL,
77:         end_time REAL NOT NULL,
78:         duration REAL NOT NULL,
79:         confidence REAL,
80:         stats TEXT,
81:         detection_params TEXT,
82:         confirmed_segment_id INTEGER,
83:         human_label TEXT,
```

```
88-             notes TEXT,  
89-             created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
```

27. user (2026-06-10T05:36:42.411Z)

```
C:/Users/avidu/Projects/badminton-highlight-indexer/run_all_tests.py  
C:/Users/avidu/Projects/badminton-highlight-indexer/run_eval.py  
C:/Users/avidu/Projects/badminton-highlight-indexer/run_phase3_eval.py  
C:/Users/avidu/Projects/badminton-highlight-indexer/run_process_and_stitch.py  
C:/Users/avidu/Projects/badminton-highlight-indexer/run_regression.py
```

28. user (2026-06-10T05:36:46.420Z)

```
local_rally_model.json EXISTS on disk (gitignored)  
docs/NEXT_STEPS.md  
docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md  
  
docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md
```

29. user (2026-06-10T05:36:56.483Z)

```
90     - `@run_regression.py::main` ? `regress_with_provider` vs baseline. Exit codes semantic:  
    **0=ok, 1=REGRESSION (CI gate), 2=missing DB**. ? `--update-baseline` runs AFTER compare  
    (snapshots even a regressing run); `::default_db_path` resolves via  
    `backend.config.load_config`.  
91  
92     - `@backend/config/__init__.py` ? ? canonical config import surface. Re-exports  
    `load_config`, `save_default_config` (from `.loader`) + `AppConfig`, `Config`, `IndexingConfig`,  
    `ValidationConfig` (from `.models`); `__all__` = exactly those 6. Use `from backend.config  
import load_config, AppConfig`.  
93     - `@backend/config/models.py` ? Pydantic v2 models; **single source of truth for  
defaults**.  
94     - `::AppConfig` ? root. Scalars `sport='badminton'`, `ai_provider='gemini'`,  
    `ai_model='gemini-2.5-flash'`; sections `validation`, `indexing`, `output`, `stitching`  
    (`default_factory`). `model_config` = {extra:'allow', validate_assignment:True}. ?  
    extra='allow' -> unknown/typo'd config.json keys silently kept (legacy tolerance).  
95     - ? `::AppConfig.get` ? legacy dict-compat shim: returns nested **sections as plain  
dicts** (`model_dump(mode='json')`, NOT the model) so chained  
    `config.get("indexing", {}).get(...)` keeps working; falls back to searching a leaf key in any  
    section (sections discovered dynamically from `model_fields`). ? "bit every validator on real  
runs" per docstring ? returning the model breaks `.get`. `::AppConfig.__getitem__` raises  
    `KeyError` for unknown top-level keys else delegates to `.get`.  
96     - `::IndexingConfig` ? segmenter/detector knobs (`motion_threshold=0.08`,  
    `min_segment_duration=3.0`, `dead_time_merge_gap=2.0`, `chunk_duration=90.0`,  
    `chunk_overlap=10.0`, `yolo_velocity_threshold=0.15`, `yolo_frame_skip=3`,  
    `ai_handoff_padding=2.0`, `ai_max_workers=5`, `log/store_yolo_candidates=True`, nested  
    `tracknet`). ? `default_segmenter='yolo_hybrid'` HERE but config.json overrides to `gemini`. ?  
    `::IndexingConfig.segmenter_must_be_registered` `@field_validator` is a **no-op pass-through** ?  
    a typo'd segmenter validates fine, only fails later at `segmenter_registry.get(...)`.  
97     - `::ValidationConfig` (`min_resolution_width=1280`, `min_resolution_height=720`, `min_f  
ps=29.9`, `min_blur_threshold=20.0`, `max_scene_cuts_per_minute=0.5`, nested `relevance`);  
    `::ValidationRelevanceConfig` (court HSV gates, `court_pixels_min_ratio=0.05`).  
    `::TrackNetConfig` (WSL invocation contract: `wsl_distro='Ubuntu'`, `repo_dir`,  
    `conda_env='TrackNetV4'`, `weights_path='', `queue_length=5`, `velocity_threshold=0.02`).  
    `::OutputConfig` (`default_highlights_name`, `db_name='sports_indexer.db'`,  
    `output_dir='output'` ? CLI `--output-dir` overrides it on the typed model in `main()`).  
    `::StitchingConfig`  
    (`begin_padding=0.5`, `end_padding=0.5`, `use_cache=False`, `min_shot_count=1`). `::Sport` Literal  
    of 5 sports. `::Config` = `AppConfig` alias.  
98     - `@backend/config/loader.py` ? load/save with defaults-merge.  
99     - `::DEFAULT_CONFIG_PATH` = Path("config.json")` (? CWD-relative, NOT repo-root).  
    `::get_built_in_defaults` = `AppConfig().model_dump(...)`.  
100    - `::load_config` ? precedence built-in-defaults -> file overrides. ? **shallow top-  
level merge** ({**defaults, **file_data}): a section in config.json REPLACES the whole
```



```

defaults dict for that section (Pydantic then re-applies field defaults for missing leaves).
Missing/unreadable/parse-error file -> fully-defaulted AppConfig + `[CONFIG WARNING]` (non-
fatal).
101     - `::save_default_config` ? writes fresh `config.json`, **overwrites** if present
(used by setup.py/--setup). `::get_default_config` ?? transitional plain-dict alias.
102     - `@config.json` ? on-disk config; mirrors `AppConfig`. ?
**indexing.segmenter='gemini'*** diverges from model default `yolo_hybrid` ? file
wins at runtime.
103     - `@setup.py` ? `::init_default_config` ? delegates to
`backend.config.save_default_config` (single-source defaults; skips if config.json exists).
`::run_pip_install` GPU-aware (`nvidia-smi` -> cu124 else cpu). `::run_diagnostics` (Python?3.8,
ffmpeg/ffprobe on PATH, optional torch+CUDA).
104
105     ### 2.1 Segmenters (under pipeline/segmenters) ?
106     - `@backend/pipeline/segmenters/base.py`
107     - `::BasePlaySegmenter` ABC ? `__init__(db, config)`; abstract
`name`, `description`, `process_video`. § `process_video(video_id, video_path, player_pool=None,
max_frames=None)` -> List[dict{id, start_time, end_time, duration, metadata}]`.
108     - `::SegmenterRegistry`, `::segmenter_registry` (singleton). ? **register a new
segmenter**: subclass `BasePlaySegmenter`, call `segmenter_registry.register(MyCls)` at module
bottom, AND import the module in `@backend/pipeline/segmenters/__init__.py` (registration is an
IMPORT side-effect). ? `register()` builds `cls(None, {})` just to read `name` ->
name/description/`__init__` MUST NOT touch `self.db`/`self.config`. ? `list_available()` re-
instantiates + swallows exceptions into "No description available." (a throwing `description` is
masked).
109     - `@backend/pipeline/segmenters/__init__.py` ? imports all 5
(`motion,gemini,yolo_hybrid,tracknet_hybrid,wasb_hybrid`) = the de-facto registration manifest;
re-exports them + `segmenter_registry` via `__all__`.
110     - `@backend/pipeline/segmenters/wasb_hybrid.py::WasbHybridSegmenter` (name
`wasb_hybrid`) ? WSL WASB substrate (MIT; ships pretrained badminton weights, the LIVE
default). Imports `WasbRunner`/`WasbConfig` from `pipeline/detectors/wasb_runner`. Config
`indexing.wasb.velocity_threshold(0.02)` + shared
`dead_time_merge_gap/min_action_window_duration/ai_handoff_padding/ai_max_workers`. ? whole-
video single pass; reuses YOLO-named keys `log_yolo_candidates/store_yolo_candidates`; needs
`{PROVIDER}_API_KEY` (missing -> []); `detection_params` has NO `queue_length` (unlike
tracknet). ? divergent shot_count: `int(shot_count)` inside try with no `is None` short-circuit
(copy-paste drift).
111     - `@backend/pipeline/segmenters/tracknet_hybrid.py::TrackNetHybridSegmenter` (name
`tracknet_hybrid`) ? copy-paste twin of wasb_hybrid; imports `TrackNetRunner`/`TrackNetConfig`
from `pipeline/detectors/tracknet_runner`; `indexing.tracknet.velocity_threshold(0.02)`;
`detection_params` adds `queue_length`. ? any P3/P4 fix must be applied to BOTH (and arguably
yolo_hybrid).
112     - `@backend/pipeline/segmenters/yolo_hybrid.py::HybridYoloSegmenter` (name
`yolo_hybrid`, no YOLO remains ? torchvision detector + supervision ByteTrack; ADR-005).
`::PERSON_CLASS_ID=1` ? (ultralitics used 0; wrong value -> zero detections).
`::lazy_load_model` (lazy `torchvision` import so module/tests load without it),
`::extract_action_windows_from_chunk`, `::make_tracker` (per-chunk ByteTrack ? track IDs
chunk-local), `::DETECTOR_FACTORIES`/`::DEFAULT_DETECTOR='fasterrcnn_resnet50_fpn'`. ?
velocity at `effective_fps = fps/frame_skip` -> `yolo_velocity_threshold` coupled to
`frame_skip`; pipelined-vs-sequential P2 (prefetch ffmpeg overlaps sequential GPU);
`yolo_tracker`/`bytetrack.yaml` is dead/back-compat. ? `gemini_*` config keys are deprecated
aliases still honored.
113     - `@backend/pipeline/segmenters/gemini.py::GeminiPlaySegmenter` (name `gemini`) ?
pure-AI, no P2. ? chunks only if `duration>chunk_duration`; `max_workers=5` hardcoded;
`max_segment_duration=90.0` hardcoded; `debug_duration` silently truncates; `parse_time` accepts
colon timestamps (warns); does NOT persist candidate segments (only hybrids feed the fine-tuning
table).
114     - `@backend/pipeline/segmenters/motion.py::MotionPlaySegmenter` (name `motion`) ?
pixel absdiff baseline. ? **metadata + serve/smash/foul/winner events are RANDOM**
(random.sample/randint/uniform/choice); only segmenter using `db.add_event` and passing
server/receiver; uses `print()`; honors `max_frames`.
115
116     ### 2.2 Detectors & windowing (WSL-quarantined; ABC abstraction complete)
117     - `@backend/pipeline/detectors/base.py` ? DetectorRunner ABC (healthcheck() ->
Tuple[bool,str] never-raises; run_predict(...) -> Optional[str] CSV path). Contract deliberately

```

matches the pre-existing tuple returns and call sites in the hybrids. Rich module docstring includes "how to add a new runner", Linux-native path notes (the de-risk goal), and guidance on threading health into reports. Re-exported from `detectors/\_\_\_init\_\_\_py`. This finishes the WSL de-risk seam (low-regret 6 / review item 1).

```

118     - `@backend/pipeline/detectors/tracknet_runner.py`
119     - `::TrackNetConfig` (+`.from_indexing_cfg`; ? key drift `wsl_distro` JSON -> `distro`
field), `::TrackNetRunner` ? runs TrackNet `predict.py` in WSL.
120     - `::to_wsl_mnt_path` (`C:\x`->`/mnt/c/x`; ? POSIX/~` passthrough, UNC unhandled),
`::TrajectoryPoint` (frame,visible,x,y).
121     - `::parse_trajectory_csv` @staticmethod** + `::trajectory_to_action_windows`
@staticmethod** = MODEL-AGNOSTIC brain, re-exported verbatim by WasbRunner. ? `weights_path`
defaults `''` -> `run_predict` returns None; predict.py only `.mp4/.avi` + hard-names
`<stem>_predict.csv`; `visible==False` resets prev (occlusion gap breaks track); `fps<=0->30`,
`frame_width<=0->1280` silent; `track_id_count` hardcoded 1; `timeout_sec=0` -> NO timeout
(hangs forever). `::stage_video` "OK" sentinel.
122     - `@backend/pipeline/detectors/wasb_runner.py::WasbRunner` (inherits DetectorRunner)
(+`::WasbConfig`, from `indexing.wasb`; ? `weights_path` HAS a real default
`wasb_badminton_best.pth.tar` ? no required guard) ? stages `wasb_infer.py` (`::INFER_WIN`) +
video into WSL, runs in `wasb` env. `::parse_trajectory_csv`/`::trajectory_to_action_windows` =
`staticmethod(TrackNetRunner.*)` (explicit rebind = the plugin-equivalence point). Output hard-
named `<stem>_wasb.csv`. ? `wasb_infer.py` re-staged each run (editing Windows copy inert until
staged); `wsl_bash` duplicated (drift risk). Both runners now satisfy the DetectorRunner ABC
for future swappability.
123     - `@backend/pipeline/detectors/wasb_infer.py` (WSL) ? `::run` core. ? imports WASB-SBDT
repo modules + torch ? NOT importable on Windows. Reboot-resumable detector cache:
`::DET_FILE='det_raw.jsonl'`, `::MANIFEST_FILE='manifest.jsonl'`, `::CACHE_VERSION=1`,
`::load_and_clean_cache` (reads longest CONTIGUOUS prefix where `w==line index`, rewrites to
heal crash-truncated tail), `::manifest_compatible` (? `frames_dir` abspath'd but `weights`
compared raw), `::atomic_write_text`/`atomic_write_trajectory`, `::dets_to_tracker` (xy MUST be
np.array), `::build_cfg` (`runner.gpus=[0]` hardcoded), `::extract_frames` (cv2-PNG SLOW),
`::default_cache_dir` (`<stem>_wasbcache` next to out). ? GPU pass resumable; CPU tracker
STATEFUL -> always fully re-run; `--fresh` ignores cache.
124     - `@backend/pipeline/detectors/rally_gate.py` ? Gate A net-crossing post-filter (pure,
no I/O/model). `::estimate_play_axis` (larger-spread axis; ? ties -> `x`),
`::count_net_crossings` (sign changes vs median net; deadband jitter zone),
`::gate_windows`/`::apply_gate` (`min_crossings=2` default; kills single-toss service feeds).
`::GatedWindow` (start,end,crossings,visible_points,kept). ? window contract here is
`(start,end)` tuples NOT the rich window dict ? caller must adapt. Consumed by calibrate_local /
distill_local / export_wasb_segments.
125
126     ### 2.3 Eval & calibration (all pure core; I/O in CLI drivers)
127     - `@backend/eval/metrics.py` ? scoring primitive (THE spine). `::interval_iou`,
`::match_intervals` (greedy, deterministic), `::score`->`::Score`(`.as_dict()` = wire shape),
`::pr_curve`, `::Match`, `::MatchResult`. ? greedy not Hungarian; empty matches -> 0.0 (not NaN)
for mean_iou/boundary errors.
128     - `@backend/eval/annotations.py` ? generic GT loader. `::load_annotations` (`start,end`
CSV; RAISES on bad rows/`start>=end`), `::parse_timestamp` (decimal or MM:SS/HH:MM:SS),
`::write_scaffold`.
129     - `@backend/eval/calibration.py` ? overfit guard. `::cross_validate` (within-video
k-fold; ? defaults `metric='recall'` deliberately), `::leave_one_video_out` (honest cross-video;
needs ?? labelled videos), `::improvement_report` (OPTIMISTIC ? selected on full GT),
`::select_best`, `::kfold_indices`, `::pct_improvement` (? from-zero -> `inf`), `::evaluate`. $
`PredictFn = Callable[[Config], List[Interval]]` = the plug-in seam. ? `generalization_gap
>=0.1` = overfit alarm.
130     - `@backend/eval/calibrate_wasb.py` ? WASB tuning CLI. `::BASELINE=(0.02,2.0,2.0)`,
`::default_grid` (45 cfgs), `::make_predict_fn` (parses trajectory once, closes over
`trajectory_to_action_windows`), `::load_golden_gts` ($`start_time,end_time` cols; ? SILENTLY
skips bad rows ? opposite of annotations.load_annotations). ? `{load_golden_gts,
make_predict_fn, BASELINE}` imported by 4 downstream drivers (calibrate_local,
export_wasb_segments, gemini_refine, audio/e1_probe).
131     - `@backend/eval/calibrate_local.py` ? distill Gemini->local thresholds (windowing +
net-crossing gate, no cloud at runtime). `::local_grid` (180 cfgs), `::make_local_predict_fn`
(adds `rally_gate.apply_gate` when `min_crossings>0`), `::build_videos` (? `SystemExit` if a
labels file yields no rallies), `::run`/`::main`. $
`LocalConfig=(velocity,merge,min_dur,min_crossings)`, `BASELINE_LOCAL=(0.02,2.0,2.0,0)`;

```

manifest JSON shared with distill_local.

```
132 - `@backend/eval/export_wasb_segments.py` ? tuned cfg -> golden/slicer CSV + comparison
+ optional clip cut. `::TUNED=(0.05,1.0,1.0)` (? from ONE video),
`::SEG_FIELDS`, `::COMP_FIELDS`, `::build_comparison`, `::seconds_to_mmss`, `::maybe_slice`
(lazy-imports rally_slicer). `--slice` requires `--video`. ? `write_segments_csv` output loads
straight into `rally_slicer.load_rally_labels` AND re-imported by gemini_refine ? schema is
load-bearing.
133 - `@backend/eval/batch.py` ? corpus aggregation (pure; orchestration is in
run_phase3_eval). `::aggregate` (micro pool TP/FP/FN, TP-weighted mean_iou, macro F1),
`::default_stages_for` (`auto`; `::FINAL_ONLY_SEGMENTERS={motion,gemini}`),
`::resolve_video_path` (normalizes collector `./data\..` Windows paths), `::format_aggregate`.
134 - `@backend/eval/harness.py` ? DB-pred scorer (no re-run).
`::STAGES=('candidates','final')`, `::get_predictions`
(candidates->`query_candidate_segments(detector=)`; final->`query_play_segments({})`); ? unknown
stage raises ValueError; reused by regression.py),
`::evaluate`->`::EvalReport`(`::StageReport`)->`::report_to_dict` (input contract for
batch.aggregate), `::format_report_text`.
135 - `@backend/eval/training.py` ? learned-segmenter data layer. `::YOLO_FEATURE_NAMES`
(5-d), `::extract_yolo_features` (? reads `row['stats']` dict from DB; missing fields default
0.0), `::label_candidates` (same IoU `match_intervals` as evaluator),
`::build_training_set`->(X,y,intervals); `feature_fn` is the substrate plug-in seam.
136 - `@backend/eval/classifier.py::LogisticRegression` ? numpy-only logreg (no sklearn).
`fit/predict_proba/predict/to_dict/from_dict/save/load`. ? `class_weight='balanced'` default;
stores feature standardization (mean/std) ? inference MUST use same feature ORDER; `_sigmoid`
clips z to [-30,30]; `lr=0.1,epochs=1000,l2=1e-3`; JSON "model":"logreg" tag.
137 - `@backend/eval/eval_partial_labels.py` ? score pipeline vs **PARTIAL human labels**
(PR #60). `::restrict_to_window` (drop preds outside `[0, last GT end]` so un-labeled-tail
detections aren't FPs ? THE partial-label methodology), `::run` (baseline/+gate/tuned/+gate ops
table + per-rally diff via `export_wasb_segments.build_comparison`), `::sweep`/`--sweep` (grid
vs labels; ? fit-on-self = optimistic), `::DEFAULT_GRID`, `::TUNED=(0.05,1.0,1.0)`. ?
`window_end` inferred from `max(GT end)` is WRONG if a video is labelled PAST its last rally ?
P1 fix = collector persists `labeled_window` in the manifest. Reuses calibrate_wasb + rally_gate
+ metrics.
138 - `@backend/eval/rally_seq_proto.py` ? **PROTOTYPE** learned per-FRAME rally segmenter =
owned-model **Gen-0 seed** (PR #61). 6 features over the WASB trajectory (`vis_rate, spd_mean,
spd_max, cross_rate, x_std, y_std`) -> `classifier.LogisticRegression`, honest LOVO.
`::build_frame_arrays`, `::features` (scipy `uniform/maximum_filter1d` windowed; ? speed NOT
fps-normalized yet ? 30 vs 60fps), `::labels_for`, `::probs_to_segments`
(smooth->threshold->merge/min-dur), `::roc_auc` (rank-based, no sklearn), `::run` (per-video AUC
+ LOVO-threshold F1 + oracle-threshold F1). ? DIRECTIONAL only (n?5, linear, threshold-transfer
unsolved); held-out per-frame AUC 0.83?0.92, **beats the heuristic on multi-court footage**
(Kushagra/gBaaddy). Manifest `output/human_lovo_manifest.json` (the n=5 human corpus) shared w/
calibrate_local.
139 - ? **MULTI-COURT / background-game confound (2026-06-08 finding):** WASB tracks EVERY
shuttle in frame incl. other courts; videos with background games (Kushagra, gBaaddy) have high
dead-time shuttle visibility (41?57% vs 18?22% clean) ? the velocity-windowing heuristic can't
find clean rally boundaries (F1 0.16/0.28 vs ~0.75 on single-court). Diagnostic = in-rally÷dead-
time visibility "contrast" (?3.6× heuristic works, ?1.9× fails). Lever = **foreground-court
isolation** (spatial gate to the prominent court, needs real court-region detection). Academy =
multi-court = the target env. See `docs/archives/quality-iterations/2026-06-multivideo-lovo/`.
140 - `@backend/eval/distill_local.py` ? distill Gemini -> a LEARNED classifier (beyond
calibrate_local's threshold ceiling). `::FEATURE_NAMES` (5 fps/res-INVARIANT:
duration,peak,mean,active_ratio,net_crossings), `::build_candidates` (recall-first proposal),
`::predict_rallies` (proba filter then `gemini_refine.merge_overlaps(gap_tol=1.0)`),
`::run`/`::main`. consts `PROPOSAL=(0.005,1.0,0.5)`, `BASELINE_LOCAL=(0.02,2.0,2.0)`. ? leave-
one-video-out needs ?2 videos; final model trained on all.
141 - `@backend/eval/gemini_refine.py` ? refine WASB candidates with Gemini over SHORT
padded clips (precision lever; eval/tuning, NOT the live pipeline).
`::merge_overlaps(gap_tol=0.0)` (reused by distill_local with gap_tol=1.0), `::refine_window` (?
FRESH genai client per worker ? shared throws socket errors), `::full_video_rallies`,
`::run`/`::main` (`--full-video` skips `--trajectory`). ? requires `GEMINI_API_KEY` (raises
SystemExit); imports `BASELINE` from calibrate_wasb +
`TUNED`/`write_segments_csv`/`seconds_to_mmss` from export_wasb_segments.
142 - `@backend/eval/regression.py` ? CI gate. `::DEFAULT_TOLERANCE=0.05`,
`::DEFAULT_BASELINE_PATH=output/regression_baseline.json`, `::regress`,
```

```

`::regress_with_provider` (GT via ?`annotation_registry.get(tier)`;
"file"->FileProvider/"vendor"->HumanVendor/"auto"->oracle),
`::save_baseline`/`::load_baselines`, `::RegressionResult`. ? imports `annotation_registry` from
**`backend.annotations`** NOT `backend.eval.annotations`; gate fires on precision OR recall drop
(F1 reported, not gated); no baseline -> `regressed=False` (silent pass).
143
144     ### 2.3a Audio (E1 probe; NOT adopted for fusion ? ADR-007)
145     - `@backend/pipeline/audio/hit_detector.py` ? classical-onset shuttle-hit detector. Pure
numpy + stdlib `wave` (NO scipy); ffmpeg only for extraction. `::HitEvent`(t,strength),
`::extract_audio` (ffmpeg -> mono 16k PCM WAV), `::load_wav`, `::onset_envelope` (pure-numpy
STFT spectral flux; `band=(lo,hi)` band-limit), `::detect_hits` (adaptive `mean+k.std`;
`k`=sensitivity knob), `::detect_hits_in_video`. ? separate signal ? WASB never modified by
this; per audio-e1-findings recall 100% but discrimination ~2.2x, band-pass null -> NOT adopted
(PR #35).
146     - `@backend/eval/audio/e1_probe.py` ? E1 GO/NO-GO diagnostic (no training/fusion).
`::cluster_hits` (groups thwacks; ? `min_hits>=2` so a lone service-feed hit isn't a rally),
`::run` (discrimination ratio + recall recovery + audio-only P/R/F1), `::main`. reuses
`hit_detector` + `calibrate_wasb.{load_golden_gts,make_predict_fn,BASELINE}` + `metrics`.
147
148     ### 2.4 Stitcher / tools / utils
149     - `@backend/pipeline/stitcher/compiler.py::VideoCompiler` ? `compile_highlights(raw,
segments:[(start,end)], out_name)`. Per-clip re-encode -> `-f concat -c copy`. `::parse_time`
(mirrors gemini.parse_time; ? unparseable -> 0.0), `::get_video_duration`. ? if duration probe
fails (0.0) NO boundary validation; `begin_padding`(0.5)/`end_padding`(1.0) defaults; temp clips
in a TemporaryDirectory under output_dir; raises ValueError/FileNotFoundError/RuntimeError; uses
`print()`. ? two-stage codec contract: clips re-encoded with IDENTICAL settings precisely so
final concat can stream-copy ? change a preset and concat may break.

```

30. user (2026-06-10T05:36:57.674Z)

C:\Users\avidu\Projects\badminton-highlight-indexer\docs\COMMERCIALIZATION.md:43:[Omitted long matching line]

C:\Users\avidu\Projects\badminton-highlight-indexer\docs\NEXT_STEPS.md:11:rally-vs-non-rally problem). **Framework = ms-swift, base = Qwen3-VL** (Gemma-4 AMBER/bench) come **later** for the C:\Users\avidu\Projects\badminton-highlight-indexer\docs\PLATFORM_ARCHITECTURE.md:260:> **The moat is *not* the model.** Anyone can download Qwen-VL/Gemma. The defensible C:\Users\avidu\Projects\badminton-highlight-indexer\docs\PLATFORM_ARCHITECTURE.md:269:> are open, Apache-2.0 models only** (Qwen-VL / InternVL / Gemma 4). Detail in the C:\Users\avidu\Projects\badminton-highlight-indexer\docs\PLATFORM_ARCHITECTURE.md:304:**Qwen3-VL-4B/8B (Apache-2.0)**; **Gemma 4 multimodal (Apache-2.0, 2026)** is a clean C:\Users\avidu\Projects\badminton-highlight-indexer\docs\PLATFORM_ARCHITECTURE.md:309:- ? **Reject for the flagship:** **Gemma 3** ? custom terms with a **viral derivative

C:\Users\avidu\Projects\badminton-highlight-indexer\docs\PLATFORM_ARCHITECTURE.md:311: permanently bound to Google's *updatable* Prohibited-Use Policy); **Gemma 4's switch to C:\Users\avidu\Projects\badminton-highlight-indexer\docs\PLATFORM_ARCHITECTURE.md:347:| **Together AI** | Managed (tune + serve) | Qwen3-VL / Gemma-3 / Llama-4; image data | verify | partial | ? adapters downloadable | managed Qwen3-VL + serving |

C:\Users\avidu\Projects\badminton-highlight-indexer\docs\PLATFORM_ARCHITECTURE.md:419: InternVL / Gemma 4 ? Apache-clean); **(2)** keep the paid API strictly as a **runtime

C:\Users\avidu\Projects\badminton-highlight-indexer\docs\PLATFORM_ARCHITECTURE.md:531: (Qwen-VL Apache-2.0 vs Gemma 3 custom license vs ?) and confirm a gated commercial

C:\Users\avidu\Projects\badminton-highlight-indexer\docs\PLATFORM_ARCHITECTURE.md:563:[Omitted long matching line]

C:\Users\avidu\Projects\badminton-highlight-indexer\docs\OWNED_MODEL_IMPLEMENTATION_PLAN.md:25: (Ratifies the prior "Apache-2.0 only" to include MIT/BSD.) Reject viral/NC/custom-restrictive: Gemma 3 (viral),

C:\Users\avidu\Projects\badminton-highlight-indexer\docs\OWNED_MODEL_IMPLEMENTATION_PLAN.md:26: **Gemma 3n** (Gemma Terms follow derivatives ? even cleaner reject), Llama 4 (700M-MAU cap), Commons-Clause repos.

C:\Users\avidu\Projects\badminton-highlight-indexer\docs\OWNED_MODEL_IMPLEMENTATION_PLAN.md:35:**Framework ? base** (the Gemma-4-vs-ms-swift clarification): a *framework* is the forge (runs SFT/GRPO, emits

C:\Users\avidu\Projects\badminton-highlight-indexer\docs\OWNED_MODEL_IMPLEMENTATION_PLAN.md:42: pre-release. **Gemma-4 stays AMBER and bench-only** ? its Apache grant is likely but its

Prohibited-Use posture is

C:\Users\avidu\Projects\badminton-highlight-indexer\docs\OWNED_MODEL_IMPLEMENTATION_PLAN.md:93: target stays reachable without re-architecting. (On-device base stays on the Qwen3-VL-4B Apache lineage ? *not* Gemma-3-270M.)

C:\Users\avidu\Projects\badminton-highlight-indexer\docs\archives\checkpoints\2026-06-strategy-foundation\README.md:25:- **Owned-model base = Apache-2.0-clean, video-capable** (Qwen2.5-VL-7B/32B, Qwen3-VL, Gemma 4); **reject Gemma 3** (viral license clause); **minors excluded** from the training pool; per-user training needs a separate opt-in.

C:\Users\avidu\Projects\badminton-highlight-indexer\docs\archives\research\OWNED_MODEL_TRAINING_STUDY.md:22:- ? **Gemma 4 stays AMBER and bench-only** (NOT "confirmed clean" as the table below says): Apache grant likely but

C:\Users\avidu\Projects\badminton-highlight-indexer\docs\archives\research\OWNED_MODEL_TRAINING_STUDY.md:37:| **Gen-2 / Option B** | 12?24+ mo, after scaffolding + healthy corpus | end-to-end multimodal VLM: (frames+audio+pose+trajectory) ? rally [start,end] (+semantics); SFT cold-start then **GRPO/RFT with a temporal-IoU-vs-golden reward** | **Qwen3-VL-4B-Instruct** (Apache-2.0, on-device) primary; **Qwen2.5-VL-7B** fallback; **Gemma 4 E4B/12B-Unified** (Apache-2.0 + native video+audio) end-state | rented 24?48 GB A100/L40S (does NOT fit 8 GB box) |

C:\Users\avidu\Projects\badminton-highlight-indexer\docs\archives\research\OWNED_MODEL_TRAINING_STUDY.md:98: Gemma 4 E2B/E4B/12B-Unified (Apache-2.0 + native video/audio), InternVL3 ?38B (MIT/Apache), ASFormer/

C:\Users\avidu\Projects\badminton-highlight-indexer\docs\archives\research\OWNED_MODEL_TRAINING_STUDY.md:99: ActionFormer/TriDet (MIT). **REJECTED:** Gemma 3 (viral Gemma Terms + PUP), Qwen2.5-VL-3B (NC research),

C:\Users\avidu\Projects\badminton-highlight-indexer\docs\archives\research\OWNED_MODEL_TRAINING_STUDY.md:102: (Commons Clause ? use `sj-li/MS-TCN2` MIT). ? **Gemma 4 size names = E2B/E4B/26B-A4B/31B/12B-Unified ? there is no plain "4B".**

C:\Users\avidu\Projects\badminton-highlight-indexer\docs\archives\research\OWNED_MODEL_TRAINING_STUDY.md:127:- **Gemma 4 "clean + native-video" rests on post-cutoff secondary sources** ? keep AMBER until a raw HF LICENSE

C:\Users\avidu\Projects\badminton-highlight-indexer\docs\archives\research\OWNED_MODEL_TRAINING_STUDY.md:128: revision is verified, given the Gemma-3 rejection. Its timestamp-emission recipe is less documented than Qwen's.

C:\Users\avidu\Projects\badminton-highlight-indexer\docs\archives\research\OWNED_MODEL_TRAINING_STUDY.md:147:3. Working Gemma-4 timestamp-emission recipe + exact checkpoint (E4B vs 12B-Unified)?

31. user (2026-06-10T05:38:02.068Z)

```

1      {
2          "sport": "badminton",
3          "ai_provider": "gemini",
4          "ai_model": "gemini-2.5-flash",
5          "validation": {
6              "min_resolution_width": 1280,
7              "min_resolution_height": 720,
8              "min_fps": 29.9,
9              "min_blur_threshold": 20.0,
10             "max_scene_cuts_per_minute": 0.5,
11             "relevance": {
12                 "court_green_lower": [
13                     35,
14                     40,
15                     40
16                 ],
17                 "court_green_upper": [
18                     85,
19                     255,
20                     255
21                 ],
22                 "court_pixels_min_ratio": 0.05
23             }
24         },
25         "indexing": {

```

```
26         "default_segmenter": "gemini",
27         "use_gpu": true,
28         "motion_threshold": 0.08,
29         "min_segment_duration": 3.0,
30         "dead_time_merge_gap": 2.0,
31         "chunk_duration": 90,
32         "chunk_overlap": 10,
33         "detector_model": "fasterrcnn_resnet50_fpn",
34         "detector_score_threshold": 0.5,
35         "yolo_velocity_threshold": 0.15,
36         "yolo_frame_skip": 3,
37         "yolo_tracker": "bytetrack.yaml",
38         "yolo_parallel_chunks": false,
39         "yolo_chunk_workers": 2,
40         "min_action_window_duration": 2.0,
```

32. user (2026-06-10T05:38:03.354Z)

Found 15 files

```
tests\test_normalize_annotations.py
src\cli\normalize_annotations.py
docs\MANUAL_INGESTION_POSITIONING.md
src\reporting\adapter.py
src\reporting\__init__.py
src\cli\curate.py
docs\INGESTION_PIPELINE.md
src\parsers\cvat\rally_cvat.py
tests\test_cvat_parser.py
docs\annotation\ROORA_BRIEF.md
README.md
tests\test_cli.py
src\cli\ingest.py
docs\DECISIONS.md
tests\test_local_import.py
```

33. user (2026-06-10T05:38:14.027Z)

No matches found

34. user (2026-06-10T05:38:15.430Z)

```
C:\Users\avidu\Projects\badminton-highlight-
indexer\docs\NEXT_STEPS.md:4: `docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md`. **Latest blow-by-blow:**
memory `current-state.md` / `session-handoff.md`.
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\NEXT_STEPS.md:42:- Roadmap:
`docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md` · Study:
`docs/archives/research/OWNED_MODEL_TRAINING_STUDY.md`
C:\Users\avidu\Projects\badminton-highlight-
indexer\docs\archives\research\OWNED_MODEL_TRAINING_STUDY.md:12:## Update (2026-06-09, post-
PR-#61-review) ? authoritative roadmap is now `docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md`
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\archives\quality-
iterations\2026-06-multivideo-lovo\README.md:116: `docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md`
(Gen-0 first), gated on n?5 + owner sign-off.
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\archives\quality-
iterations\2026-06-multivideo-lovo\README.md:253: + `docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md`,
after scaffolding).
```

35. assistant (2026-06-10T05:39:30.566Z)

All evidence gathered. Compiling the structured findings now.

36. user (2026-06-10T05:41:30.768Z)

Structured output provided successfully

37. assistant (2026-06-10T05:41:39.769Z)

Audit complete. Structured findings returned.